

Increasing Flexibility in the EU Energy System

Technologies and policies to enable
the integration of renewable
electricity sources



Increasing Flexibility in the EU Energy System

Technologies and policies to enable
the integration of renewable
electricity sources

Abstract

This study explores the current and future flexibility needs in the electricity system as well as existing and potential solutions to efficiently integrate renewable energy and facilitate electrification of end-uses. Key findings highlight the importance of flexible electricity generation, demand response and energy storage solutions for daily, weekly, and seasonal flexibility provision. However, challenges persist due to inadequate market rules and regulatory frameworks, and economic issues. The study recommends additional policy measures to enhance system flexibility, complementing ongoing policy initiatives at the EU and national level.

This document was provided by the Policy Department for Transformation, Innovation and Health for the Committee on Industry, Research and Energy (ITRE).

This document was requested by the European Parliament's Committee on Industry, Research and Energy.

AUTHORS

João GORENSTEIN DEDECCA, Trinomics
Mohammad ANSARIN, Trinomics
Csinszka BENE, Trinomics
Timo VAN DELZEN, Trinomics
Luc VAN NUFFEL, Trinomics
Henjo JAGTENBERG, Trinomics

ADMINISTRATORS RESPONSIBLE

Kristi POLLUVEER
Matteo CIUCCI

EDITORIAL ASSISTANT

Marleen LEMMENS

LINGUISTIC VERSIONS

Original: EN

ABOUT THE EDITOR

Policy departments provide in-house and external expertise to support European Parliament committees and other parliamentary bodies in shaping legislation and exercising democratic scrutiny over EU internal policies.

To contact the Policy Department or to subscribe for email alert updates, please write to:

Policy Department for Transformation, Innovation and Health

European Parliament

L-2929 - Luxembourg

Email: ecti-poldep-b@europarl.europa.eu

Manuscript completed: March 2025

Date of publication: March 2025

© European Union, 2025

This document is available on the internet at:

<http://www.europarl.europa.eu/supporting-analyses>

DISCLAIMER AND COPYRIGHT

The opinions expressed in this document are the sole responsibility of the authors and do not necessarily represent the official position of the European Parliament.

Reproduction and translation for non-commercial purposes are authorised, provided the source is acknowledged and the European Parliament is given prior notice and sent a copy.

For citation purposes, the publication should be referenced as: Gorenstein Dedecca J., Ansarin, M., Bene, C., van Delzen, T., van Nuffel, L., Jagtenberg, H., 2025, *Increasing Flexibility in the EU Energy System - Technologies and policies to enable the integration of renewable electricity sources*, Publication for the Committee on Industry, Research and Energy, Policy Department for Transformation, Innovation and Health, European Parliament, Luxembourg.

© Cover image used under licence from Adobe Stock

CONTENTS

LIST OF FIGURES	4
LIST OF TABLES	5
LIST OF BOXES	5
LIST OF ABBREVIATIONS	6
EXECUTIVE SUMMARY	8
1. INTRODUCTION	10
1.1. Why is flexibility important?	10
1.2. Objectives of the study	10
1.3. Structure of the report	11
2. BACKGROUND AND STATE OF PLAY OF FLEXIBILITY IN THE EU ENERGY SYSTEM	12
2.1. Framework for defining and characterising flexibility	12
2.2. Role of flexibility to address challenges in the EU electricity system	21
3. SOLUTIONS TO MEET THE INCREASING FLEXIBILITY NEEDS OF THE EU ENERGY SYSTEM	34
3.1. Flexible power generation	34
3.2. Energy storage	37
3.3. Demand-side flexibility	41
3.4. Flexibility indirectly provided by domestic electricity networks and interconnectors	45
3.5. Comparison of flexibility solutions	49
4. POLICIES AND MEASURES TO FURTHER ENABLE FLEXIBILITY	56
4.1. Current policy and regulatory framework for flexibility	56
4.2. Energy system flexibility in recent EU strategic documents	62
4.3. Additional policies and measures suggested in the literature	64
5. KEY FINDINGS AND RECOMMENDATIONS	68
Recommendations	70
REFERENCES	73

LIST OF FIGURES

Figure 1: Evolution of share of renewable energy sources in the EU electricity mix (TWh)	14
Figure 2: Final electricity demand per sector in the TYNDP 2024 scenarios (TWh)	15
Figure 3: Installed power capacity in the EU27 for the TYNDP 2024 scenarios (GW)	17
Figure 4: Evolution of energy flexibility needs in Europe	20
Figure 5: Evolution of flexibility needs per driver in Europe – 2022-2050	21
Figure 6: Average day-ahead price and volatility in European wholesale markets	22
Figure 7: Negative price events in wholesale spot markets per year in bidding zones of different regions	23
Figure 8: Price convergence on day-ahead markets in selected regions as percentage of hours in 2017-2023.	24
Figure 9: Example of activation behaviours for different frequency services in response to a frequency disturbance	25
Figure 10: Annual volume of activated balancing energy (MWh)	26
Figure 11: Annual average contracted reserve capacity (MW)	27
Figure 12: Average prices for contracted reserve capacity (€/MWh)	28
Figure 13: Average prices for contracted reserve capacity per MS for aFRR services in 2023	29
Figure 14: Evolution of redispatching needs and costs in the European power system	30
Figure 15: Monthly quantities (GWh) traded in GOPACS - 2020 to Feb 2025	32
Figure 16: Overview of barriers that hinder flexibility provision by distributed energy resources and new entrants or small actors	44
Figure 17: Number of Member States implementing measures to improve consumers' awareness on demand response - 2022	45
Figure 18: Correlation between wind and solar energy generation	47
Figure 19: Use of short- and long-duration flexibility resources in different grid conditions	50
Figure 20: Capability of flexibility solutions for different timeframes	51
Figure 21: Flexibility solutions' contributions towards 2050 in Europe according to the IEA	53
Figure 22: Flexibility solutions' contributions towards 2050 in the EU, Germany and Italy according to the JRC	54
Figure 23: Contribution of demand-side measures to the total flexibility needs in Europe in 2030	55

LIST OF TABLES

Table 1:	Summary of main underlying drivers of flexibility needs in different timeframes	19
Table 2:	Illustrative case studies of IDRA in Europe	42
Table 3:	Contributions of flexibility solutions to long-term (2030-2050) total flexibility needs according to the IEA, JRC and ACER/EEA	52
Table 4:	Key elements related to flexibility in the EU Regulatory Framework	57
Table 5:	Policies and measures proposed in the literature to facilitate the deployment of flexibility solutions	65

LIST OF BOXES

Box 1:	Belgian TSO Elia's integrated analysis of the adequacy and flexibility needs in electricity system	13
Box 2:	GOPACS - Cooperation between the Dutch electricity grid operators to procure flexibility services for local congestion management	31
Box 3:	System-friendly renewable electricity generation assets can reduce flexibility needs and/or participate in flexibility provision	37
Box 4:	Grid-scale and home batteries	40
Box 5:	Industrial Demand Response Aggregators	42
Box 6:	Flexible connection agreements	46

LIST OF ABBREVIATIONS

ACER	Agency for the Cooperation of Energy Regulators
BESS	Battery Energy Storage System
BRP	Balancing Responsible Party
CAES	Compressed Air Energy Storage
CCGT	Combined Cycle Gas Turbine
CCS	Carbon Capture and Storage
CEP	Clean Energy Package
CESA	Continental Europe Synchronous Area
CRM	Capacity Remuneration Mechanism
DSO	Distribution System Operator
EEA	European Environment Agency
EMDR	Electricity Market Design Reform
ENTSO-E	European Network of Transmission System Operators for Electricity
ERAA	European Resource Adequacy Assessment
EV	Electric Vehicle
FCR	Frequency Containment Reserve
FRR	Frequency Restoration Reserve
FNA	Flexibility Needs Assessment
IEA	International Energy Agency
LDES	Long Duration Energy Storage
NECP	National Energy and Climate Plan
NRA	National Regulatory Authority
OCGT	Open Cycle Gas Turbine

PHS	Pumped Hydropower Storage
PV	Photovoltaic
RES	Renewable Energy Sources
RR	Replacement Reserve
TSO	Transmission System Operator
TYNDP	Ten-Year Network Development Plan

EXECUTIVE SUMMARY

Background

Flexibility in the energy system is acknowledged as being crucial to the energy transition. As electricity is more difficult to store than other energy vectors (e.g., gas, oil or thermal energy), **flexibility is particularly a critical issue in the electricity system**, which is the focus of this study. Nonetheless, resources in or at the interface with other energy sectors such as the gas/hydrogen and heat sectors, can provide significant contributions to the electricity system's flexibility. Therefore, to cost-efficiently cover the flexibility needs in the electricity system, a **multi-vector approach (sector coupling)** is required.

The flexibility needs of the EU electricity system will significantly increase in the current and coming decades, while the relative and absolute contribution of historical sources of flexibility will decrease. New solutions will thus have to be developed and upscaled to fulfil the flexibility needs in the 2030-2050 timeframe.

Besides developing and implementing adequate flexibility solutions at the supply and demand side, the EU will need to **reinforce and modernise the cross-border and domestic transmission and distribution networks**. Such investments will not only reduce the flexibility needs but also enable cost-efficient provision and (cross-border) sharing of flexibility resources.

Objectives of the study

This study aims to provide a comprehensive overview of the flexibility needs and possible technology options to cost-efficiently increase the available flexibility in the EU energy system. The study mainly focuses on the capacity of the energy system to sustain a growing share of renewable energy sources and prevent or reduce renewable electricity production curtailment. Based on the key findings, policy recommendations are formulated.

The study provides an answer to the following key questions:

- What is the status of flexibility in the EU electricity system, including key challenges, opportunities and ongoing policy initiatives?
- How will the different flexibility needs and challenges evolve in the future?
- What flexibility solutions are currently being implemented or considered and how do they impact the electricity system?
- Which additional policy measures are required to increase flexibility in the EU electricity system?

Key Findings and Recommendations

- **There is high uncertainty about the exact potential and cost of each flexibility solution, and flexibility needs will substantially vary per Member State.** Nonetheless, a mix of different solutions will be needed in all Member States.
 - Demand response and energy storage (mobile and stationary batteries) will become major contributors of (sub-)daily and weekly flexibility. Flexible operation (demand response) of electrolyzers may also provide some short-term flexibility.

- For seasonal flexibility, hydrogen storage and other novel long-duration energy storage solutions are expected to contribute (complementing RES curtailment and reservoir/pumped storage hydropower).
- While the **seasonal flexibility needs** will be smaller than the daily/weekly ones, they may constitute a significant challenge as there are currently **fewer mature solutions**.
- **The deployment of new flexibility solutions, in particular small-scale assets connected to distribution grids, still faces significant challenges, due to inadequate market design rules and regulatory provisions.** The increasing number of negative clearing prices in several EU wholesale electricity markets illustrates that the flexibility market is not yet adequately reacting to market price signals.
- **There is a need to continue showcasing the benefits of decarbonising and integrating the EU energy system, as well as adequately considering the distributional impacts.** The costs to ensure sufficient system flexibility and expand and modernise the electricity networks will be significant, while the recent energy crisis has brought to the fore affordability and competitiveness concerns. Policies related to for example providing adequate electricity price and network tariff signals to consumers require the consideration of impacts on affordability and fairness.
- **Several flexibility-related policy measures are under implementation or development.** These derive from the ongoing implementation of the Clean Energy package and EU electricity market design reform provisions as well as the finalisation of the demand response network code.
- The concluding chapter details a number of **recommendations** regarding the:
 - Implementation of the **National Flexibility Needs Assessments** (FNAs) and the related reporting by ACER, including ensuring coherence between the FNAs for the different Member States;
 - Improvement of **integrated infrastructure planning**, including considering spatial planning aspects;
 - Ensuring consistency and avoiding overlaps between the European and National Resource Adequacy Assessments and the FNAs;
 - Removal of entry barriers to and promotion of cross-border participation of new and small flexibility resources in capacity remuneration mechanisms, network operator flexibility procurement, and non-fossil flexibility support mechanisms;
 - Consideration of the flexibility value in the **design of renewable energy support schemes**;
 - Improvement of **energy data management and flexibility solutions** transparency, by developing national flexibility resource registers and improving related statistics;
 - Addressal of **other factors influencing the deployment of flexibility resources**, including the roll-out of low-voltage smart meters and provision of market price signals to consumers/prosumers;
 - Development of adequate regulation for **flexible connection agreements**.

1. INTRODUCTION

1.1. Why is flexibility important?

The EU energy system faces new challenges due to several ongoing developments, in particular the **increasing share of renewable energy sources (RES)**, which is targeted to reach at least 42.5% (and possibly up to 45%) of the overall energy use in 2030¹. According to projections undertaken by the European Commission, the electricity sector should, depending on the scenario, reach a share of 69-83% of electricity generated from renewable energy sources by 2030². The massive development of variable RES (mainly solar and wind energy) poses challenges to the energy system, mainly due to their variability and potential to cause network congestion and system imbalances. This is the main driver for additional flexibility needs.

However, while the flexibility needs are significantly increasing, **flexibility offered by conventional thermal power plants is expected to decrease due to the ongoing and planned phase out of gas and coal-fuelled power plants**. Promising solutions to bridge the gap between flexibility availability and needs are other forms of flexible power generation, energy storage and demand response. These options are gradually being implemented, but the demand response potential is still largely untapped. The lack of adequate flexibility in the system is illustrated by the increasing number of hours with negative electricity spot prices in several EU wholesale markets and rising congestion management costs.

Policy makers, regulators, network operators, and market players are already working on developing and implementing several flexibility-related policy measures. These derive from the ongoing implementation of the Clean Energy package and EU electricity market design reform provisions as well as the finalisation of the demand response network code. Flexibility has also been addressed in the **Letta and Draghi reports**^{3,4}, and more recently in the European Commission's **Competitiveness Compass**⁵. Continuous improvement of the EU electricity market design will be necessary, including with the adoption of new measures – while balancing the need not to compromise solid foundations nor risk a proliferation of incompatible national approaches.

1.2. Objectives of the study

This study aims to provide a comprehensive overview of the flexibility needs and possible technology options to cost-efficiently increase the available flexibility in the EU energy system. The study mainly focuses on the capacity of the energy system to sustain a growing share of renewable energy sources

¹ European Commission, n.d., *Renewable Energy Directive*. Available at: https://energy.ec.europa.eu/topics/renewable-energy/renewable-energy-directive-targets-and-rules/renewable-energy-directive_en.

² European Commission, 2022, *Non paper on complementary economic modelling undertaken by DG ENER analysing the impacts of overall renewable energy target of 45% to 56% in the context of discussions in the European Parliament on the revision of the Renewable Energy Directive*. Available at: https://energy.ec.europa.eu/system/files/2022-06/2022_06_20_RED_non-paper_additional_modelling.pdf.

³ Letta, E., 2024, *Much more than a market*. Available at: <https://www.consilium.europa.eu/media/ny3j24sm/much-more-than-a-market-report-by-enrico-letta.pdf>.

⁴ Draghi, M., 2024, *The future of European competitiveness*. Available at: https://commission.europa.eu/document/download/ec1409c1-d4b4-4882-8bdd-3519f86bbb92_en?filename=The%20future%20of%20European%20competitiveness_%20In-depth%20analysis%20and%20recommendations_0.pdf.

⁵ European Commission, 2025, *A Competitiveness Compass for the EU. COM(2025) 30 final*. Available at: https://ec.europa.eu/commission/presscorner/detail/en/ip_25_339.

and prevent or reduce renewable electricity production curtailment⁶. Based on the key findings, policy recommendations are formulated.

The study provides an answer to the following key questions:

- What is the status of flexibility in the EU electricity system, including key challenges, opportunities and ongoing policy initiatives?
- How will the different flexibility needs and challenges evolve in the future?
- What flexibility solutions are currently being implemented or considered and how do they impact the electricity system?
- Which additional policy measures are required to increase flexibility in the EU electricity system?

The study addresses both short-term solutions and policies (expected to have an impact by 2030) and long-term solutions (measures with effects projected for 2040 and 2050).

1.3. Structure of the report

The remainder of this report is structured as follows:

- Chapter 2 details the current and future flexibility needs in the EU energy system, including an analysis of emerging challenges and potential impacts on system stability.
- Chapter 3 evaluates existing flexibility solutions being implemented or considered, such as energy storage and demand response, and their effectiveness in addressing system needs.
- Chapter 4 explores policies and measures (under implementation or discussion) to enhance flexibility in the EU electricity system in the short to long term.
- Chapter 5 provides policy recommendations based on the study's findings, aimed at increasing flexibility in the EU energy system to support RES integration and system stability.

⁶ Renewable energy curtailment refers to the deliberate reduction or restriction of the output of renewable energy sources, such as wind or solar power, that would otherwise be produced and supplied to the grid.

2. BACKGROUND AND STATE OF PLAY OF FLEXIBILITY IN THE EU ENERGY SYSTEM

KEY FINDINGS

- Flexibility addresses the variability of electricity generation and consumption patterns as well as constraints in grid availability, minimising system costs and facilitating the integration of RES.
- Flexibility can be provided through services to network operators and market parties (explicit flexibility), or as a reaction to market price signals (implicit flexibility). Flexibility is not a standalone product or service, but a capability of various assets or operators in the power system.
- Timeframes for flexibility range from sub-hourly to inter-annual needs. Currently, the main need for flexibility is for short-term markets.
- In addition to their role on wholesale markets, flexibility solutions can also provide ancillary and congestion management services to network operators, as well as voltage regulation, inertia and system restoration.

2.1. Framework for defining and characterising flexibility

Flexibility in the energy system is increasingly acknowledged as being crucial to the energy transition. As electricity is more difficult to store than other energy vectors (e.g., gas, oil or thermal energy), flexibility is particularly a critical issue in the electricity system. Flexibility is needed for a broad range of applications, each with various dimensions (regarding **timeframes**, **geographical scope**, and **beneficiaries**)⁷. In this section we first provide a definition and characterisation of flexibility and then discuss the main drivers behind the flexibility needs.

2.1.1. What is (not) flexibility in the electricity system?

The Electricity Regulation (EU) 2019/943 defines flexibility as the ability of the electricity system to adjust to the variability of generation and consumption patterns and to the grid availability, across relevant market timeframes⁸. These timeframes range from **seconds or minutes (sub-hourly needs)** to **days (daily or weekly needs)** and **months or years (seasonal or inter-annual needs)**.

The main direct and indirect purposes of flexibility provision are:

- **Ensuring electricity system stability and security of supply,**
- **Facilitating the deployment of intermittent renewable energy sources (RES), and**
- **Minimising system costs.**

Flexibility is not a single specific product or service: instead, it is the capability of different assets or operators within the power system to respond to variations in electricity supply or demand and to resulting imbalances, grid congestion or other needs of market participants or network operators.

⁷ Magnus Energy, 2024, *Disentangling flexibility and its needs assessment methodology*. Available at: <https://www.magnusenergy.com/insights/working-paper-flexibility-needs-assessment>.

⁸ Regulation (EU) 2024/1747 of the European Parliament and of the Council of 13 June 2024 amending Regulations (EU) 2019/942 and (EU) 2019/943 as regards improving the Union's electricity market design, OJ L, 2024/1747, 26.6.2024, Art. 2(79). Available at: <https://eur-lex.europa.eu/eli/reg/2024/1747/oj>.

Flexibility providers employ this capability to provide specific flexibility services such as **balancing services (explicit flexibility)**, or **to react to price signals reflecting variations in demand, supply, and grid availability (implicit flexibility)**.

Electricity system flexibility and adequacy are approached separately in legislation, but they are interrelated. Adequacy can be defined as the “overall adequacy of the electricity system to supply current and projected demand levels for electricity”⁹. Or, as phrased by the System Operations Guideline¹⁰, “the ability of in-feeds into an area to meet the load in that area”. In other words, it refers to the ability of the system to cover peak load, ensuring sufficient capacity to avoid loss of load. So, while adequacy focuses on ensuring the availability of sufficient capacity to meet demand at any moment, flexibility involves the system's adaptability to any fluctuation in supply and demand. The interaction is important as solutions such as flexible power generation, energy storage or demand response can cover both flexibility and adequacy needs (to different extents).

While adequacy policies (such as Capacity Remuneration Mechanisms, CRMs) mainly focus on specific capacity-related needs to cover peak load, flexibility is multi-dimensional in terms of its goals, timeframes, and use cases. Flexibility solutions, such as dispatchable power generation, energy storage and demand response, address challenges over diverse timescales and use cases, making their role in the energy transition increasingly vital. Flexibility services are often procured with medium to long contract durations (capacity reservations), but their activation periods are typically short. Flexibility needs are generally defined and procured in upward or downward directions, depending on the actual market and system situation, or on the occurrence of congestion or a voltage issue at specific locations.

Box 1: Belgian TSO Elia's integrated analysis of the adequacy and flexibility needs in electricity system

To cost-efficiently respond to the major challenges resulting from the energy transition and properly consider the **interaction between system flexibility and adequacy**, Belgian law mandates Elia to conduct every two years a **ten-year adequacy and flexibility analysis for the Belgian electricity system**. The latest report presents this analysis for the 2024-2034 period¹¹.

The report concludes that the increasing share of renewable energy in the electricity mix and the expected electrification of end-uses in sectors such as mobility, building heating and industry create **additional capacity needs that can to some extent be reduced by promoting flexible demand**. As demand response allows to reduce peak load and adapt the consumption pattern to the variability of renewable energy sources, it reduces the capacity needs and hence directly contributes to security of supply.

Due to the energy transition, the flexibility needs and solutions are substantially evolving. While flexibility services were in the past mainly provided by dispatchable power generators to TSOs, – to ensure system stability – Distribution System Operators (DSOs) have now increasingly also localised needs for flexibility, to prevent or reduce congestion or voltage issues in their grids¹². Additionally,

⁹ See Article 1 of ACER, 2020, *ACER Decision on the ERAA methodology: Annex I Methodology for the European resource adequacy assessment*. Available at: https://www.acer.europa.eu/sites/default/files/documents/Individual%20Decisions_annex/ACER%20Decision%202022%20-%20Annex%20I%20Detailed%20ERAA%202021%20review_0.pdf.

¹⁰ Art. 2(68) of *Commission Regulation (EU) 2017/1485 of 2 August 2017 establishing a guideline on electricity transmission system operation (System Operations Guideline, SOGL)*. Available at: <https://eur-lex.europa.eu/eli/reg/2017/1485/oj/eng>.

¹¹ Elia, 2023, *Adequacy & flexibility study for Belgium (2024-2034)*. Available at: <https://www.elia.be/en/electricity-market-and-system/adequacy/adequacy-studies>.

¹² Magnus Energy, 2024, *Disentangling flexibility and its needs assessment methodology*. Available at: <https://www.magnusenergy.com/insights/working-paper-flexibility-needs-assessment>.

flexibility services are increasingly being offered via short-term markets to enable balancing responsible parties to adjust their portfolios. To ensure cost-efficient procurement and provision of flexibility services, including by assets connected to distribution networks, enhanced coordination between the transmission and distribution levels is needed.

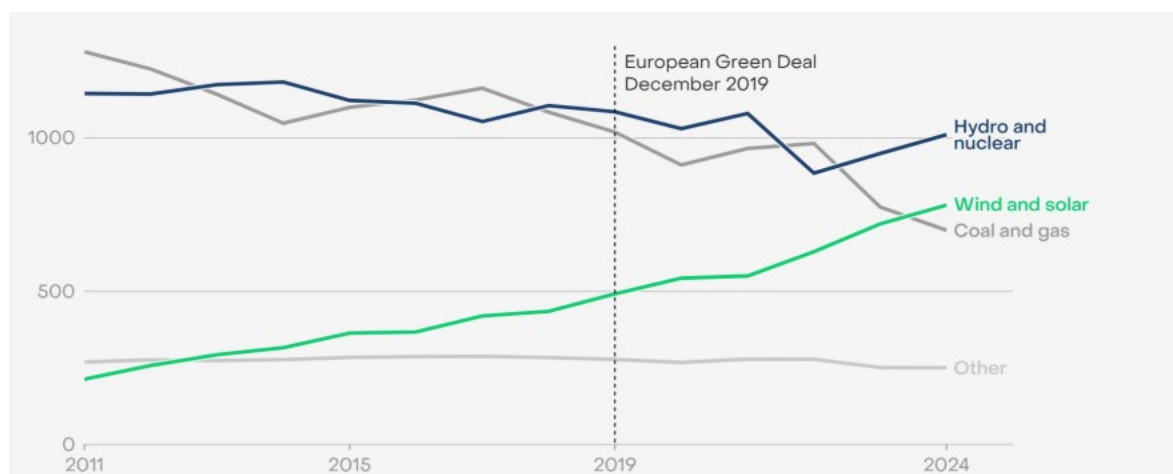
2.1.2. Drivers for current and future flexibility needs

To ensure system stability, electricity supply and demand must always be balanced. Consequently, the flexibility needs of the power system are influenced by the uncertainty and variability of both power generation and consumption. Flexibility needs are anticipated to further rise across all EU Member States, mainly driven by the developments described below.

a. Increasing share of variable renewable energy sources in the electricity mix

The increasing integration of variable RES in the EU electricity system, mainly wind and solar photovoltaic (PV) energy, necessitates flexibility sources such as dispatchable generation, storage, or demand response to maintain the supply-demand balance. The development of variable RES is expected to strongly increase in view of achieving the 2030 energy and climate targets and the 2050 decarbonisation goal¹³, thereby driving up the flexibility needs of the power system. Between 2019 and 2024, the share of wind and solar energy in the EU electricity mix has risen from 17% to 29%¹⁴; this evolution is presented in Figure 1.

Figure 1: Evolution of share of renewable energy sources in the EU electricity mix (TWh)



Source: Ember, 2025, European Electricity Review 2025. Available at: https://ember-energy.org/app/uploads/2025/01/EER_2025_22012025.pdf.

Note: 'Other' includes other fossil, bioenergy and other renewables.

According to an EMBER assessment of the National Energy and Climate Plans (NECPs), the share of renewable energy in the overall electricity generation mix would increase to 66% by 2030¹⁵. This would mean falling short of the REPowerEU objective of 72%¹⁶. Solar PV capacity in the EU would grow to 623

¹³ European Commission, 2025, *2050 long-term strategy*. Available at: https://climate.ec.europa.eu/eu-action/climate-strategies-targets/2050-long-term-strategy_en.

¹⁴ Ember, 2025, *European Electricity Review 2025*. Available at: <https://ember-energy.org/latest-insights/european-electricity-review-2025/>.

¹⁵ Ember, 2023, *Draft NECPs show EU just falling short of REPowerEU*. Available at: https://ember-energy.org/app/uploads/2023/11/Report_Draft-NECPs-show-EU-just-falling-short-of-REPowerEU.pdf.

¹⁶ DG Energy, 2022, *Non-paper on complementary economic modelling of the impacts of the RES target in RED*. Available at: https://energy.ec.europa.eu/system/files/2022-06/2022_06_20_RED_non-paper_additional_modelling.pdf.

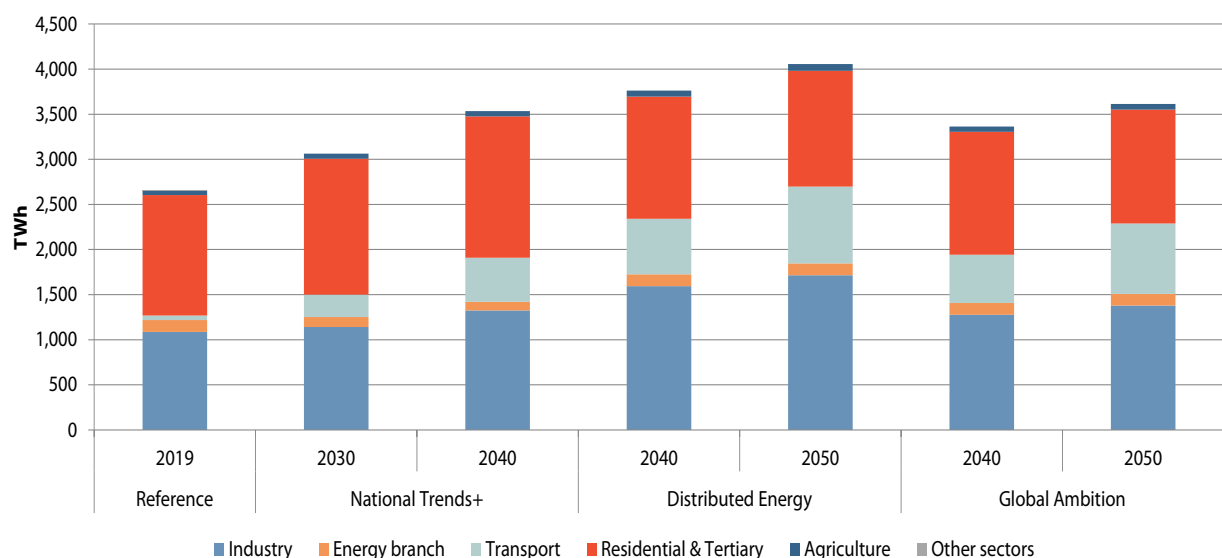
- 672 GW by 2030, which means more than tripling the 2022 capacity of 195 GW. Wind energy capacities are set to more than double, jumping from 204 GW in 2022 to 450 GW by 2030¹⁷.

Deviations between forecasted and actual variable RES generation and electricity load (i.e. forecast errors) significantly impact the flexibility needs of the power system as well¹⁸. These deviations may lead to imbalances in portfolios of Balancing Responsible Parties (BRPs)¹⁹, who can reduce their imbalances through transactions in the short-term market. The residual imbalances at system level are then addressed by the TSOs via procurement of balancing services in real-time. The impossibility to predict the output of variable RES with 100% certainty, combined with fluctuating electricity demand amplify the need for a flexible and highly responsive power system, capable of adapting to these uncertainties.

b. Electrification and rising electricity demand

The EU's 2030 climate targets demand significant electrification, anticipating a 30% increase in total electricity generation. Figure 2 presents the final electricity demand per sector for the different Ten-Year Network Development Plan (TYNDP) 2024 scenarios developed by the European Networks of Transmission System Operators (ENTSOs), highlighting the significant increase in total electricity demand for all scenarios towards 2050, driven especially by increasing electricity demand for transport but also from the industrial sector (while residential and tertiary demand would increase towards 2030/2040, but fall subsequently to levels similar to current consumption).

Figure 2: Final electricity demand per sector in the TYNDP 2024 scenarios (TWh)



Source: Authors' own elaboration based on ENTSO-E & ENTSG, 2025, TYNDP 2024 - Final Scenarios Report. Available at: <https://2024.entsoe-tyndp-scenarios.eu/>.

Electrification is hence on the one hand a key driver for increased flexibility needs in the power system, but on the other hand electrified applications (e.g. heat pumps, EVs) can be important sources of flexibility. As several main end-use applications such as heating and road transport are expected to gradually shift from easily storable fuels like oil and gas to electricity, which is more difficult to store,

¹⁷ Ember, 2023, *Draft NECPs show EU just falling short of REPowerEU*. Available at: <https://ember-energy.org/app/uploads/2023/11/Report-Draft-NECPs-show-EU-just-falling-short-of-REPowerEU.pdf>.

¹⁸ ENTSO-E, 2024, *System Flexibility Needs for the Energy Transition*. Available at: <https://www.entsoe.eu/system-flexibility/>.

¹⁹ Balancing Responsible Parties are parties which are responsible for the injections and offtake in one or more points of the electricity system, and which are charged by electricity TSOs for any differences between the declared and actual net position (that is, the BRP's position considering its network injection and offtake).

the electricity system faces new challenges. The electrification of these end-use sectors will increase the overall electricity demand and peak load, while also altering demand patterns requiring adequate flexibility resources.

While the supply side (wind and solar energy) is becoming increasingly weather-dependent, the electrification of the energy use for heating will also to some extent lead to more weather (and temperature) dependent electricity demand. The ENTSOs TYNDP 2024 scenarios foresee that fully-electric heat pumps could in 2050 represent 50-63% of the deployed heating technologies in the residential sector, depending on the scenario.²⁰ The expected massive adoption of electric heat pumps and e-boilers will lead to higher peak demand, especially during morning and evening hours, and in winter periods, when one of the main supply technologies (PV) is not or only limitedly available.

This development will put pressure on the grid (capacity) and on the system in general (peak generation capacity and flexibility). However, electrified heating and road transport will also be able to provide flexibility: hybrid heat pumps can be installed where appropriate (at least during the transition period), electric heat pumps can be coupled with thermal storage to enable flexible electricity offtake (i.e. withdrawal) from the grid, electric vehicles (EVs) can be charged during off-peak hours and deliver power back to the grid (V2G) during peak hours. Enabling these developments will require smart energy management systems and a highly flexible and integrated energy system, where several technologies and energy vectors (electricity, heat, methane, hydrogen) are used to cover the overall energy needs at least cost while also contributing to the climate goals.

c. Decommissioning of dispatchable power plants

As Europe progresses towards carbon neutrality, the power system will gradually have to reduce its dependence on coal- and gas-fuelled thermal power generation, which currently still plays a major role in providing flexibility and ancillary services. The declining share of gas-powered electricity in the EU electricity mix since Russia's invasion of Ukraine has been a major factor in reducing the EU gas consumption: according to Ember, the total gas demand in the block fell by 20% in the 2018-2023, with the power sector accounting for about a third of this decline. The availability of coal-fired power plants to provide flexibility is also declining; the share of coal in the EU power generation mix has decreased from 16% in 2019 to 9.8% in 2024. About half of the remaining EU coal-fired power plants would have to be decommissioned by 2030, with Poland and Germany²¹ postponing the phase-out until the 2040s^{22,23}. This is reflected in the installed capacities of the TYNDP 2024 scenarios, where the National Trends+ scenario foresees that fossil-based installed capacity would represent only 13% of total installed capacity already in 2030 (Figure 3). This further phasing out of conventional dispatchable power plants will significantly increase the flexibility needs, which will have to be covered by new (and clean) flexible power generation, active demand response, energy storage, sector coupling solutions and optimised grid management.

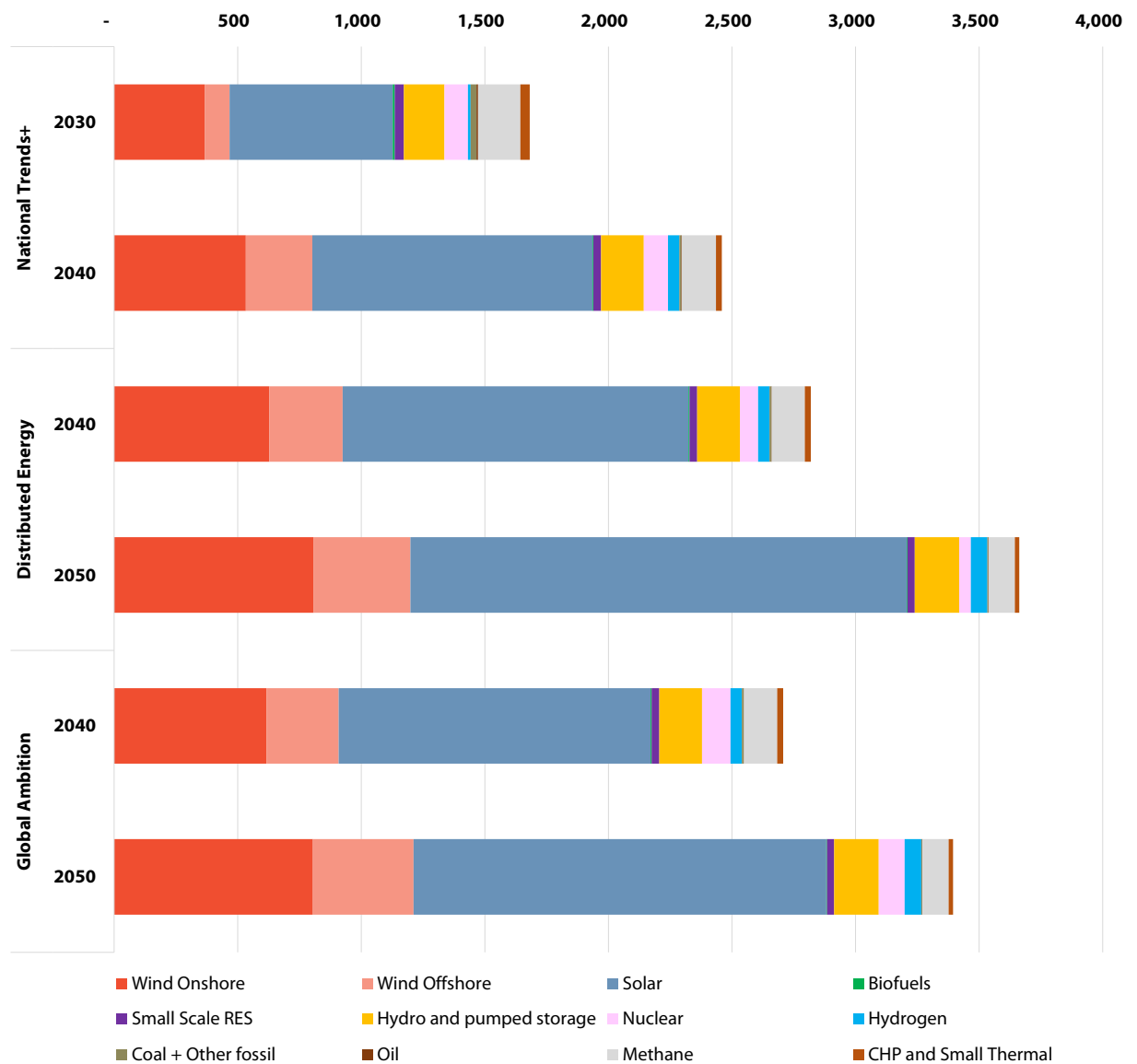
²⁰ ENTSO-E & ENTSG, 2025, *TYNDP 2024 - Final Scenarios Report*. Available at: <https://2024.entsos-tyndp-scenarios.eu/>.

²¹ Wehrmann, B., 2024, Germany delays gas plant decision, 2030 state coal phase-out uncertain, Clean Energy Wire. Available at: <https://www.cleanenergywire.org/news/germany-delays-gas-plant-decision-2030-state-coal-phase-out-uncertain>.

²² Ember, 2024, *Clean flexibility is the brain managing the clean power system*. Available at: <https://ember-energy.org/latest-insights/clean-flexibility-is-the-brain-managing-the-clean-power-system/>.

²³ Ember, 2025, *European Electricity Review 2025*. Available at: <https://ember-energy.org/latest-insights/european-electricity-review-2025/>.

Figure 3: Installed power capacity in the EU27 for the TYNDP 2024 scenarios (GW)



Source: Authors' elaboration based on ENTSO-E & ENTSOG, 2025, TYNDP 2024 - Final Scenarios Report. Available at: <https://2024.entsos-tyndp-scenarios.eu/>.

d. Security of electricity supply

The EU is expected to face increasing challenges to the security of energy supply resulting from extreme weather events, potential international conflicts, and cyber threats. Such events in a specific country or region can jeopardise the electricity supply and lead to widespread network outages. Moreover, as power generation is becoming more weather-dependent, the availability of sufficient (adequacy) and flexible resources is increasingly important in ensuring security of energy supply in the medium to long term.

Extreme weather events can impact electricity generation as well as transmission & distribution assets alike, but each event (such as extreme heat waves, droughts and severe storms and floods) affects each

asset category differently²⁴. The IEA²⁵ has reported on a record number of major power disruptions in 2024 globally, citing extreme weather conditions as key driver of adequacy risks, and calling for multi-metric approaches in their mitigation. The likelihood and impact of these events occurring in Europe will vary per European region²⁶.

Offshore wind energy farms in the North Sea and Baltic Sea regions, for instance, are located in areas that are not only vulnerable to severe weather but also potential targets for sabotage²⁷. And various power generation as well as transmission and distribution asset categories in Southern European regions are vulnerable to climate change impacts²⁸. This combination of factors increases the risk of disruptions in the energy supply, highlighting the need for robust security measures and flexible energy solutions to maintain system stability and resilience.

The current understanding of security of electricity supply is still mostly adequacy-related (i.e. having enough generation and transmission/distribution capacity to meet peak demand). The security of electricity supply level in the EU is measured through a combination of metrics that assess the adequacy and reliability of the power system. The European Resource Adequacy Assessment (ERAA)²⁹ plays a crucial role in this evaluation by providing a comprehensive analysis of resource adequacy across the EU. While the reliability of the electricity supply is in general quite high, ACER's 2024 'Security of electricity supply'³⁰ report highlights that affordability has become a major concern, with electricity prices still higher than the pre-crisis levels. The report underscores the need for flexible assets, such as batteries and demand-side response systems, to reduce the high price peaks resulting from the variable output of RES-based electricity generation. The cost of capacity mechanisms, which are one of the market tools designed to ensure sufficient resources are available to meet peak demand, increased by 40% in 2023, reaching EUR 7.4 billion, and are for 2024 estimated at EUR 6.5 billion. This rise in costs, coupled with the fact that 85% of the long-term contracts for 2035 are directed towards fossil fuel-based electricity generators, highlights the urgent need for improved flexibility measures that also support the decarbonisation of the electricity sector.

2.1.3. Flexibility needs and timeframes

The above-mentioned drivers have an impact on the flexibility needs (and provision) in the electricity system in various timeframes:

- Flexibility at the **sub-hourly** level is essential for maintaining the real-time balance between supply and demand. This involves two key aspects: reacting to unexpected events like unplanned (forced) outages and maintaining stable system performance against forecast deviations.

²⁴ Eurelectric, 2022, *The Coming Storm - Building electricity resilience to extreme weather*. Available at: <https://resilience.eurelectric.org/>.

²⁵ IEA, 2025, *Electricity 2025*. Available at: <https://iea.blob.core.windows.net/assets/0f028d5f-26b1-47ca-ad2a-5ca3103d070a/Electricity2025.pdf>.

²⁶ European Environmental Agency, Trinomics et al., 2022, *Adaptation challenges and opportunities for the European energy system*. Available at: <https://www.eea.europa.eu/en/analysis/publications/adaptation-in-energy-system>.

²⁷ As shown by the undersea electricity and data cable damage under the Baltic Sea in 2024. EEAS, 2024, Joint Statement by the European Commission and the High Representative on the Investigation into Damaged Electricity and Data Cables in the Baltic Sea. Available at: <https://www.eeas.europa.eu/eeas/joint-statement-european-commission-and-high-representative-investigation-damaged-electricity-and-en>.

²⁸ European Environmental Agency, Trinomics et al., 2022, *Adaptation challenges and opportunities for the European energy system*. Available at: <https://www.eea.europa.eu/en/analysis/publications/adaptation-in-energy-system>.

²⁹ ACER, 2024, *European Resource Adequacy Assessment*. Available at: <https://www.acer.europa.eu/electricity/security-of-supply/european-resource-adequacy-assessment>.

³⁰ ACER, 2024, *Security of EU electricity supply*. Available at: https://www.acer.europa.eu/sites/default/files/documents/Publications/Security_of_EU_electricity_supply_2024.pdf.

- **Daily** flexibility needs are influenced by three main factors. First, the integration of solar PV, which follows a daily generation cycle. Second, daily demand patterns, which increasingly vary and typically show higher demand during certain parts of the day. Third, new uses such as electric vehicles and heat pumps, which can increase flexibility needs if not managed properly.
- **Weekly** flexibility needs are shaped by the weekly demand pattern (working days versus weekends) and wind power regimes, which typically last for several days.
- **Seasonal** flexibility needs are driven by several factors. On the supply side, these include variations in solar power generation, which is higher in the summer, while wind power generation is in general higher in the winter. The hydropower availability also varies between dry and wet (or glacial) seasons. On the demand side, seasonal electricity demand is higher in winter due to heating needs, which are expected to further increase by the electrification of heating in buildings.

Inter-annual variability also influences flexibility needs on the seasonal and weekly timeframe. Year-to-year variability in weather patterns affects RES generation (wind and solar energy availability), hydro generation (variable rainfall), and demand (colder winters and hot summers increase electricity demand). Climate change can alter weather patterns, exacerbating inter-annual variability with events such as droughts or other extreme weather conditions. However, inter-annual variability affects particularly resource adequacy needs, although one needs to consider it also to estimate flexibility needs in shorter timeframes (seasonal and weekly).

Table 1: Summary of main underlying drivers of flexibility needs in different timeframes

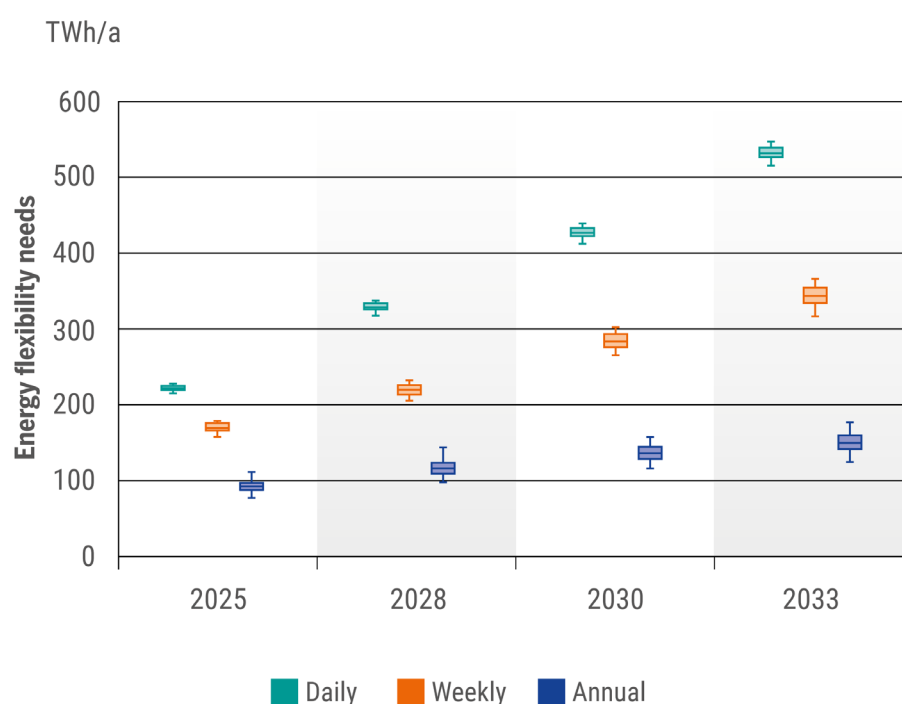
	Solar PV production	Wind power production	Hydropower production	EVs	Heat Pumps	Other electricity demand
Sub-hourly	✓ Forecast errors & variability (ramps)	✓ Forecast errors & variability (ramps)				✓ Forecast errors & variability (ramps)
Daily	✓ Daily solar cycle (day vs night)			✓ Daily demand patterns	✓ Daily demand patterns	✓ Daily demand patterns
Weekly		✓ Wind power regimes		✓ Working days vs weekends		✓ Working days vs weekends
Seasonal	✓ Seasonal irradiation cycle (summer > winter)	✓ Seasonal wind cycle (summer < winter)	✓ Seasonal inflow patterns		✓ Thermo-sensitivity	✓ Thermo-sensitivity
Inter-annual	✓ Inter-annual solar irradiance variability	✓ Inter-annual wind regimes variability	✓ Inter-annual inflow variability			✓ Inter-annual temperature variability

Source: Adapted from Trinomics & Artelys, 2023, Power System Flexibility in the Penta region – Current State and Challenges for a Future Decarbonised Energy System³¹.

³¹ Available at: <https://www.benelux.int/wp-content/uploads/2023/03/Penta-Flexibility-Final-report-03-2023.pdf>.

Quantifying the flexibility needs is not an easy task, and highlights the complementarity between adequacy and flexibility solutions. ENTSO-E's latest projections for Europe's flexibility needs indicate that about 220 TWh of flexibility is needed each year on a daily basis in 2025³². This volume grows linearly over time up to 530 TWh per year in 2033. The flexibility needs on a weekly and annual basis are lower than the needs on a daily basis. Similar studies also conclude that the daily flexibility needs are highest but provide slightly different numbers and projections based on differing methodologies³³. For example, a 2023 ACER/EEA study³⁴ shows that towards 2030 especially daily (and sub-daily) flexibility needs would increase substantially (from 153 TWh in 2021 to 362 TWh in 2030)³⁵. Weekly flexibility is estimated at about two-thirds of daily needs at 242 TWh and seasonal flexibility about half at 168 TWh. As a comparison, total EU electricity generation in 2024 was around 2742 TWh³⁶. But flexibility needs will increase across the board - Figure 4 indicates energy flexibility needs in the EU electricity system are projected to almost double across all timeframes by 2033.

Figure 4: Evolution of energy flexibility needs in Europe



Source: ENTSO-E, 2024, *System Flexibility Needs for the Energy Transition*. Available at: <https://www.entsoe.eu/system-flexibility/>.

Moreover, the total flexibility needs depend on the actual importance of the drivers above in each system. For example, daily needs are largely influenced by the variability in PV output in South-European countries, while weekly needs are more impacted by wind patterns and hydro-electricity production cycles as well as seasonal demand patterns in North-European countries. Currently, demand variations are the main driver of flexibility needs in Europe's electricity system, despite growing wind energy and solar PV penetration. By 2030, the increased share of wind energy and solar

³² ENTSO-E, 2024, *System Flexibility Needs for the Energy Transition*. Available at: <https://www.entsoe.eu/system-flexibility/>.

³³ JRC, 2023, *Flexibility requirements and the role of storage in future European power systems*. Available at: <https://publications.jrc.ec.europa.eu/repository/handle/JRC130519>.

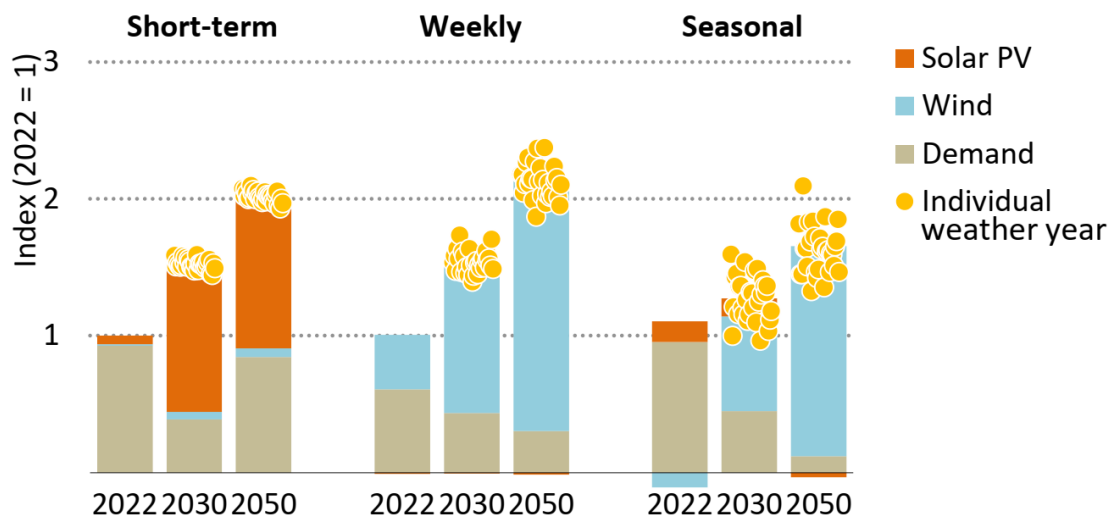
³⁴ ACER and EEA, 2023, *Flexibility solutions to support a decarbonised and secure EU electricity system*. Available at: <https://acer.europa.eu/document/acer-eea-report-flexibility-solutions-support-decarbonised-and-secure-eu-electricity-system>.

³⁵ The study assumes that demand saving and demand response are not part of the total flexibility needs and hence could lower the total needed volume.

³⁶ Ember, 2025, *European Electricity Review 2025*. Available at: <https://ember-energy.org/latest-insights/european-electricity-review-2025/>.

PV generation will substantially increase these flexibility requirements. Solar PV will become the primary driver of short-term flexibility needs by 2050 (Figure 5).

Figure 5: Evolution of flexibility needs per driver in Europe – 2022-2050



Source: IEA, 2023, Managing the Seasonal Variability of Electricity Demand and Supply³⁷.

Note: License: CC BY 4.0.

2.2. Role of flexibility to address challenges in the EU electricity system

As indicated in section 2.1.1, flexibility is not a single specific product or service, with solutions being able to provide various flexibility services. Hence, flexibility challenges in the EU are reflected across a variety of markets and services. The following section presents each challenge separately (while Chapter 3 presents individual flexibility solutions and their capabilities).

2.2.1. Flexibility to reduce electricity price volatility and avoid negative prices in wholesale markets

Investments in and smart use of flexibility resources can substantially improve price formation on wholesale electricity markets. Operators of flexible assets (e.g. storage, demand response) can adapt their grid off-take or injection volumes depending on market price levels. This market arbitrage allows to take advantage of price fluctuations within or between bidding zones. Flexible assets in the market acting on arbitrage opportunities allow to reduce price volatility and lower average prices, hence improving market efficiency³⁸.

Currently, the main need for flexibility is for **short-term markets**. These markets trade electricity hourly (or sub-hourly) either on a day-ahead basis or intra-day, i.e. before or during the specific day that it will be delivered. Day-ahead market price volatility remains quite high, although average prices as well as volatility have dropped since the energy crisis 2022-2023 peaks (Figure 6).

³⁷ Available at: <https://www.iea.org/reports/managing-the-seasonal-variability-of-electricity-demand-and-supply>.

³⁸ ACER, 2024, *Progress of EU electricity wholesale market integration*. Available at: https://www.acer.europa.eu/sites/default/files/documents/Publications/ACER_2024_MMR_Market_Integration.pdf.

Figure 6: Average day-ahead price and volatility in European wholesale markets

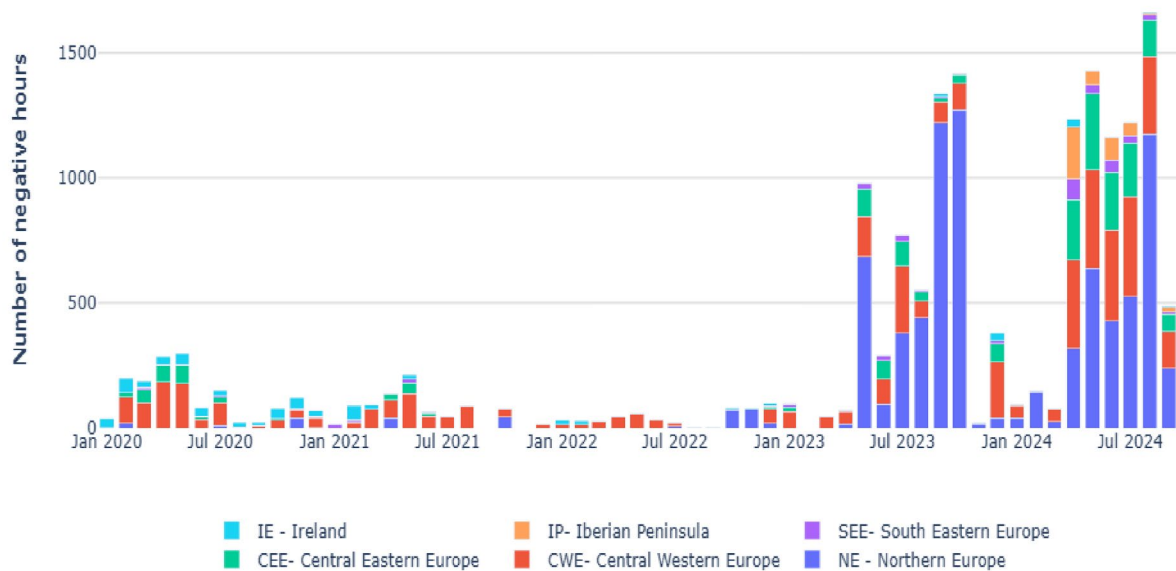


Source: ACER, 2024, *Progress of EU electricity wholesale market integration - 2024 Market Monitoring Report*. Available at: <https://acer.europa.eu/monitoring/MMR>.

The increasing temporal volatility of electricity prices is also illustrated by **negative price events**. Such events are caused by high production from solar/wind energy sources coinciding with low electricity demand (e.g. weekend days in summer period), in some cases amplified by inadequately designed renewable energy support mechanism design, cross-border congestion and/or technical limits in the power system (such as power plant must-run or ramping constraints). These factors limit the potential or incentives to reduce the power generation level, leading to negative wholesale spot prices. In recent years, negative price events have become a significant issue in many markets, including in the Central and Western Europe and Northern Europe regions (Figure 7), increasing not only in frequency but also magnitude³⁹.

³⁹ Eurelectric, 2024, *Understanding ultra-low and negative power prices: causes, impacts and improvements*. Available at: <https://www.eurelectric.org/wp-content/uploads/2024/11/Eurelectric-explainer-on-negative-prices.pdf>.

Figure 7: Negative price events in wholesale spot markets per year in bidding zones of different regions



Source: European Commission, 2024, *Quarterly report on European electricity markets*, Volume 17⁴⁰.

Intra-day markets have far lower volumes of deliveries compared to day-ahead markets, and the role of flexibility is hence lower in absolute terms. However, intra-day markets are expected to become more relevant, as the share of intermittent renewable electricity further increases. For example, ENTSO-E estimated that intra-day flexibility needs arising from forecast errors would increase by 2030 for all five analysed Member States⁴¹. Balancing responsible parties (BRPs) use the short-term markets to adjust (balance) their portfolio and hence avoid (high) imbalance costs charged by TSOs. BRPs can use own flexibility assets to optimise their portfolio internally, or resolve portfolio imbalances by transactions on the short-term markets, and thus avoid higher residual imbalance costs.

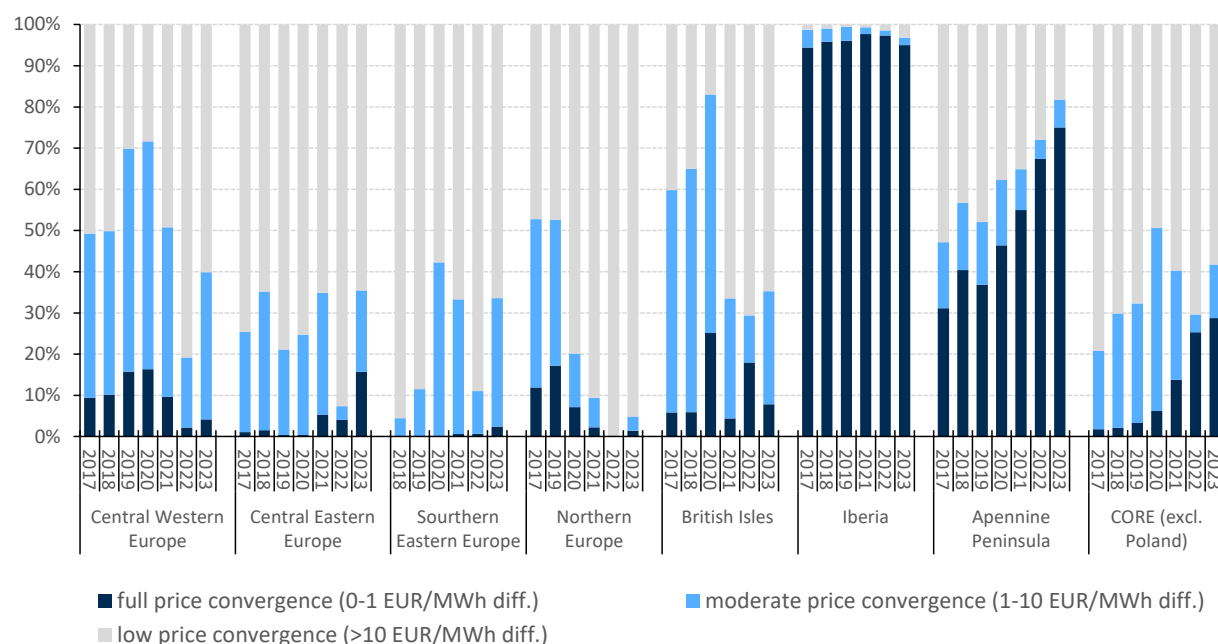
In addition to addressing temporal price variations, flexibility can also impact **locational variations in electricity market prices** (see Figure 8). In the EU's short-term markets, price differences between bidding zones continue to exist due to cross-zonal congestion, both within certain regions as shown in the figure, and especially across regions. Flexibility assets also improve the correlation of prices across bidding zones by arbitraging on these markets.

Flexibility solutions can also help address risks in **forward electricity markets**, for example allowing counterparties to hedge their profile risks in power purchase agreements (i.e. the long-term risk arising from a mismatch between actual and contractual production or offtake profiles). Flexibility solutions could also hedge the peak power needs of forward market participants, for example providing the peak power necessary to complement a baseload forward electricity contract. With these products market participants commit to delivering electricity over a long-time span in the future during periods of peak electricity demand. However, forward markets in the EU remain underdeveloped for most bidding zones, reducing the demand for such services.

⁴⁰ Available at: https://energy.ec.europa.eu/data-and-analysis/market-analysis_en.

⁴¹ ENTSO-E, 2024, *System Flexibility Needs for the Energy Transition*. Available at: <https://www.entsoe.eu/system-flexibility/>.

Figure 8: Price convergence on day-ahead markets in selected regions as percentage of hours in 2017-2023.



Source: Adapted from Trinomics et al., 2024, *Study on energy prices and costs - Evaluating impacts on households and industry: 2024 edition*. Available at: <https://op.europa.eu/s/z4gx>.

Note: Price convergence indicates here the share of hours in which zonal prices were within a certain range across all bidding zones in the region.

Regions include the bidding zones of the following countries: Central Western Europe - AT, DE/LU, BE, NL, FR and CH. Central Eastern Europe - CZ, SK, HU, RO, PL and SI. Southern Eastern Europe - BG, GR, HR and RS. Northern Europe - DK, EE, FI, LV, LT, SE, NO. British Isles - UK and IE. Iberia - ES and PT. Apennine Peninsula - IT and MT.

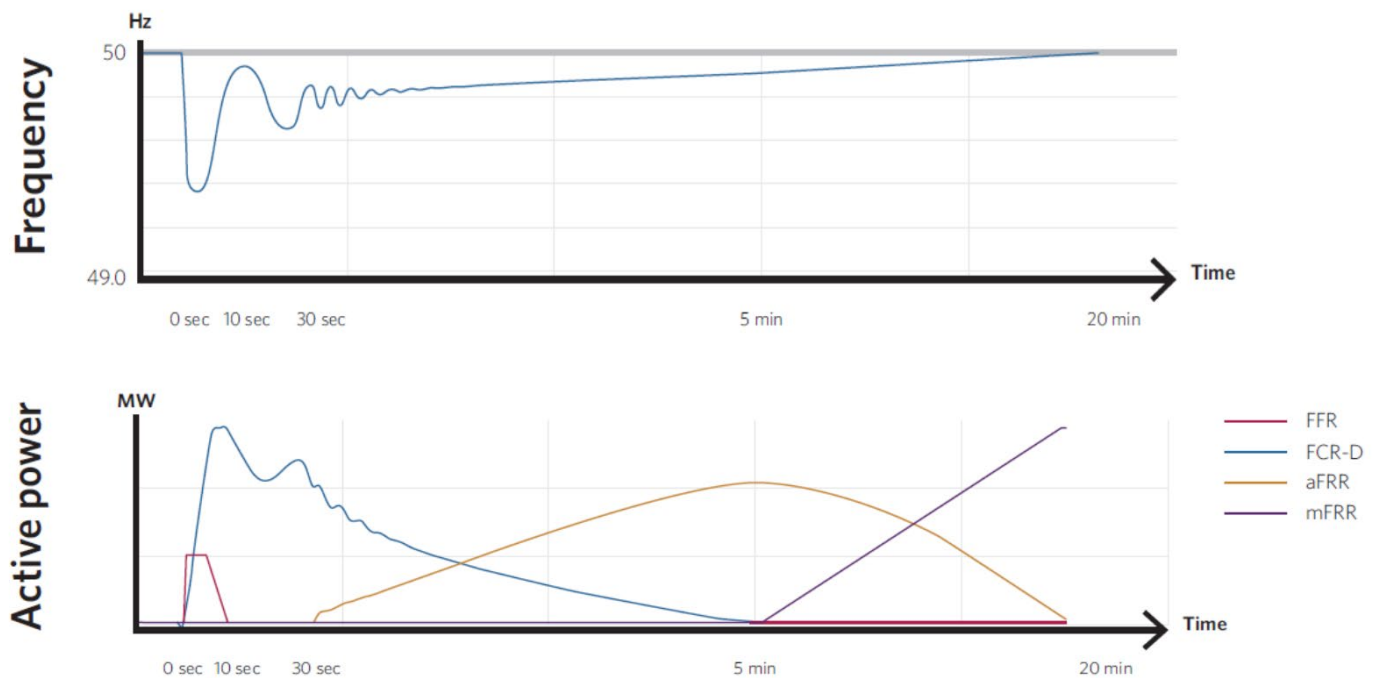
2.2.2. Flexibility to provide ancillary and congestion management services

In addition to their role on wholesale markets, flexibility solutions can also provide ancillary and congestion management services to network operators. Ancillary services include balancing and voltage regulation; other services, for example system restoration and inertia services can also be provided, but are not the main services procured by system operators, and can be highly specific to individual systems.

a. Balancing services procured by Transmission System Operators

TSOs use flexibility services (capacity and energy) offered by market operators to address residual imbalances in the system and hence ensure system stability. Frequency deviations are first contained by using Frequency Containment Reserves (FCR). Following this, three service markets are used for frequency restoration purposes in the EU, namely automated frequency restoration reserve (aFRR), manual frequency restoration reserve (mFRR), and replacement reserve (RR) markets. The aFRR market typically operates most rapidly, and in cases where it is unable to restore frequency spikes, mFRR services are engaged manually (or in a less automated fashion than aFRR services). In a few cases in specific countries, RR services may also step in afterwards for longer-term frequency restoration. Lastly, in recent years and in a few countries, Fast Frequency Response services have been developed to counteract frequency disturbances more immediately (these services are discussed later in the context of inertia services).

Figure 9: Example of activation behaviours for different frequency services in response to a frequency disturbance

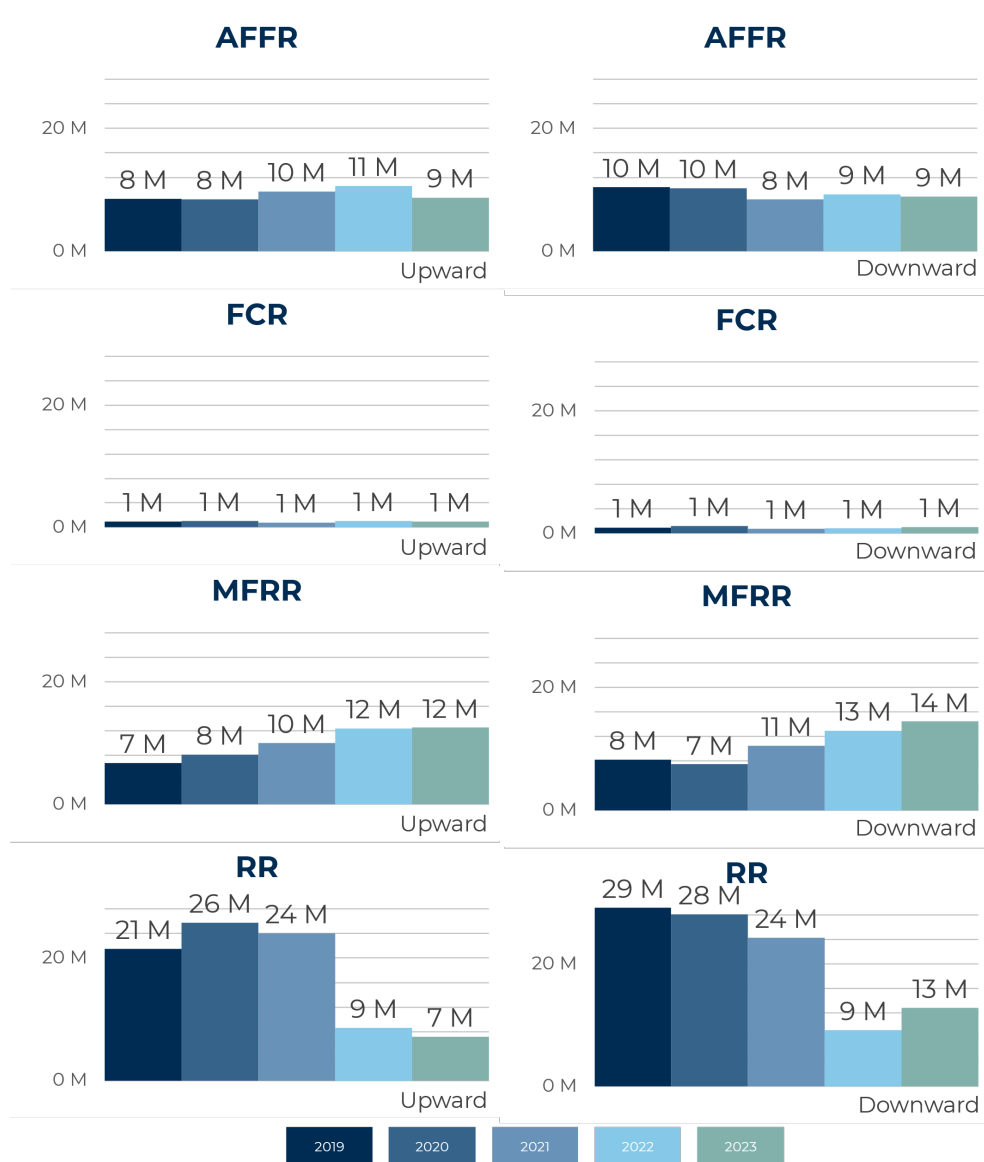


Source: ENTSO-E, 2024, *Overview of Frequency Control in the Nordic Power System*. Available at: <https://www.entsoe.eu/publications/system-operations-reports/>.

Overall, the size of these balancing markets differs considerably in the EU (Figure 10)⁴². FCR activations are rather low in energy and capacity, and step in initially to only contain frequency fluctuations. aFRR and mFRR activations are more frequent, and reserve capacities for these services are also larger. RR services were rather extensively used in the past, but have in recent years diminished to below the levels of aFRR and mFRR services. Otherwise, activations and reserve capacities do not appear to have changed significantly in other markets, excluding mFRR which is being more frequently used. Volumes of activated balancing energy across various services have been quite high in Spain, France, Italy, Poland, and Greece.

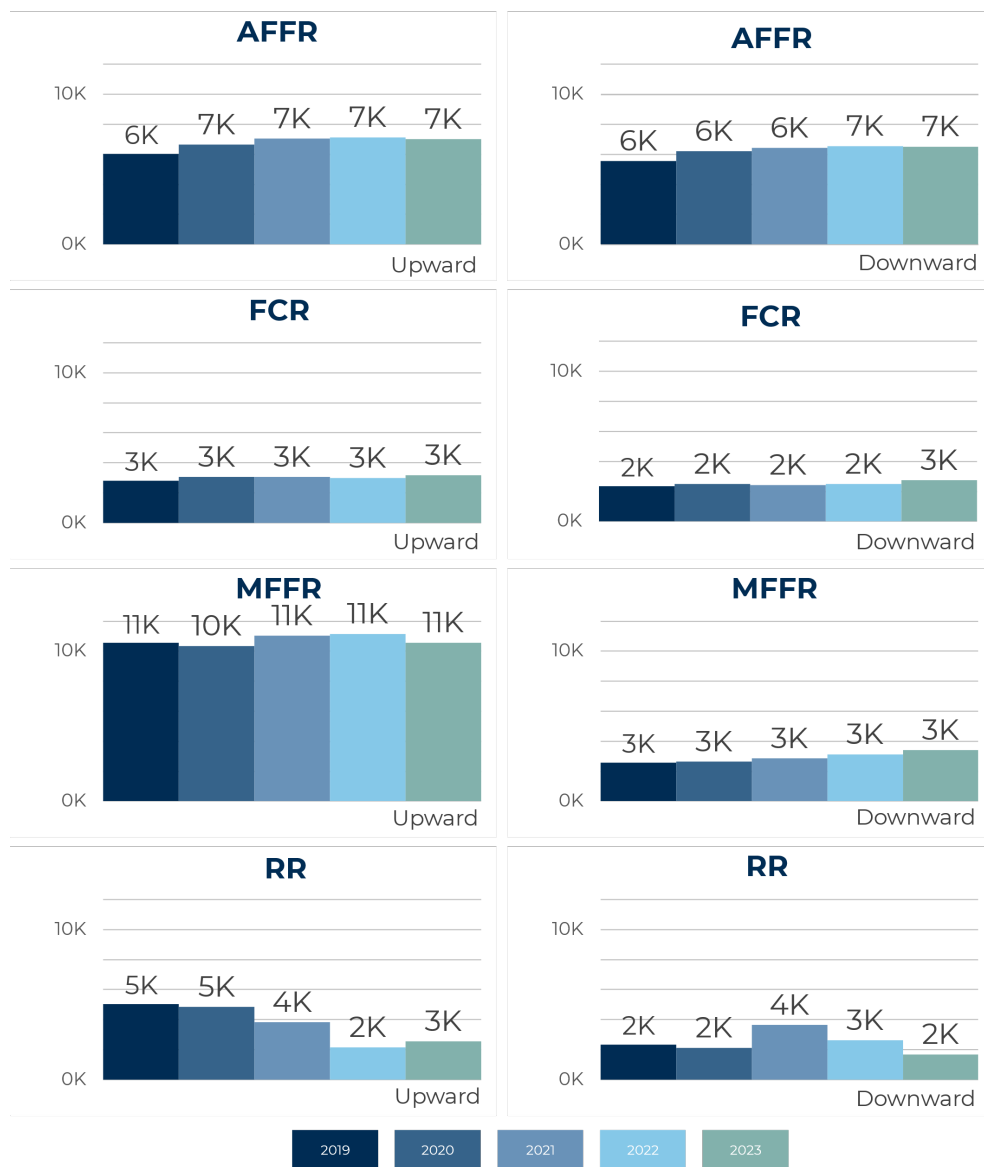
⁴² We note significant data limitations here, discussed further on page 36 of ACER's 2024 Market Monitoring Report: Progress of EU electricity wholesale market integration.

Figure 10: Annual volume of activated balancing energy (MWh)



Source: Authors' adaptation from ACER, 2024, Online dashboard for ACER 2024 Market Monitoring Report. Available at: <https://www.acer.europa.eu/media/charts/progress-eu-electricity-wholesale-market-integration-2024-market-monitoring-report>.

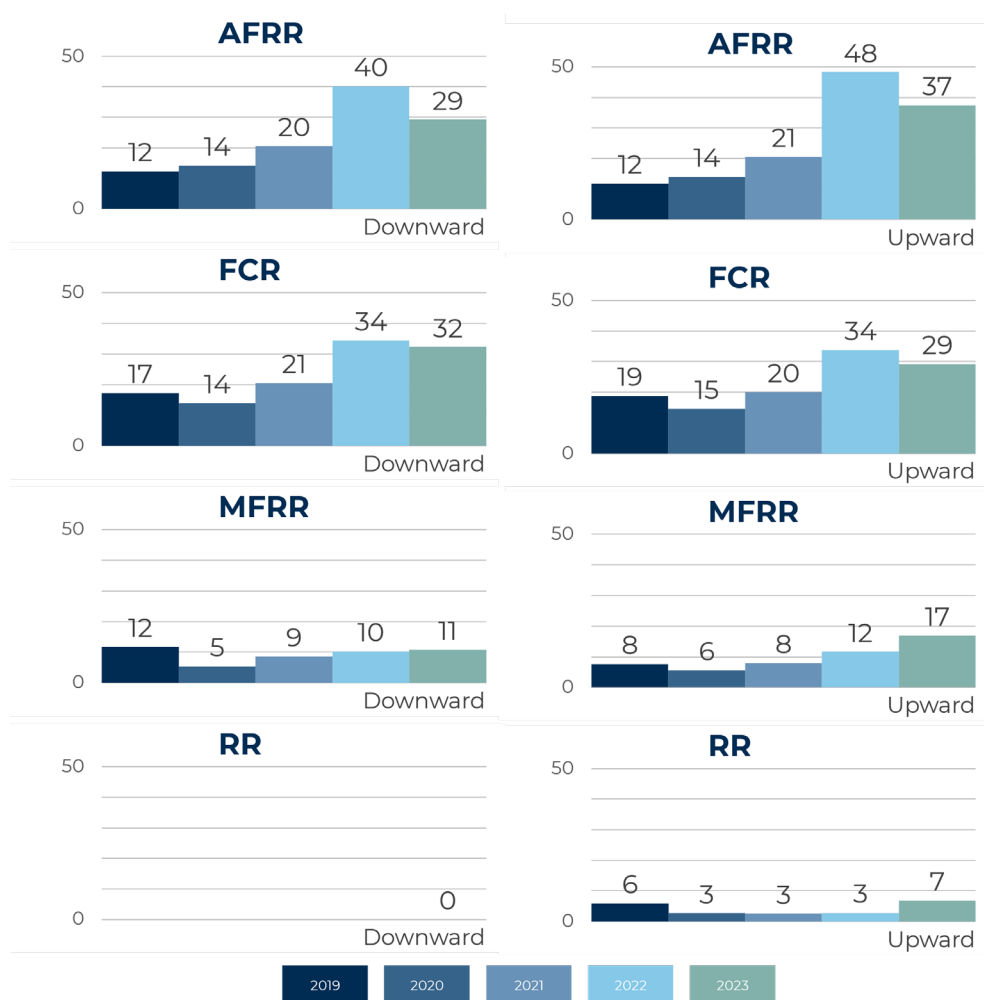
Figure 11: Annual average contracted reserve capacity (MW)



Source: Authors' adaptation from ACER, 2024, Online dashboard for ACER 2024 Market Monitoring Report. Available at: <https://www.acer.europa.eu/media/charts/progress-eu-electricity-wholesale-market-integration-2024-market-monitoring-report>.

While the balancing markets' size presented above show mixed trends, costs have a clear upward trend across all balancing timeframes (Figure 12). For all markets, we notice a steady growth in average prices for contracted reserve capacities. The price rise is especially high for aFRR services, where in 2023 upward/downward reserve capacity was contracted at 29 and 37 €/MWh, more than doubling their 2019 values.

Figure 12: Average prices for contracted reserve capacity (€/MWh)

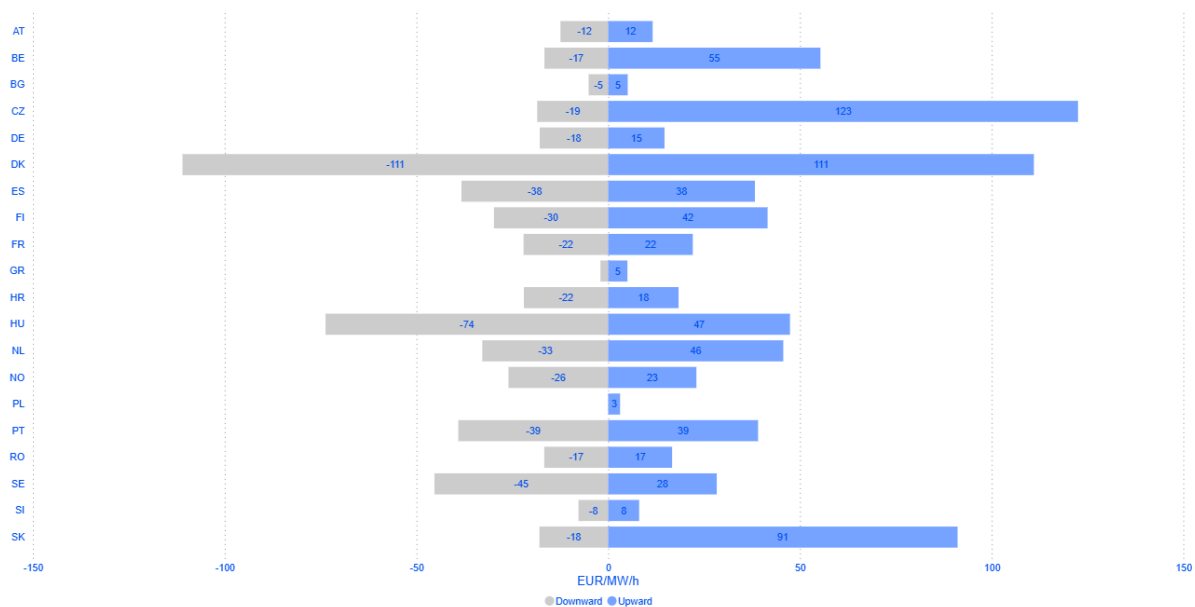


Source: Authors' adaptation from data of ACER, 2024, Online dashboard for ACER 2024 Market Monitoring Report. Available at: <https://www.acer.europa.eu/media/charts/progress-eu-electricity-wholesale-market-integration-2024-market-monitoring-report>.

A major challenge in balancing markets are the persisting high price spreads between national balancing markets (Figure 12). These spreads are significantly higher than those in coupled day-ahead markets (see Figure 8 above, for example). EU balancing platforms like PICASSO (aFRR), MARI (mFRR) and TERRE (RR) enable TSOs to exchange balancing energy, providing BRPs access to more competitive offers from neighbouring markets. Progress on the expansion plans for these platforms has been slower than expected, however. For example, as of November 2024, only the TSOs from Austria, Czechia, Germany, Italy, Denmark, the Netherlands, Belgium, and Slovakia were operational PICASSO members⁴³.

⁴³ ENTSO-E, 2024, *PICASSO*. Available at: https://www.entsoe.eu/network_codes/eb/picasso/.

Figure 13: Average prices for contracted reserve capacity per MS for aFRR services in 2023



Source: ACER, 2024, Online dashboard for ACER 2024 Market Monitoring Report. Available at: <https://www.acer.europa.eu/media/charts/progress-eu-electricity-wholesale-market-integration-2024-market-monitoring-report>.

b. Flexibility in the context of congestion management

TSOs and DSOs are responsible for ensuring that requests for new connections or reinforcements are addressed in a timely manner and that all connected grid users including generators, storage operators and consumers/prosumers have continuous access to the grid. This access must be without constraints resulting from a lack of grid capacity, unless contractually specified. However, in several EU Member States, grid capacity constraints hinder the timely connection of new network users, or restrict injections, withdrawals, or cross-zonal flows. For different reasons, congestion is increasingly occurring in EU grids. For example, congestion is now frequently being observed in the Netherlands⁴⁴, Poland⁴⁵, and Spain⁴⁶.

A typical measure used by network operators to prevent or reduce congestion is **redispatching power generation**, by reducing the injection at a specific location and increasing the injection at another location. Another option is to **reduce local demand (peak shaving)**. Flexibility offered by storage and demand response can provide such congestion management services, by off-taking or injecting power into the grid where needed.

Congestion management costs have been growing in recent years. In 2023, congestion management costs in the EU amounted to €4.3 billion, presenting an increased volume but decreased costs (due to lower energy prices) compared to 2022. Germany in particular has a highly active congestion

⁴⁴ ACM, 2024, *ACM publishes overview of measures against grid congestion*. Available at: <https://www.acm.nl/en/publications/acm-publishes-overview-measures-against-grid-congestion>.

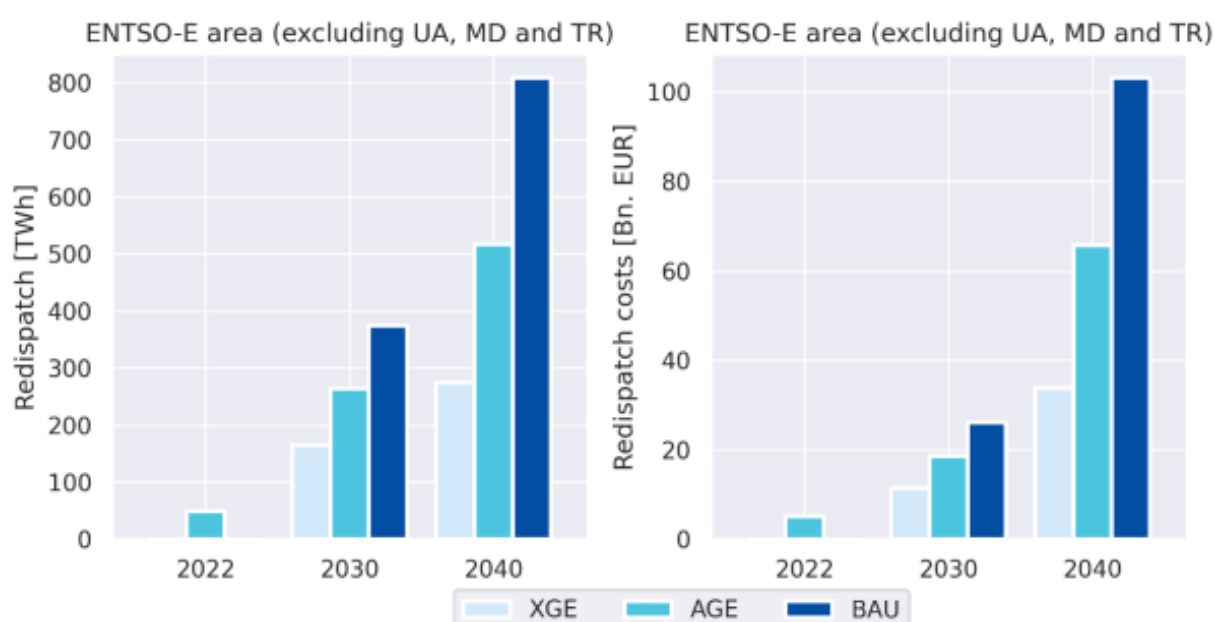
⁴⁵ PV Magazine, 2024, *Polish grid operator switches off gigawatts of PV*. Available at: <https://www.pv-magazine.com/2024/03/27/polish-grid-operator-switches-off-gigawatts-of-pv>.

⁴⁶ Pvtech, 2024, *Transmission grid congestion could hinder renewable power deployment in Spain* – DNV. Available at: <https://www.pv-tech.org/transmission-grid-congestion-could-hinder-renewable-power-deployment-spain-dnv/>.

management regime, with over €2.5 billion of costs representing over 30 TWh of remedial actions in 2023⁴⁷.

Predictions on future congestion management needs and costs expect this growth to continue in the coming years. A JRC analysis on congestion management redispatching volumes and costs shows that the costs for congestion management are expected to balloon to over €100 billion and renewable electricity curtailment to 310 TWh by 2040 in a business-as-usual scenario. Reductions in congestion management costs are possible if investments in grid expansion are accelerated and underlying drivers of reduced cross-zonal transmission capacity availability such as loop flows are addressed, among other factors.

Figure 14: Evolution of redispatching needs and costs in the European power system



Source: JRC, 2024, Future-proofing the European power market: Redispatch and congestion management. Available at: <https://publications.jrc.ec.europa.eu/repository/handle/JRC137685>.

Note: Colours indicate scenarios with low (dark blue), medium (blue), and high (light blue) levels of grid expansion.
UA: Ukraine; MD: Moldova ; TR : Turkey.

Addressing these challenges is further complicated by the long lead times required for investments in grid capacity expansion. These delays stem from complex planning and approval processes, rising equipment and construction costs, and high inflation rates, all of which compound the difficulties faced by grid operators. The lengthy timelines for grid development underscore the critical need for innovative solutions that can help bridge the gap while long-term investments are implemented.

⁴⁷ ACER, 2024, *Transmission capacities for cross-zonal trade of electricity and congestion management in the EU - 2024 Market Monitoring Report*. Available at: https://www.acer.europa.eu/monitoring/MMR/crosszonal_electricity_trade_capacities_2024.

Box 2: GOPACS - Cooperation between the Dutch electricity grid operators to procure flexibility services for local congestion management⁴⁸

While structural congestion should be addressed by investments in additional grid capacity, occasional local congestion can cost-efficiently be prevented or reduced by congestion management actions undertaken by grid operators. To this end, in the Netherlands the TSO TenneT and six regional DSOs have created the GOPACS platform to jointly procure flexibility services from market parties. This platform has an intermediary role and is connected with the market platforms for intraday trading managed by ETPA (Energy Trade Platform Amsterdam) and EPEX SPOT.

Assets with flexibility potential - such as heat pumps, e-boilers, cold/heat storage, batteries, and dispatchable power generation assets – are qualified for providing congestion management services. When a system operator foresees congestion in a specific section of its grid which needs to be remediated, it publishes an announcement on the platform indicating the characteristics of the congestion management services needed. Once bids are submitted for fulfilling the announced needs, GOPACS clears the market and communicates to asset operators the accepted bids and specific activation needs.

The GOPACS initiative was possible thanks to the market-based approach to congestion management in the Netherlands. This approach can as such be replicated in other Member States that have a market-based approach towards grid congestion but may need to be adapted in countries that have regulated redispatch rules (e.g. Germany). The project was enabled by the creation of a new framework for grid congestion management in the Electricity Network Code. This new code was approved in 2022 and contains specific provisions that facilitate coordinated procurement of flexibility by system operators:

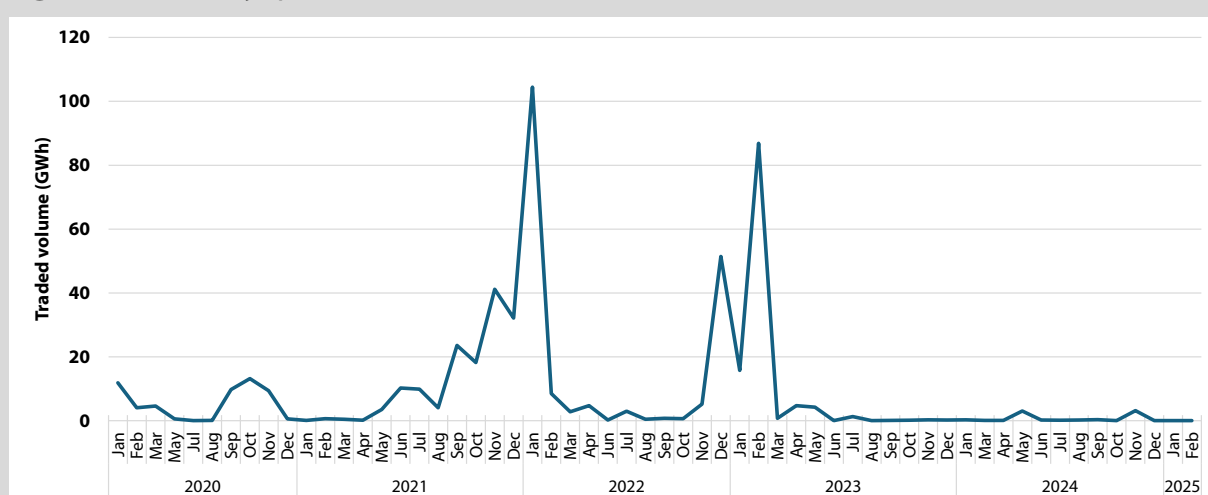
- Coordinated measures and direct communication between DSOs and TSO that can use each other's options for redispatch needs.
- Introduction of the concept of Congestion Management Service Provider (CSP), which is an aggregator for capacities smaller than 1 MW.
- Allows DSOs to procure long-term capacity products, whether based on reservation and/or activation.

At present about 20 market parties are offering flexibility services to grid operators via the GOPACS platform. The graph below shows the volumes traded monthly, with more significant trading volumes in 2022-2024, in the range of 114-181 GWh annually, and a lower trading level in 2024. GOPACS represents an important feature of the Dutch electricity market design removing barriers for the entry of flexibility resources. The figure shows that the effective utilisation of the platform significantly varies over time, depending on actual congestion management needs of network operators and offers from flexibility asset operators. It must be noted that other regulatory barriers remain in the Netherlands, such as the need to improve access of independent aggregators to the spot markets⁴⁹.

⁴⁸ Own elaboration based on expert knowledge and the following: GOPACS, s.d., main website. Available at: <https://www.gopacs.eu/en/>. EPEX Spot, 2024, EPEX SPOT & Local Flexibility - GOPACS developments. Zambotti, 2024, ETPA Whitepaper – GOPACS. Available at: <https://www.etpa.nl/en/whitepapers/gopacs>.

⁴⁹ SmartEn, 2024, Map on Wholesale Markets: Enabling demand-side flexibility in Europe. Available at: <https://smarten.eu/reports/2024-smarten-map-on-wholesale-markets-enabling-demand-side-flexibility-in-europe/>.

Figure 15: Monthly quantities (GWh) traded in GOPACS - 2020 to Feb 2025



Source: Authors' elaboration based on GOPACS, 2025, GOPACS performance. Available at: <https://app.gopacs.eu/public/performance-metrics>.

c. Other services

A few other services can also be provided by flexibility assets.

Similar to frequency, the voltage of power lines must be maintained within adequate ranges. **Voltage regulation** refers to two actions; first, the correction of voltage deviations due to higher local demand or infeed on a specific line, and second, maintaining stable phase between active and reactive power. Both these services can be provided by some types of flexibility assets.

Inertia in the electricity system refers to the inertia of synchronous rotating machines which are connected via generators to the electricity system, often in the form of thermal power plants. Disturbances in the electricity system are counteracted on the supply side by the inertia of these devices, which limits the extent and speed of frequency fluctuations. These impacts happen instantaneously or almost, after which other frequency containment tools can come into effect (aFRR, mFRR, and RR, as discussed in Section 2.2.2 a).

Inertia is not yet a major problem in most power grids that can still rely on thermal power plants. However, as wind and solar energy plants do not have inertia, and as thermal power plants will gradually be decommissioned, the missing inertia will in the medium to long term become a major issue.

In the near term, the inertia provision is not expected to become problematic in most of the EU, which is connected to Continental Europe's Synchronous Area (CESA). However, for areas outside of CESA, such as the Irish or British systems, lower available system inertia can become a problem for the grid. In recent years, some countries have already begun setting up service markets to this end. The Nordic power system, for example, has begun Fast Frequency Reserve (FFR) services since 2020, with 300 MW procured between the Nordic countries (excluding Iceland). FFR services react almost instantaneously to frequency fluctuations, mimicking the traditional role of inertia by preventing harsh swings in frequency due to system, load, or generation disruptions⁵⁰. Some flexibility assets can offer

⁵⁰ ENTSO-E, 2024, *Overview of Frequency Control in the Nordic Power System*. Available at: <https://www.entsoe.eu/publications/system-operations-reports/>.

(quasi)inertia services, but as the need for this service remains rather low, this is not discussed further in this report.

System restoration is another challenge for power grids. Most thermal plants cannot start without an initial amount of electricity to supply energy for some critical elements. However, many flexibility assets can begin injecting energy into the grid without the need of an initial burst of energy to start – i.e. they have “black start capability”. Maintaining some of this capacity in the energy system is important for restoring power in cases of emergency. Nonetheless, the need for it is lower than that of other services described earlier.

3. SOLUTIONS TO MEET THE INCREASING FLEXIBILITY NEEDS OF THE EU ENERGY SYSTEM

KEY FINDINGS

- Flexibility solutions differ in the optimal timescale they can be used for, and a mix of solutions will be necessary to meet all needs across the different timeframes.
- Short-term flexibility solutions, such as battery storage and demand response, typically have high upfront costs per unit of capacity but activation costs for these short-term solutions are comparatively low.
- Long-term flexibility solutions such as pumped hydro, flow batteries or power-to-x-to-power focus on minimising the cost per kWh of flexibility effectively provided, and ensuring low energy losses over extended periods.
- Estimating the actual potential and costs of different power system flexibility solutions poses significant challenges, due to the diversity of solutions, and lack of comprehensive and transparent information on operational performance and costs.
- Demand response, electrolyzers, hydropower (pumped-hydro and reservoirs), and battery storage are all expected to play a central role in meeting flexibility needs towards 2050. In the short to medium term, flexible thermal power plants will still play a major role.
- But there are in the literature still significant differences in the assessment of the possible contribution of the different solutions to flexibility. Furthermore some studies do not explicitly present disaggregated results for key solutions, including demand response, and study assumptions vary regarding the role of flexibility from unabated fossil fuel-based power generation.

There are many different solutions which are able to cover the required flexibility needs in the electricity system. While in the past flexibility was mainly provided by dispatchable power generation and pumped hydro, a shift can be observed towards storage and demand side solutions. In this chapter the main flexibility options with high potential are discussed. Where possible, quantitative assessments are added to provide more insights. However, as the definitions and approaches across studies are not harmonised, the quantitative information is highly dependent on market and other factors. It therefore needs to be interpreted carefully and is mainly illustrative.

3.1. Flexible power generation

Unabated fossil energy fuelled power plants still deliver most of the flexibility to the EU electricity system, but their share in the electricity mix is decreasing and they will gradually be phased out in the next decades. The flexibility of the current nuclear power park is limited, and only some renewable electricity generation technologies are dispatchable or flexible. Dispatchable generation assets deliver flexibility to the power system by adapting their output on demand. Depending on their ramp up/down rates, they can provide flexibility in very short timeframes. The main flexibility solutions on the generation side are:

- **Flexible (dispatchable) fossil energy fuelled power generation:** according to the IEA, about 50% of the flexibility needs in the EU was in 2023 covered by fossil energy based dispatchable power generation. The main technology used are open and combined cycle gas turbines

(OCGTs and CCGTs, respectively), which deliver the majority of flexibility currently, while the share of coal fired power plants in the EU is declining⁵¹. The flexibility delivered by these 'traditional' dispatchable power generation plants will decrease significantly due to the ongoing energy transition, with most scenario studies estimating the decrease will intensify after 2030 when more novel flexibility solutions have matured and become more competitive.

- **Combustion engines** can also provide short-term flexibility, due to their fast start-up times, lower minimum output, ability to ramp up quickly and capacity to use different fuels, albeit at higher operational costs than gas turbines⁵². However, the environmental impact of these engines is a concern, leading to innovations aimed at using cleaner fuels such as biogas or synthetic fuels to mitigate their carbon footprint.

Some scenarios also foresee a role for **abated gas-fired turbines** using Carbon Capture and Storage (CCS) for providing flexibility on different timescales, but estimates on costs and suitability of the technology differ. The impact assessment study for the 2040 climate target foresees a limited role for abated gas turbines in the power generation mix⁵³.

- **Dispatchable renewable electricity generation** - several solutions are implemented or being considered:
 - **Hydropower:** hydropower installations equipped with water reservoirs can respond quickly to (unexpected) fluctuations in supply and/or demand and provide short and long-term flexibility. Hydropower reservoirs provided in 2022 about 8% of the EU's total electricity generation. While being the largest renewable dispatchable generation source in the EU currently, the potential for expansion is limited. Hydropower plants without reservoirs (run-of-river installations) delivered in 2022 about 3% of total generation but have limited flexibility capability⁵⁴. Additionally, pumped hydro storage (PHS) delivers the majority of current EU and global storage capacity and volume, and has some but still limited potential for growth⁵⁵. PHS is discussed separately in the section 3.2 on storage below.
 - **Renewable or low-carbon hydrogen-fuelled turbines and engines:** dispatchable turbines and engines using green hydrogen produced with renewable electricity via electrolyzers can in the future provide flexibility. Existing methane-fuelled turbines could be converted to allow for the use of pure or a mix of hydrogen with methane. Due to the low round-trip efficiency of using renewable or low-carbon hydrogen for (flexible) electricity generation and the resulting high costs, some experts do in general not see a large role for this flexibility pathway, also given the economic value of using clean hydrogen is larger in other applications. In their 2023 study Trinomics and Artelys

⁵¹ IEA, 2024, *Managing seasonal variability of electricity demand and supply*, Figure 12. Available at: <https://www.iea.org/reports/managing-the-seasonal-variability-of-electricity-demand-and-supply>.

⁵² Energy Brainpool, 2017, *Flexibility Needs and Options for Europe's Future Electricity System*. Available at: <https://www.eugine.eu/category/brochures/>.

⁵³ European Commission, 2024, *SWD/2024/63 final: Impact assessment 2040 climate target*. Available at: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52024SC0063>.

⁵⁴ JRC, 2023, *Hydropower and pumped hydropower storage in the European Union*. Available at: https://setis.ec.europa.eu/hydropower-and-pumped-hydropower-storage-european-union-0_en.

⁵⁵ JRC, 2023, *Hydropower and pumped hydropower storage in the European Union*. Available at: https://setis.ec.europa.eu/hydropower-and-pumped-hydropower-storage-european-union-0_en.

estimated for the Pentalateral⁵⁶ region a substantially larger role for the flexible operation of electrolyzers than for hydrogen-fuelled turbines (PtoH2toP) in providing flexibility⁵⁷. The potential role of electrolysis for providing flexibility is discussed separately in section 3.3.

- **Biomass or biogas fuelled power generation:** currently, in some EU Member States, a substantial part of renewable dispatchable power generation comes from biomass or biogas fired Combined Heat and Power (CHP) plants or other turbine/engine types. Biomass offers several options for storage, controlled use, and conversion to energy carriers including electricity⁵⁸. A 2023 JRC study estimates that part of the current methane-fuelled turbines could be replaced by biomass-fuelled turbines and could mainly play a role in delivering weekly flexibility. Moreover, some biomass power plants could partly substitute decommissioned coal fired power plants⁵⁹. Similarly to fossil fuel based power plants, the flexibility of bioenergy has several dimensions: short term flexibility to support the electricity system by providing ancillary services and congestion management, and long term (seasonal) flexibility by storing (dry) biomass and using it in existing assets when required.
- **Renewable electricity generation curtailment:** although this option often has a negative connotation, it can be a cost-effective way of delivering local flexibility (e.g. for grid congestion management) in times of high solar radiation. However, structural grid congestion should not be addressed by curtailment of renewable electricity injection into the grid but rather by appropriate investments in grid expansion, co-located storage solutions or other measures.
- **System-friendly renewable electricity generation:** this solution consists of measures taken in the design, siting or operation of renewable electricity generating assets, enabling them to have a smoother (or more aligned to residual load) generation profile, thus reducing flexibility needs, and/or to provide flexibility services⁶⁰. Further details are provided in Box 3 below.

According to a cost–benefit analysis and comparison of different flexibility options undertaken for a current and outlook scenario in Germany, the flexibilisation of existing CHPs stands out as a particularly attractive option from an economic perspective. Similarly, the flexibilisation of wind turbines and photovoltaic systems appears to be an appropriate solution in the outlook scenario; their participation in flexibility markets would be enabled by pooling⁶¹.

⁵⁶ Pentalateral region consists of Belgium, Germany, France, Luxembourg, the Netherlands, Austria, and Switzerland.

⁵⁷ Trinomics & Artelys, 2023, *Power System Flexibility in the Penta region*. Available at: <https://www.benelux.int/nl/etude/power-system-flexibility-in-the-penta-region-current-state-and-challenges-for-a-future-decarbonised-energy-system-2/>.

⁵⁸ IEA, 2021, *Technologies for Flexible Bioenergy*. Available at: <https://www.ieabioenergy.com/wp-content/uploads/2021/08/Task-44-Technologies-for-Flexible-Bioenergy.pdf>.

⁵⁹ JRC, 2023, *Flexibility requirements and the role of storage in future European power systems*. Available at: <https://publications.jrc.ec.europa.eu/repository/handle/JRC130519>.

⁶⁰ Trinomics & Artelys, 2022, *Study on flexibility options to support decarbonization in the Energy Community*.

⁶¹ Ackermann, L., et al., 2024, *Cost-benefit analysis and comparison of grid-stabilizing energy flexibility options and their applications in relation to the German energy system*. Available at: <https://www.sciencedirect.com/science/article/pii/S1755008424000346>.

Box 3: System-friendly renewable electricity generation assets can reduce flexibility needs and/or participate in flexibility provision

Examples of system-friendly RES include the on-site combination of renewable energy generation with storage, solar trackers for solar PV farms, new operational strategies (e.g. reserving part of the generation capacity for provision of upward balancing energy), or enhancing the capability of the renewable electricity generation assets in view of providing ancillary services such as synthetic inertia or fault ride-through⁶².

By reflecting electricity supply scarcity and system constraints, well-designed and properly functioning electricity markets should naturally incentivise project developers to design and build system-friendly renewable electricity generating facilities, as developers have an interest in maximising their revenues by focusing on producing during periods with high electricity prices. But in practice, support schemes may still not properly incentivise system-friendly investment and operational decisions. We notice for example that several RES operators continue to produce and inject electricity during moments of negative electricity spot prices. Even if support schemes do not remunerate renewable electricity generation when negative electricity prices occur, their design may still not incentivise operators to operate flexibly. Moreover, in order to choose for a system-friendly design, operators need to be able to forecast electricity prices throughout the project lifetime and to accurately predict that their net revenues would be higher with a system-friendly design than with a 'conventional' design.

Therefore, both due to the need for improvement of electricity markets and support schemes' design and the need for RES operators to accurately forecast highly uncertain future electricity prices, insufficient incentives may at present exist for operators to choose for system-friendly instead of conventional designs. The implementation of well-designed CfDs can be an appropriate measure to incentivise system-friendly design and operation of renewable electricity plants. The design of CfDs or other support schemes should provide system-beneficial investment incentives and should not hinder the participation of resources in electricity and ancillary services' markets. RES generators should instead be incentivised to respond to market signals and market prices should signal/reflect profitable investments in flexibility assets.

3.2. Energy storage

Energy storage will play an increasing role in providing flexibility to the electricity system. Much of the short term (up to a few days) flexibility capacity needs in the future is expected to be provided by batteries, especially given the significant cost decreases in recent years⁶³. The most relevant storage technologies to cover flexibility needs are:

- **Stationary batteries:** battery energy storage systems (BESS) constitute an important focus of current electricity storage solutions. BESS enable to support flexibility activities such as peak shaving, self-consumption optimisation, and backup electricity production in the event of

⁶² Power system synthetic inertia refers to the emulation of the inertial response of traditional synchronous generators by power electronic devices and control systems to stabilise the grid during disturbances. Fault ride-through capability is the ability of power generation systems to remain connected and operational during and after short-term voltage disturbances on the grid.

⁶³ See e.g. IEA, 2024, *Batteries and Secure Energy*, Available at: <https://iea.blob.core.windows.net/assets/cb39c1bf-d2b3-446d-8c35-aae6b1f3a4a0/BatteriesandSecureEnergyTransitions.pdf>; Bloomberg BNEF, 2024, *Lithium-Ion Battery Pack Prices See Largest Drop Since 2017, Falling to \$115 per Kilowatt-Hour*.

supply deficits. BESS are increasingly being installed in stand-alone utility-scale facilities, or behind the meter in industrial, commercial and residential buildings, as well as co-located with renewable energy installations, and are used by the concerned market operators to balance their own portfolio and to provide flexibility services to network operators and other market parties. Lithium-ion batteries, with round-trip efficiencies of up to 90%, are common for applications lasting seconds to hours, with the largest installation being around 3 GWh⁶⁴. Due to the scarcity of lithium, other options are being explored such as sodium-ion (Na-ion), sodium-sulphur (Na-S), and metal-air batteries. While currently BESS provide up to a few hours of storage, based on cost reductions forecasts, about twelve hours of discharge capacity could be achieved by 2050⁶⁵. Moreover, alternative chemistries are being developed to enable storage for even longer durations, for example flow batteries, which can reach round-trip efficiencies of 50–80%, store energy for up to 100 hours, and have large-scale pilot and even some commercial projects commissioned. Metal anode batteries have a lower technological maturity, but could store energy for up to 200 hours with a round-trip efficiency of 40–70%⁶⁶.

- **Mobile batteries (EVs):** given the expected widescale adoption of EVs in Europe, scheduled charging/discharging of EVs based on the actual system and market conditions (reflected by variable grid tariffs and/or dynamic electricity prices) can substantially contribute to system flexibility. Vehicle-to-grid (V2G) technology allows EVs to operate as a two-way power flow to both take off and feed electricity into the grid (while EV smart charging is rather a form of demand response). Europe could leverage its growing park of battery electric vehicles and ensure that a significant share of the fleet is capable of providing V2G services. While the flexibility potential of EVs is high, complexities such as technological barriers, privacy concerns regarding data communication, high investment costs for V2G-enabled charging infrastructure, the dynamics of stakeholder participation and regulatory frameworks remain pivotal challenges to overcome⁶⁷.
- **Pumped-Hydro Storage (PHS):** more than 95% of the currently installed storage capacity and energy (volume) comes from PHS. Not all hydropower reservoirs have a PHS design: about 30% of the installed capacity in the EU consists of PHS systems (either closed-loop or mixed systems)⁶⁸. PHS plays a vital role in delivering flexibility on all timescales (sub-hourly to seasonal), but the potential for increasing the PHS capacity through new facilities or transforming existing facilities in the EU is limited, also due to environmental considerations⁶⁹.
- **Other Medium and Long-Duration Energy Storage technologies:** next to batteries and pumped-hydro storage (PHS), energy can be stored in various other forms including chemical, thermal, mechanical, or electrochemical. Medium-duration energy storage mainly refers to

⁶⁴ Xoserve, 2023, 'Blog: Energy storage—the missing piece of the net zero puzzle?'. Available at: <https://www.xoserve.com/news/blog-energy-storage-the-missing-piece-of-the-net-zero-puzzle>.

⁶⁵ WindEurope, 2024, *Maximising the power of wind through grid flexibility*, based on *Power for 127 Hours: The Economics of Long-Duration Energy Storage*. Available at: <https://environment-review.yale.edu/power-127-hours-economics-long-duration-energy-storage>.

⁶⁶ Science and Technology Committee House of Lords UK, 2024, *Long-duration energy storage: get on with it*. Available at: <https://publications.parliament.uk/pa/ld5804/ldselect/ldstech/68/68.pdf>.

⁶⁷ Ondřej, Š., et al., 2024, *Electric vehicles as facilitators of grid stability and flexibility: A multidisciplinary overview*. Available at: <https://wires.onlinelibrary.wiley.com/doi/10.1002/wene.536>.

⁶⁸ JRC, 2023, *Hydropower and pumped hydropower storage in the European Union*. Available at: https://setis.ec.europa.eu/hydropower-and-pumped-hydropower-storage-european-union-0_en.

⁶⁹ Gimeno-Guiterrez, M., et al., 2015, *Assessment of the European potential for pumped hydropower energy storage based on two existing reservoirs*. Available at: <https://www.sciencedirect.com/science/article/abs/pii/S096014811400706X>.

technologies with storage duration capability of between 4 and 24 hours or a few days at most, while long-duration energy storage technologies should be capable to store energy for several days, to weeks, months or even years⁷⁰. The most promising technologies are:

- **Hydrogen** can be used for longer-duration energy storage over weeks and months. It can be stored under moderate pressure in underground salt caverns (including repurposed natural gas storage sites) or in converted depleted gas fields and used for flexible (peak) electricity generation. While the technology could reach relatively low energy storage costs, a main disadvantage concerns the comparatively low round-trip efficiency of 30–40% of the overall process, including hydrogen production, storage and use in power plants. The combination of long storage times with low energy efficiency means that this option is most useful for infrequent cycling and longer-term storage.
- **Compressed air energy storage (CAES)** involves storing air at high pressure in tanks or underground caverns in view of using it to drive electricity generators. Commercial CAES facilities have been commissioned, especially in China, achieving a round-trip energy efficiency of up to 40–70%. Variations include advanced CAES, which re-uses the heat generated during compression to achieve higher efficiencies, and liquid-air energy storage, which has achieved maturity for small systems and a round-trip energy efficiency of around 55%.
- **Thermal energy storage:** while small-scale heat storage in tanks is frequently used as short-term flexibility option e.g. in combination with heat pumps, technologies are available to store heat for example in molten salts or phase-change materials and use it to provide medium to long-term (seasonal) flexibility.
- **Renewable or low-carbon synthetic fuels** can be produced and stored similarly to fossil fuels currently and converted back into electricity with a round-trip energy efficiency of around 30%. Storage on a TWh scale is technically possible, but these technologies are less mature (TRL 6–7) and are projected to be more expensive than using the hydrogen pathway⁷¹.

⁷⁰ Science and Technology Committee House of Lords UK, 2024, *Long-duration energy storage: get on with it*. Available at: <https://publications.parliament.uk/pa/ld5804/ldselect/ldsctech/68/68.pdf>.

⁷¹ Science and Technology Committee House of Lords UK, 2024, *Long-duration energy storage: get on with it*. Available at: <https://publications.parliament.uk/pa/ld5804/ldselect/ldsctech/68/68.pdf>.

Box 4: Grid-scale and home batteries

Stationary large-scale storage systems are increasingly expected to cover short-term flexibility needs resulting from the development of solar PV and wind energy. Such utility-scale batteries are very well suited to absorb intra-day and daily fluctuations in the output of variable electricity generation installations. Home batteries can also contribute to system flexibility, but in general they primarily aim to increase self-consumption of on-site generated solar power.

Stationary battery storage grew 11 times from 2018 to 2023, reaching 86 GW globally⁷². This growth is expected to continue, with Bloomberg New Energy Finance predicting 411 GW by 2030, and the IEA estimating 680 GW in its Net Zero Scenario. This would surpass pumped hydro (160 GW) as the main flexibility option globally, helping diversify the energy storage fleet. Most new installations are in China, the US, Europe, and South Korea.⁷³

In Europe, according to EASE and LCPDelta the installed capacity in grid-scale batteries would reach 57 GW by 2030. The European TSOs expect even a higher growth and assume in their latest European Resource Adequacy Assessment that the EU grid-scale battery capacity would reach 74 GW by 2030. If this becomes a reality, battery storage would surpass the expected 2030 pumped hydro total installed capacity of 53 GW. Battery installed capacity could increase even further in case co-location with utility-scale solar PV parks becomes more widespread. According to TYNDP 2024 scenarios, grid-scale batteries could reach up to 142 GW in 2030 in the EU in case of widespread co-location with solar PV parks, assuming a 2.8 hour battery duration. The Impact Assessment for the EU 2040 Climate Target expects 100 GW of battery storage by 2030, with installed capacity reaching 135 to 200 GW by 2040⁷⁴. Notwithstanding the high expected growth rates, massive development could still be hindered by the limited availability of the necessary resources for the construction of battery storage on a large scale.

The current technology of stationary batteries is suitable for **short-term storage of electricity**, but not for storage during weeks or months. Flow batteries could cover such longer storage periods, but this technology has not yet reached commercial maturity. Considering the capacity constraints of stationary batteries, they can to some extent contribute to addressing electricity supply shortage during Dunkelflaute events (period of low wind and solar power generation due to a combination of overcast skies and calm weather, often occurring during winter months), but additional back-up or peak generation capacity would anyhow be required.

Both utility-scale and home batteries can be used to balance portfolios of BRPs and to participate in short-term electricity markets. They can also provide ancillary services to TSOs such as Primary Control Reserve and Secondary Reserve capacity and serve as black start units. Finally, batteries can provide flexibility to avoid bottlenecks in transmission and distribution grids.

⁷² SIA Partners, 2024, *Stationary battery storage*. Available at: <https://www.sia-partners.com/en/insights/publications/stationary-battery-storage>.

⁷³ Next, 2023, *FlexAbility: How Battery Storage is Advancing the Energy Transition*. Available at: <https://www.next-kraftwerke.com/energy-blog/flexibility-battery-storage>.

⁷⁴ Ember, 2024, *Making Clean Power Flexy*. Available at: <https://ember-energy.org/latest-insights/making-clean-power-flexy/>.

3.3. Demand-side flexibility

Demand-side flexibility, also referred to as demand-side management, involves programs, policies and technologies that aim to adapt the amount of electricity consumed during times of peak demand (high spot prices) or peak supply (low or negative spot prices), often by using load shifting or shaving enabled by financial incentives or other measures.

According to a report from the European association for digital and decentralised energy solutions (SmartEn) and DNV, activating the full demand-side flexibility potential from buildings, electric vehicles and industry could provide significant economic and climate benefits⁷⁵.

Demand response solutions can vary significantly in scale, complexity and ease of deployment:

- **Industrial demand response** typically involves large-scale facilities with significant electricity consumption, where automated systems and advanced controls can adjust operations and processes to respond to grid tariffs or market price signals, or provide explicit flexibility services. This sector often leverages sophisticated energy management systems and has a higher potential for demand flexibility due to the magnitude of its energy use. Box 5 further illustrates the industrial demand response aggregators.
- **Commercial demand response** includes a range of businesses such as retail stores, office buildings, and warehouses. These solutions focus on adjusting heating, ventilation, air conditioning (HVAC) systems, lighting, and other non-essential loads to manage energy use depending on the electricity grid and market situation. Commercial entities often employ smart building technologies and energy management platforms to coordinate these adjustments.
- **Residential demand response solutions** are more decentralised, involving individual households that adjust their energy consumption through smart appliances, thermostats, and home energy management systems. While the impact of each residential unit is smaller compared to industrial or commercial users, the aggregated effect across many households can be significant. Residential programs often rely on incentives and dynamic pricing to encourage participation, with a focus on simplicity and ease of use for consumers.
- **Hydrogen electrolyzers** can in the future present a significant potential for demand response. By adjusting their operational schedules in response to electricity price signals, electrolyzers can consume electricity during periods of low demand or high renewable generation, thus providing flexibility to the grid (balancing, congestion management and in some cases voltage control services). This flexible operation of electrolyzers can help manage the system and may reduce the overall cost of hydrogen production. Alkaline and proton-exchange membrane electrolyzers can be turned on or off relatively quickly, making them suitable for both short and medium-term demand response, although provision of faster balancing services (FCR, and for alkaline electrolyzers also aFRR) from a cold start is more difficult. Solid oxide electrolyzers are currently less able to provide balancing, voltage control or congestion management services, in particular to the current small plant size⁷⁶.

⁷⁵ SmartEn & DNV, 2022, *Demand-side flexibility: Quantification of benefits in the EU*. Available at: <https://www.dnv.com/publications/demand-side-flexibility-quantification-of-benefits-in-the-eu-232342/>.

⁷⁶ Frontier Economics & ENTSO-E, 2022, *Potential of P2H technologies to provide system services*. Available at: <https://www.entsoe.eu/2022/06/28/entso-e-publishes-a-study-on-flexibility-from-power-to-hydrogen-p2h2/>.

Box 5: Industrial Demand Response Aggregators

Industrial Demand Response Aggregators (IDRAs) are specialised intermediaries that enable industrial facilities to participate collectively in demand response (DR) programs. By bundling the load reduction capabilities of multiple industrial sites, these aggregators create a sizable resource that can be used for demand response.

Making large-scale industrial production processes more flexible is challenging. Industrial demand response plays a key role in flexibility provision, as many companies have unused potential to adjust their energy use for demand response with clear economic and sustainability benefits. However, many industries have not yet fully utilised this potential, which limits its broader adoption.

The table below shows illustrative case studies based on real commercial applications of **IDRAs in Europe**. These examples show how different industrial sectors have leveraged flexible equipment, ranging from the food industry to a manufacturing company and a warehouse.

These case studies demonstrate how IDRAs can tailor solutions to a wide range of industrial settings. By identifying available flexibility, aggregating it into marketable assets (often through virtual power plants), and providing the necessary market access, these projects have enabled industrial players to reduce the pressure on the electricity system while lowering their energy costs and generating revenues. The case studies highlight different levels of impact, from minimising the **curtailment of RES** during peak demand to more advanced strategies like **energy trading**. This reflects the broad scope and potential of demand response in industrial settings.

Table 2: Illustrative case studies of IDRAs in Europe

	Bamboo Energy & Europastry ⁷⁷	Next-Kraftwerke and Peleman industries ⁷⁸	Flextricity and Norish ⁷⁹
Country	Spain	Germany	England
Industry	Food industry	Printable Products Manufacturing	Multi-temperature Warehousing
IDRA activity	Identified flexibility and provided market access	Aggregated energy storage systems into a virtual power plant for FCR services	Identified flexibility and provided market access
Flexible equipment	Refrigerator	Energy storage	Cooling plant
Outcomes	Participation in flexibility markets by the refrigerator, with a potential revenue of 50-120 k€/year based on electricity prices.	Increased self-consumption by an additional 8%-15%, reducing reliance on grid capacity by 5-20%.	Generated an annual average revenue of €22.230 since 2008. Reduced CO ₂ emissions by an estimated 300-750 tonnes/MW annually.

Source: Ranaboldo, M., et al., 2024, *A comprehensive overview of industrial demand response status in Europe*. Available at: <https://www.sciencedirect.com/science/article/pii/S1364032124005239>.

⁷⁷ Bamboo energy, 2022, *Europastry*. Available at: <https://bambooenergy.tech/en/europastry/>.

⁷⁸ Next-kraftwerke, 2023, *Buffering the energy crisis with wind power and storage by Peleman industries*. Available at: <https://www.next-kraftwerke.com/vpp/case-studies/battery-bess-fcr>.

⁷⁹ Flextricity, 2024, *Load management helps Norish Cold Storage reduce GB carbon emissions and generate revenue*. Available at: <https://www.flextricity.com/resources/case-studies/load-management-helps-norish-cold-storage-reduce-gb-carbon-emissions-and-generate-revenue>.

The main conditions for a successful development of demand response are:

- **Energy management systems** including smart meters, smart appliances and adequate communication tools represent the first step to unlock the full flexibility potential of energy users. Adequate regulation that enables more complex use cases (e.g. activation of demand response by independent operators other than BRPs) is also key to foster active market participation of demand response. A preliminary condition to enable the efficient activation of demand response is a full rollout of smart meters. Despite the critical role of smart meters in providing flexibility via dynamic pricing and consumer participation, deployment of low-voltage smart meters remains limited in 10 EU Member States⁸⁰.
- The implementation of adequate **network tariff structures** is another important factor to support flexibility at demand side, by offering time-of-use tariffs that are higher during (system-coincident) peak demand times and lower during off-peak times. DSOs and TSOs can also use grid tariff signals to incentivise grid users to offer locally available flexibility to prevent or reduce occasional grid congestion. Network tariffs that include time-differentiated elements can decrease total network charges to flexible network users, as well as more broadly reduce tariffs to all network users, by deferring (some) network reinforcements. However, the (potentially negative) impact on users which are not able to react to time-differentiate signals needs to be considered. Spain and Finland among others have implemented such tariffs⁸¹.
- Next to time-of-use grid tariffs, **dynamic electricity retail pricing** can enhance demand-side flexibility by providing financial incentives to household and commercial end-users. Passing on variable electricity prices linked to day-ahead wholesale prices to end-users incentivises them to use electricity appliances at times when prices are low, which is usually during high generation and low demand. Demand response balances out the load profile and contributes to reducing price peaks and occurrence of negative spot prices on the wholesale market, which can be detrimental to the business case of renewable electricity generation projects. However, retail consumers are in general quite risk-averse, and may also be reluctant to switch to variable price contracts due to high search and transaction costs and the fact that fixed-price contracts are offered as the regulated default option, while the availability of variable-price contract options may still be limited. This in turn can hinder the provision of adequate incentives for the provision of implicit flexibility by these consumers⁸².

Several barriers for deployment of demand response persist in many Member States, as highlighted in Figure 16. These include the lack of appropriate legal frameworks to allow market access, restrictive requirements to provide balancing services, lack of incentives to participate in balancing markets. Governmental interventions in retail electricity prices, such as regulated or capped retail prices or subsidising energy consumption costs, often dilute price signals, reducing consumers' motivation to adjust their energy consumption pattern. Without adequate price signals the potential for demand response remains to a large extent untapped⁸³.

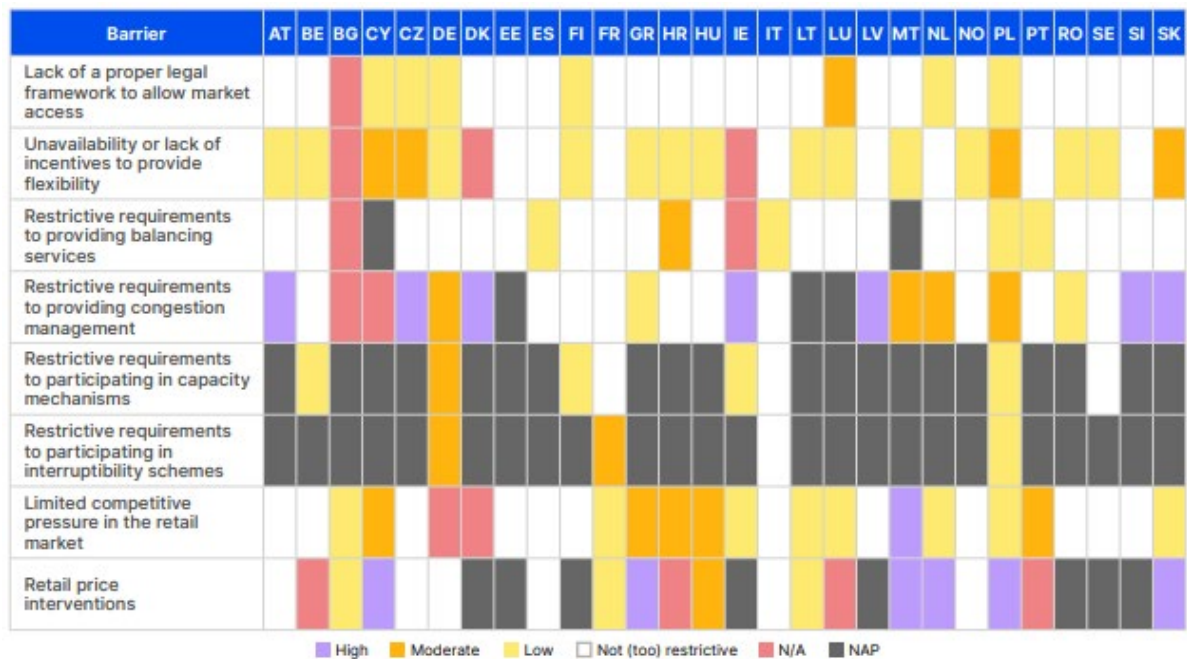
⁸⁰ ACER & CEER, 2024, *Energy retail - Active consumer participation is key to driving the energy transition: how can it happen?* Available at: https://www.acer.europa.eu/sites/default/files/documents/Publications/ACER-CEER_2024_MMR_Retail.pdf.

⁸¹ ACER & CEER, 2024, *Energy retail - Active consumer participation is key to driving the energy transition: how can it happen?* Available at: https://www.acer.europa.eu/sites/default/files/documents/Publications/ACER-CEER_2024_MMR_Retail.pdf.

⁸² ACER & CEER, 2024, *Energy retail - Active consumer participation is key to driving the energy transition: how can it happen?* Available at: https://www.acer.europa.eu/sites/default/files/documents/Publications/ACER-CEER_2024_MMR_Retail.pdf.

⁸³ ACER, 2023, *Demand response and other distributed energy resources: what barriers are holding them back? 2023 Market Monitoring Report*. Available at: <https://acer.europa.eu/monitoring/MMR>.

Figure 16: Overview of barriers that hinder flexibility provision by distributed energy resources and new entrants or small actors

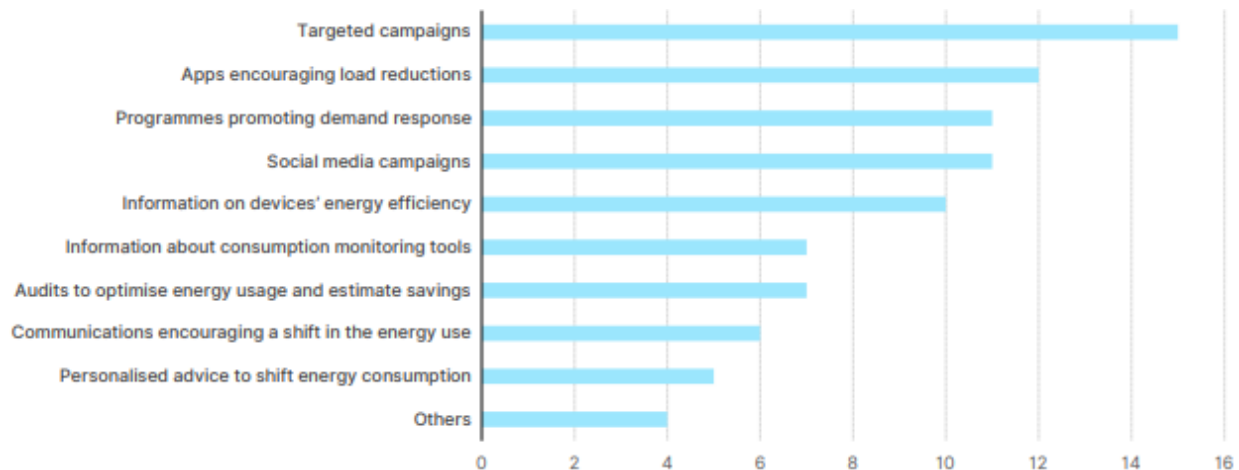


Source: ACER and CEER, 2024, Energy retail - Active consumer participation is key to driving the energy transition: how can it happen?⁸⁴

While supply-side measures like support schemes may be essential for enabling investments in flexibility resources, the demand side should also be properly stimulated. Education campaigns, clear advice and easy-to-use apps empower consumers to actively participate in demand response programs. Figure 17 highlights that several Member States have made significant efforts in this area. Targeted campaigns are the most common, with 15 Member States using this measure. Digital tools, such as apps to optimise energy use and programs promoting demand response, are also widely implemented. Social media campaigns and detailed information about energy-efficient devices further enhance consumer engagement.

⁸⁴ Available at: https://www.acer.europa.eu/sites/default/files/documents/Publications/ACER-CEER_2024_MMR_Retail.pdf.

Figure 17: Number of Member States implementing measures to improve consumers' awareness on demand response - 2022



Source: ACER and CEER, 2024, *Energy retail - Active consumer participation is key to driving the energy transition: how can it happen?*⁸⁵

3.4. Flexibility indirectly provided by domestic electricity networks and interconnectors

The flexibility needs to be provided by assets at the supply or demand side (including storage) can be reduced by **timely investments in and adequate operation of distribution and transmission grids**. Flexibility solutions such as demand response, storage and flexible connection agreements can to some extent be used to address occasional congestion and delay or reduce the need for grid investments, but grid operators should anticipate future grid use needs and timely invest in grid expansion to avoid any structural congestion and to timely connect new grid users at the demand or supply side. In addition to reducing flexibility needs, **transmission and distribution networks act as a link between flexibility options**. Conversely, if grid capacities or availability are too low, congestion can be a limiting factor for some flexibility providers.

⁸⁵ Available at: https://www.acer.europa.eu/sites/default/files/documents/Publications/ACER-CEER_2024_MMR_Retail.pdf.

Box 6: Flexible connection agreements

Historically, connection agreements between grid users and grid operators were “firm”: grid users were provided with unconstrained access at a pre-specified MW capacity. However, grid users often do not need the entire contractual capacity, or derive diminishing value from additional capacity. To optimally use the available grid capacity, grid operators have now in several EU Member States the possibility to provide capacities that are not “firm” but “flexible”. Such flexible connection agreements (FCAs) allow to connect new grid users to grid sections with limited capacity while also unlocking benefits for the grid operator.

FCAs are in the EU legislation particularly highlighted as a solution for grid congestion. In the electricity market design reform, a paragraph (6(a)) has been added to Directive 2019/944 describing and mandating a framework for FCAs.

FCAs can take many forms and differ on various aspects. These aspects include the level of firmness, the time profile, the (share of the) connection capacity that is deemed flexible, and the type of connections (shared or individual):

- Firm connections can be limited **during specific timeslots**, for example when the grid encounters significant congestion, with a short up-front warning (e.g., 1 day before). The most advanced version of this setup includes dynamic operating envelopes, such as those implemented in Australia⁸⁶.
- Connections can also be **on a time-of-use basis**, limited at specific hours of the day (possibly in specific months) to a lower capacity.
- **Some connections can be fully flexible**, whereby the entire capacity of the connection can be disconnected if needed. **Shared connections** can be considered, whereby the maximum capacity is shared between different grid users.

In addition to these aspects, NRAs and grid operators that aim to introduce FCAs must also decide on additional aspects: **connection fee design**, **financial compensation** for lack of firm access, **geographical scope**, **type of grid users**, and **contract length**. The extent to which FCAs are successful in addressing grid congestion, and the amount of burden they place (and on whom) is highly dependent on these choices.

There are several examples of FCAs across Europe. These are developed in different jurisdictions under different geographical and political conditions, with differing reasoning for the introduction of FCAs. There are multiple examples in the CWE region, such as in the Netherlands, Belgium, France, and Austria. In Finland, where connections are generally set as firm, the DSO provides a FCA whereby the grid user reduces its consumption if and when the DSO instructs. There is no financial compensation for this constrained off-take, but the user is given grid access sooner than would have otherwise been possible. In Hungary, the regulatory codes allow for FCAs for a limited number of grid users (new power plants and storage units) if this option provides economic benefits. The concerned regulation (adopted in 2021) was aimed at facilitating the expected high increase in solar power installations to be connected to the DSO grid, but it is not clear whether such FCAs have effectively been implemented in this country⁸⁷. Likewise, Estonia’s DSO (Elering) has recently begun offering flexible connections to its grid users⁸⁸.

⁸⁶ ARENA, 2022, *DEIP Dynamic Operating Envelopes Workstream: Outcomes Report*. Available at: <https://arena.gov.au/knowledge-bank/deip-dynamic-operating-envelopes-workstream-outcomes-report/>.

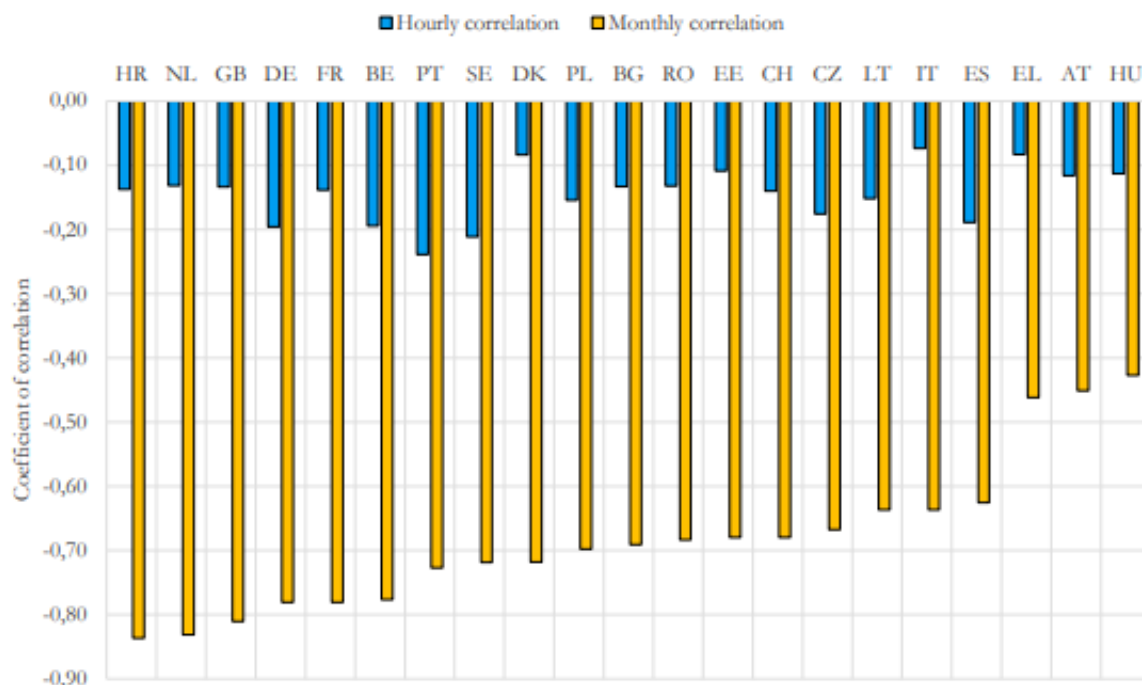
⁸⁷ CEER, 2023, *CEER Paper on Alternative Connection Agreements*. Available at: https://www.ceer.eu/wp-content/uploads/2024/04/ACA_2023.pdf.

⁸⁸ Elering, s.d., *Flexible connections*. Available at: <https://elering.ee/en/flexible-connections>.

The flexibility-related roles of transmission networks depend on the existence of a robust **intrazonal network** to balance supply and demand over larger areas, as well as **cross-border interconnections** to enable the exchange of electricity/flexibility across national borders. Transmission networks play a critical role in reducing flexibility needs, by connecting regions with different production technologies and consumption patterns.

Interconnections allow to optimise the use of renewable energy and flexibility resources, leveraging both technological and geographical complementarities. **Technological complementarity** at the supply side (e.g. different generation profiles depending on the technology and location) will to some extent reduce the variability of electricity production and lead to lower flexibility needs. The combined integration of wind and solar PV energy can for instance reduce short-term variability by up to 30%⁸⁹. Seasonal complementarities between wind and solar energy further enhance system resilience by balancing periods of low generation from one resource with higher output from the other. For example, wind energy generation peaks during the winter when solar production is low, while solar energy output is highest in the summer. This inverse relationship is highlighted in Figure 18, which shows negative correlations between wind and solar energy generation across European countries, with monthly correlations ranging from -0.43 to -0.84. These values are significantly more negative than the hourly correlations, which range from -0.07 to -0.24⁹⁰. The stronger negative correlation on a monthly basis reflects the seasonal complementarity of wind and solar energy. This complementarity reduces the need for flexibility resources, as periods of low generation from one resource are over the season compensated by higher production from another resource.

Figure 18: Correlation between wind and solar energy generation



Source: EEA et al., 2023, Flexibility solutions to support a decarbonised and secure EU electricity system.⁹¹

⁸⁹ ACER & EEA, 2023, *Flexibility solutions to support a decarbonised and secure EU electricity system*. Available at: https://www.acer.europa.eu/sites/default/files/documents/Publications/EEA-ACER_Flexibility_solutions_support_decarbonised_secure_EU_electricity_system.pdf.

⁹⁰ EEA et al., 2023, *Flexibility solutions to support a decarbonised and secure EU electricity system*. Available at: <https://www.eionet.europa.eu/etcs/etc-cm/products/flexibility-solutions-to-support-a-decarbonised-and-secure-eu-electricity-system>.

⁹¹ Available at: <https://www.eionet.europa.eu/etcs/etc-cm/products/flexibility-solutions-to-support-a-decarbonised-and-secure-eu-electricity-system>.

Geographical complementarity relies on variations in renewable energy output profiles across different regions. While wind and solar energy resources are not evenly distributed across Europe, interconnections (and for large zones transmission lines) allow to utilise these regional differences to balance supply and demand more cost-efficiently. For example, while one European region may at a certain moment experience low wind speeds or minimal sunlight, another region may have better conditions. Additionally, interconnections help mitigate localised weather phenomena, such as Dunkelflaute, which may affect specific areas but are less likely to occur simultaneously across the entire European continent. By enabling electricity exchanges across large regions, interconnections and domestic transmission lines not only stabilise the grid but also reduce the overall system's flexibility requirements and associated costs.

To accelerate network development, addressing a major issue faced across the EU for an efficient integration of renewable electricity into the system, the European Commission has in November 2023 launched an ambitious **action plan to overhaul the EU power grids** and estimated that a budget of €584 billion would be required by 2030 for the modernisation of the electricity grids⁹².

The plan is based on the objective to double the **cross-border electricity transmission capacity** to approximately 180 GW, with the aim of leveraging cross-border energy infrastructure allowing to decrease electricity generation costs by €9 billion annually by 2040⁹³. These investments should contribute to reaching the interconnection targets set at minimum 15% of the national electricity production capacity in 2030⁹⁴. According to ENTSO-E's draft 2024 TYNDP infrastructure gaps report, the electricity TYNDP project portfolio with a cross-border impact scheduled to be commissioned by 2030 will total 161 GW, and by 2040/2042 would total 241 GW. This will leave still a 28 GW gap in 2040 to reach the economically optimal cross-border transmission capacity estimated by ENTSO-E⁹⁵. In its recent Market Monitoring Report, ACER estimates the cross-border transmission network capacity gap in 2040 to total 37 GW⁹⁶.

However, investments in transmission capacity overall and cross-border capacity in particular represent a minority share of total needed grid investments. **Distribution grids constitute at present the majority and will continue to drive electricity network investments towards 2050.** A recent study by Trinomics, Artelys and LBST estimates that distribution grids will require €730 billion in investments between 2024-2040, representing nearly half of all EU energy infrastructure investment needs in that period (considering not only electricity networks but also electricity storage, hydrogen and CO₂ transport infrastructure)⁹⁷.

This shows that grid expansion is quite costly – and moreover requires long lead times for construction. Therefore, **a complementary option to reduce the required grid expansion** is to use grid-enhancing technologies and flexibility resources to increase the effectively available network capacity and

⁹² European Commission, 2023, *Commission sets out actions to accelerate the roll-out of electricity grids*. Available at: https://ec.europa.eu/commission/presscorner/api/files/document/print/en/ip_23_6044/IP_23_6044_EN.pdf.

⁹³ European Commission, 2023, *EU action plan for grids*. Available at: <https://data.europa.eu/doi/10.2775/919160>.

⁹⁴ European Commission, n.d., *Electricity interconnection targets*. Available at: https://energy.ec.europa.eu/topics/infrastructure/electricity-interconnection-targets_en.

⁹⁵ ENTSO-E, 2025, *Opportunities for a more efficient European power system by 2050: Infrastructure Gaps Report TYNDP 2024 - Version for public consultation*. Available at: <https://tyndp.entsoe.eu/>.

⁹⁶ ACER, 2024, *Electricity infrastructure development to support a competitive and sustainable energy system - 2024 Monitoring Report*. Available at: https://www.acer.europa.eu/monitoring/MMR/electricity_infrastructure_2024.

⁹⁷ Trinomics, 2024, *Advancing cross-system solutions to address electricity network challenges*. Available at: https://www.acer.europa.eu/sites/default/files/documents/Publications/Consultancy_study_Electricity_Network_Solutions_2024.pdf.

utilisation, deferring network expansion⁹⁸. The least cumbersome resources are energy storage systems, which can be installed to support the grid by smoothening the load profile. Another way to reduce grid expansion needs involves advanced controls to improve communication between grid elements, enabling, for example, automatic control of generators, automatic activation of demand response, or advanced power flow control. These tools are already in use at the transmission grid level but should increasingly also be used in distribution grids. The integration of variable renewable energy installations, especially PV, is mainly taking place in distribution grids, which is leading to increasing bidirectional exchanges between voltage levels. This near-load generation enables and requires flexible technologies to adapt to changing requirements⁹⁹.

To increase the efficient use of cross-border transmission networks, Regulation (EU) 2019/943 requires TSOs by the end of 2025 to allocate at least **70% of the nominal cross-zonal transmission capacities for short-term trading**, enabling market participants to exchange energy and hence reduce residual flexibility needs at national (or local) level¹⁰⁰. In its latest monitoring report, ACER indicates that the 70% target has not yet been reached at most borders, and even the intermediate targets set in action plans and derogations are not met for all hours. Loop flows in the CORE¹⁰¹ region as well as constraints introduced to maintain the system's operational security limits are main causes. Furthermore, data is often not available to calculate availability at specific borders for a significant number of hours¹⁰².

3.5. Comparison of flexibility solutions

The previous section details several flexibility solutions across the categories of flexible power generation, energy storage and demand-side flexibility, as well as flexibility indirectly provided by electricity networks. However, **flexibility solutions differ significantly in their potential as well as technical and economic characteristics**.

Both short- and long-term flexibility solutions are essential, as illustrated in Figure 19¹⁰³. Flexibility needs in terms of required capacity and energy are in general high on the short-term and decrease for longer time horizons. As mentioned in Section 2.1, especially daily (and sub-daily) flexibility needs will increase substantially, mainly to cover variations in solar PV generation, but cost-efficient seasonal flexibility is a major challenge as well.

⁹⁸ Trinomics, 2024, *Advancing cross-system solutions to address electricity network challenges*. Available at:

https://www.acer.europa.eu/sites/default/files/documents/Publications/Consultancy_study_Electricity_Network_Solutions_2024.pdf.

⁹⁹ Henning Schuster & E-Bridge, 2017, *Sichere und effiziente Koordinierung von Flexibilitäten im Verteilnetz: Beitrag zur weiteren Ausgestaltung der Rolle des Verteilnetzbetreibers in der Energiewende*. Available at: https://e-bridge.de/wp-content/uploads/2022/11/E-Bridge_Abschlussbericht-Sichere-und-effiziente-Koordinierung-von-Flexi....pdf.

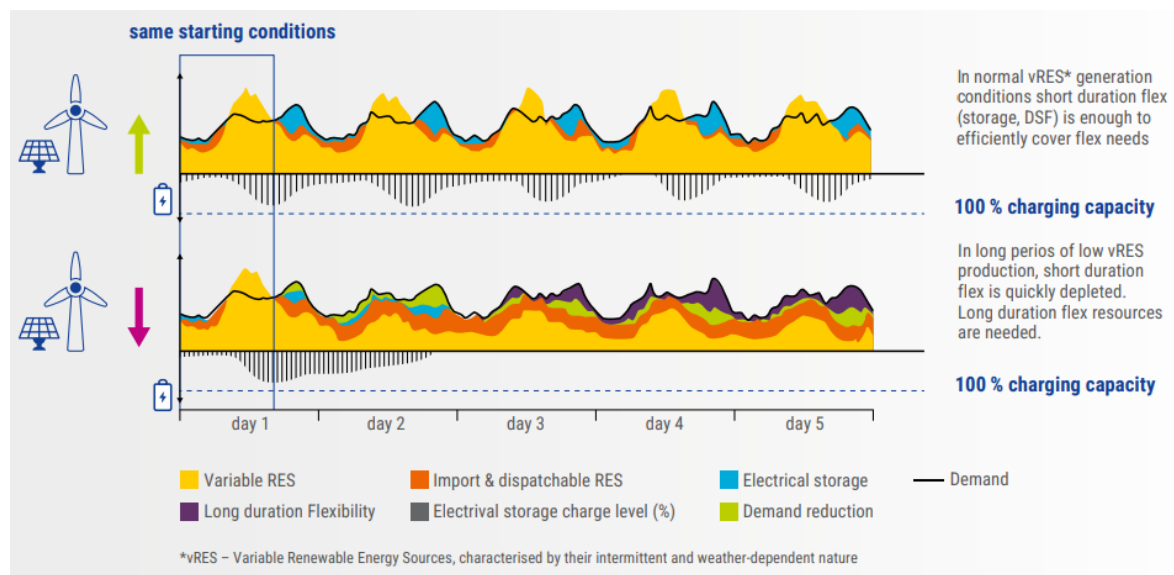
¹⁰⁰ Regulation (EU) 2019/943 of the European Parliament and of the Council of 5 June 2019 on the internal market for electricity (recast). Available at: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A02019R0943-20240716>.

¹⁰¹ CORE is the region for calculating cross-zonal capacities of the borders between the bidding zones of Austria, Belgium, Croatia, the Czech Republic, France, Germany, Hungary, Luxemburg, the Netherlands, Poland, Romania, Slovakia and Slovenia.

¹⁰² ACER, 2024, *Transmission capacities for cross-zonal trade of electricity and congestion management in the EU - 2024 Market Monitoring Report*. Available at: https://www.acer.europa.eu/monitoring/MMR/crosszonal_electricity_trade_capacities_2024.

¹⁰³ ENTSO-E, 2024, *System Flexibility Needs for the Energy Transition*. Available at: <https://www.entsoe.eu/system-flexibility/>.

Figure 19: Use of short- and long-duration flexibility resources in different grid conditions



Source: ENTSO-E, 2024, *System Flexibility Needs for the Energy Transition*. Available at: <https://www.entsoe.eu/system-flexibility/>.

Flexibility solutions differ in the optimal timescale they can be used for, as shown in Figure 20. The key characteristics of solutions for the different timescales and services differ. In general, to efficiently cover **short-term flexibility needs**, solutions must have **fast response times** to quickly respond to fluctuations in supply and demand and **high discharge rates** to deliver high capacity/volumes during a short time period. Since flexibility solutions have (sub) daily activation frequencies, the **costs per activation** and the **cycling capacity** are key metrics. Short-term flexibility solutions, such as battery storage and demand response, typically have high upfront costs per unit of capacity, due to the capital-intensity of the flexibility assets themselves (e.g. batteries) or the associated infrastructure (e.g. metering and control systems to enable demand response). However, the activation costs for these short-term solutions are comparatively low. Moreover, while for smart (unidirectional) charging of EVs, upfront costs may be limited, the upfront costs to enable bi-directional charging costs are more important.

Figure 20: Capability of flexibility solutions for different timeframes

	Sub-hourly	Daily	Weekly	Seasonal
OCGT	●	●	●	●
Engines	●	●	●	●
CCGT	●	●	●	●
Hydrogen turbines	As for OCGT/CCGT			
Hydropower reservoir	●	●	●	●
RES curtailment	●	●	●	●
System-friendly RES	●	●	●	●
Stationary batteries	●	●	●	●
Pumped-hydro	●	●	●	●
LDES (P2G2P)	●	●	●	●
Industrial response	●	●	●	●
Electric vehicles (V2G)	●	●	●	●
Heat pumps	●	●	●	●
Electric boilers	●	●	●	●
Data centers	●	●	●	●
Electrolysers	●	●	●	●

Ability to provide flexibility

● Yes

● Under conditions / Not mature

● No

Source: Authors' own elaboration, based on Trinomics & Artelys, 2024, *Power System Flexibility in the Penta region*¹⁰⁴.¹⁰⁴ Available at: <https://www.benelux.int/nl/etude/power-system-flexibility-in-the-penta-region-current-state-and-challenges-for-a-future-decarbonised-energy-system-2/>.

To cover longer-term flexibility needs, such as weeks or seasons, the short-term solutions need to be backed up by long-duration resources with greater capacity. For providing **long-term flexibility** cost-effectiveness at scale is important, with a focus on minimising the cost per kWh of flexibility effectively provided, and ensuring low energy losses over extended periods (in the case of storage), while the cost per cycle is less relevant. Examples of long-term flexibility solutions include large installations such as pumped hydro, utility-scale flow batteries and power-to-x-to-power solutions.

Estimating the actual potential and costs of different power system flexibility solutions poses significant challenges. These arise from the diversity of technical and economic characteristics discussed in the previous sections. Furthermore, uncertainty about the effective development and share of variable renewable energy sources in the electricity mix introduces additional complexity in predicting the potential deployment and cost of flexibility resources. Accurate data collection and analysis are often hindered by the lack of comprehensive and transparent information on the operational performance and costs associated with these solutions. Additionally, the evolving technological landscape (e.g. development of flow batteries) and the uncertainty regarding future market conditions and regulatory frameworks further complicate the estimation process.

Nonetheless, a few studies have attempted in recent years to estimate the contribution of different flexibility solutions to the EU energy system. The comparison in Table 3 qualitatively summarises the contribution of the different solutions according to key studies, while the subsequent figures present key results for each individual study. From the analysis it is apparent that:

- **A mix of solutions will be key for meeting flexibility needs for each timeframe.** Demand response, hydropower (pumped-hydro and reservoirs), battery storage and electrolyzers are all expected to play a central role in meeting flexibility needs towards 2050. In the short to medium term, flexible thermal power plants will still play a major role;
- **There are still significant differences in the assessment of the solutions' contribution** towards flexibility, with specific solutions making a minor to major contribution, depending on the study;
- **Some studies do not explicitly present disaggregated results for key solutions**, including demand response;
- **Study assumptions vary regarding the role of unabated fossil fuel-based power generation solutions** and whether they will still constitute an important part of the flexibility mix towards 2050.

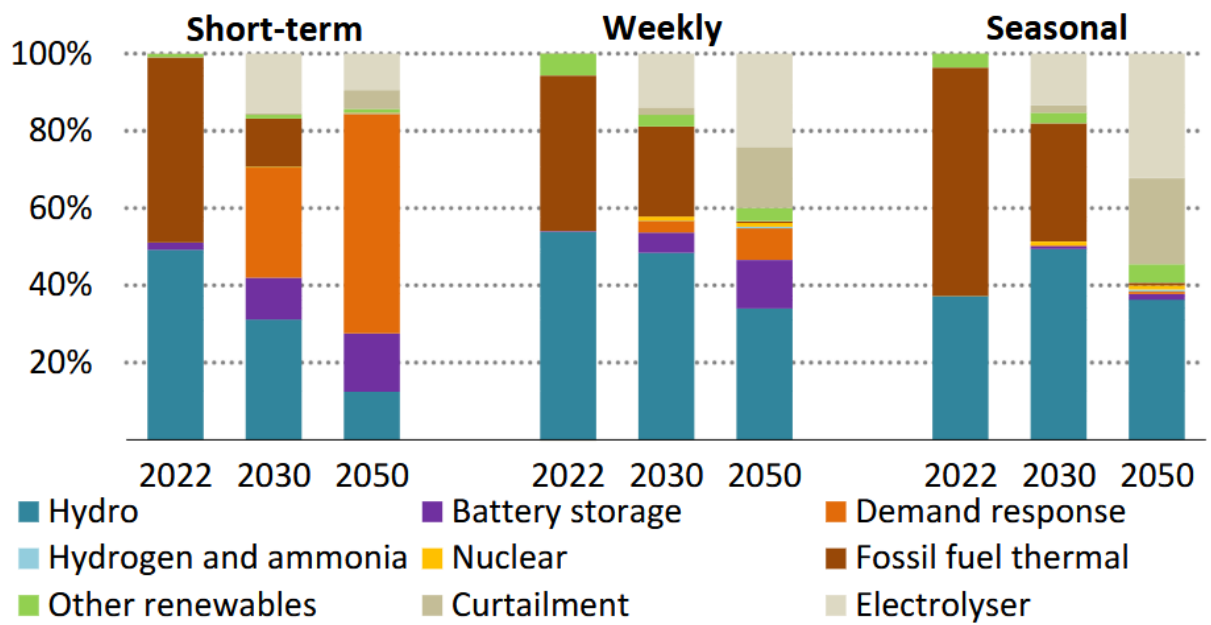
Table 3: Contributions of flexibility solutions to long-term (2030-2050) total flexibility needs according to the IEA, JRC and ACER/EEA

	Daily/short-term	Weekly	Seasonal/monthly
CCGT	None to significant	None to significant	None to significant
Electrolysers	Significant	Relevant to Major	Relevant to Major
Hydro	Significant	Significant to Major	Relevant to Major
Battery storage	Significant	Relevant to Significant	
Demand response	Major	Relevant	
RES curtailment		Relevant	Relevant

Source: Authors' own elaboration.

Note: **Relevant** is ~ 5-15% of total flexibility needs; **Significant** is ~15-25% of total flexibility needs; **Major** is >25% of flexibility needs.

Figure 21: Flexibility solutions' contributions towards 2050 in Europe according to the IEA



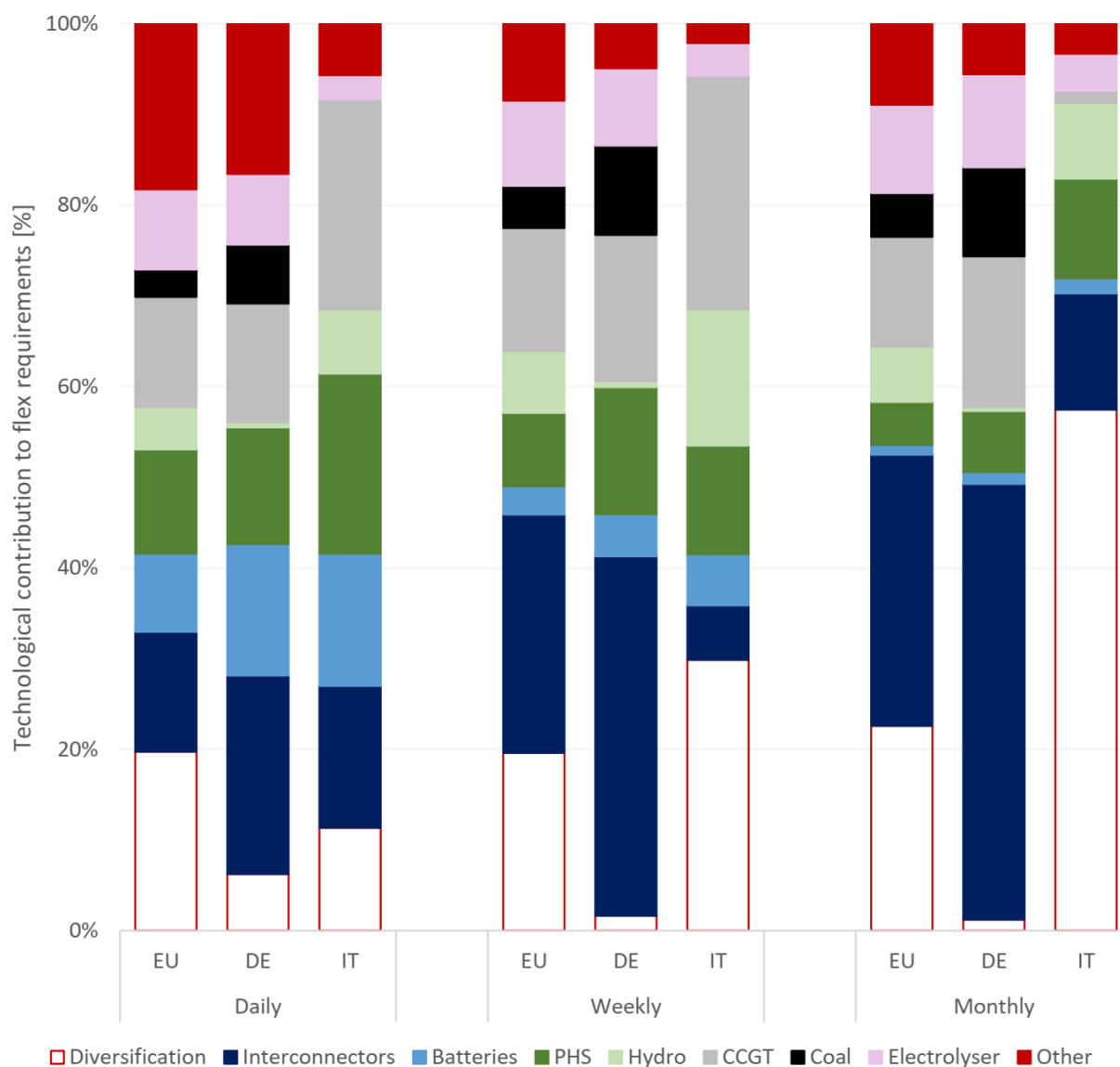
IEA. CC BY 4.0.

Source: IEA, 2024, Managing seasonal variability of electricity demand and supply¹⁰⁵.

Note: License: CC BY 4.0.

¹⁰⁵ Available at: <https://www.iea.org/reports/managing-the-seasonal-variability-of-electricity-demand-and-supply>.

Figure 22: Flexibility solutions' contributions towards 2050 in the EU, Germany and Italy according to the JRC



Source: JRC, 2023, *Flexibility requirements and the role of storage in future European power systems*. Available at: <https://publications.jrc.ec.europa.eu/repository/handle/JRC130519>.

Note: Diversification is where the interactions between different technologies helps meeting the flexibility needs due to differences in their profiles.

Figure 23: Contribution of demand-side measures to the total flexibility needs in Europe in 2030



Source: EEA et al., 2023 in ACER/EEA, 2023, Flexibility solutions to support a decarbonised and secure EU electricity system¹⁰⁶.

Note: From the original source: "bars represent changes in daily, weekly, and seasonal flexibility needs for demand reduction and shifting compared with the average results with the reference demand. Values correspond to the averages of all simulations performed per demand scenario".

¹⁰⁶ Available at: <https://acer.europa.eu/document/acer-eea-report-flexibility-solutions-support-decarbonised-and-secure-eu-electricity-system>.

4. POLICIES AND MEASURES TO FURTHER ENABLE FLEXIBILITY

KEY FINDINGS

- The EU’s electricity market framework, with the Electricity Market Design Reform (EMDR) as major component, is increasingly shaped by the need to adapt electricity markets to the variability of renewable energy and changing consumption patterns.
- Key elements of the EU regulatory framework include the Clean Energy Package, the Flexibility Needs Assessment (FNA) and the Demand Response Network code. These are complemented with national objectives and support schemes, and regulatory obligations on network operators.
- Flexibility is becoming an increasingly central topic in EU level policy discussions on competitiveness and decarbonisation, being a key element in various strategic documents published by the Commission and other EU entities.
- Additional policies and measures can be considered to complement existing or planned policies. Measures suggested in the literature focus on investment planning and operational considerations, cross-border issues, the need for additional flexibility services and marketplaces, and the potential of support mechanisms to make flexibility solutions financially viable.

Adequate electricity market design and policy/regulatory measures are essential to provide a level-playing field between traditional and new (non-fossil) flexibility solutions, and to stimulate the deployment of flexibility solutions to cost-efficiently meet the increasing needs. This chapter analyses the policy aspects related to flexibility, covering policies and measures implemented or under implementation, the role of flexibility in key recent EU strategy documents, and additional policies and measures suggested in the literature.

4.1. Current policy and regulatory framework for flexibility

The EU’s electricity market framework is increasingly shaped by the need to incorporate flexibility as a cornerstone for cost-efficiently achieving renewable energy integration and security of supply. A major legal component to enable this transition is provided by the Electricity Market Design Reform (EMDR), enacted through Regulation (EU) 2024/1747, adopted on 13 June 2024. The EMDR aims to adapt the electricity system and markets to the variability of renewable energy and changing consumption patterns.

The EMDR builds on earlier regulatory milestones, including Regulation (EU) 2017/1485, which establishes a guideline on electricity transmission system operation. Furthermore, an important step forward was taken on 8 May 2024, when the EU DSO Entity and ENTSO-E submitted a joint proposal for the Network Code on Demand Response to ACER. This new Network Code will play a crucial role in ensuring the integration of demand-side flexibility resources into the electricity system.

Key elements related to flexibility in the EU regulatory framework are listed in the following table. It provides an overview of the initiatives which have been or are being implemented. Further suggestions for improvement of the regulatory framework which are under discussion are listed in section 4.3.

Table 4: Key elements related to flexibility in the EU Regulatory Framework

Framework element	Description
Clean Energy Package	The Clean Energy Package (CEP) is designed to create a more flexible and decarbonised electricity system. Through Electricity Regulation (EU) 2019/943 and Electricity Directive (EU) 2019/944, the CEP strengthens market-based price signals for flexibility resources, such as demand response, energy storage and interconnections. The CEP establishes measures to enhance short-term market functioning, consumers' participation and system operators' coordination to integrate flexibility into the electricity system efficiently.
Flexibility Needs Assessment	Article 19e of Electricity Regulation (EU) 2019/943 mandates TSOs and DSOs to prepare national flexibility needs reports every two years, projecting needs over 5-10 years. These reports assess the seasonal, daily and hourly flexibility needs, including the role of demand response and energy storage. Electricity TSOs and DSOs provide data, coordinated with gas and hydrogen operators, and submit this report to a designated national authority/NRA for adoption. ENTSO-E and the EU DSO Entity provide standardised methodologies. ACER will have to review the national FNAs and provide cross-border recommendations on the national reports. As part of this mandate, ACER should estimate the flexibility needs in the electricity system at Union level and its projected economically available potential. For this, ACER is setting up an EU-wide flexibility assessment platform. The platform will support data-driven recommendations for regulatory and policy frameworks, support cross-border coordination in flexibility markets and identify investment needs for a resilient electricity system.
Indicative National Objectives	Article 19f of Regulation (EU) 2019/943 requires EU Member States to set indicative objectives for non-fossil flexibility within six months of submitting the flexibility reports. These objectives, reflected in the NECPs, outline the contributions from demand response and energy storage. The European Parliament oversees the process, ensuring alignment with the 2030 and 2050 energy and climate targets. To further strengthen coherence across Member States, the European Commission is tasked with developing a Union strategy on flexibility, based on the national flexibility needs reports and indicative national objectives.
Support Mechanisms for non-fossil flexibility	EU Member States can implement support schemes to address investment gaps and achieve flexibility objectives, as per Article 19g and 19h of revised Regulation (EU) 2019/943. These schemes may include subsidies for non-fossil flexibility capacity and encourage cross-border participation where feasible. Alternative approaches, like removing market barriers, can also support the deployment of demand response and energy storage. Additionally, raising consumer awareness for demand response is a critical component of achieving flexibility objectives.
Flexibility related to regulation of transmission system operators	Flexibility in transmission systems is regulated under Commission Regulation (EU) 2017/1485, which emphasises the role of TSOs in ensuring that the electricity system remains in balance, also with the increasing integration of variable renewable energy sources. TSOs play a central role in managing ramping up/down capabilities, ensuring smooth adjustments in supply or consumption. Additionally, to maintain the system's stability, they rely on reserve capacity like FCR, FRR and RR.

Framework element	Description
Demand Response Network Code	In May 2024, the EU DSO Entity and ENTSO-E submitted a joint proposal for the NC DR to ACER. This Code establishes a harmonised framework across EU Member States to integrate demand-side flexibility into the energy system. The NC DR creates market-based processes for the cost-efficient procurement of flexibility services by DSOs and TSOs, harmonises rules to enable resource aggregation and includes amendments to other electricity network codes. The NC DR is currently undergoing a review process by ACER, which includes incorporating feedback from a public consultation.

Source: Authors' own elaboration.

4.1.1. Clean Energy Package

The Clean Energy Package (CEP) is a cornerstone of the EU's energy policy framework, shaping the electricity market to support the transition to a more flexible and decarbonised energy system. Through Electricity Regulation (EU) 2019/943¹⁰⁷ and Electricity Directive (EU) 2019/944¹⁰⁸, the CEP introduces key measures to improve the electricity system and to ensure that flexibility is integrated into the energy system.

Both the Electricity Regulation and Directive emphasise that electricity prices should be determined by supply and demand, sending market-based signals for investments in flexibility resources such as demand response, energy storage and interconnections. A well-functioning market should allow for real-time adjustments, ensuring that flexibility resources can respond dynamically to changes in generation and consumption. To facilitate this, balancing markets, spot markets and redispatch mechanisms must be structured to encourage participation from flexibility providers.

A key aspect of increasing flexibility is enabling consumer participation in electricity markets. The Electricity Directive grants consumers/prosumers the right to engage in dynamic price contracts and sell their flexibility directly or through aggregators. Smart metering infrastructure is essential for this, allowing consumers to adjust their usage based on price signals. Regulatory frameworks must ensure that smart meters and grid digitalisation initiatives provide actionable insights for consumers/prosumers and system operators alike.

Network operators play a crucial role in integrating flexibility into the grid. TSOs must provide transparency on forecasted and actual demand, availability of flexibility resources and network conditions to facilitate efficient market participation.

Grid tariff structures must also support flexibility. Network tariffs should be cost-reflective, transparent and non-discriminatory, ensuring that they provide incentives for the efficient use of flexibility services. Distribution network tariff methodologies specifically, should where appropriate include performance targets to incentivise DSOs to procure flexibility services and promote smart grids¹⁰⁹ and smart metering.

¹⁰⁷ Regulation (EU) 2019/943 of the European Parliament and of the Council of 5 June 2019 on the internal market for electricity, OJ L, 2019/943 14.6.2019. Available at: <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32019R0943>.

¹⁰⁸ Directive (EU) 2019/944 of the European Parliament and of the Council of 5 June 2019 on common rules for the internal market for electricity and amending Directive 2012/27/EU. OJ L, 2019/944 158/125. Available at: <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32019L0944>.

¹⁰⁹ Definition of smart grids by the IEA: "Smart grids are electricity networks that use digital technologies, sensors and software to better match the supply and demand of electricity in real time while minimizing costs and maintaining the stability and reliability of the grid." Available at: <https://www.iea.org/energy-system/electricity/smart-grids>.

The Clean Energy Package highlights the importance of ensuring that at least 70% of the interconnection capacity is effectively available for cross-border trading, reinforcing the need for flexibility mechanisms that operate beyond national markets. Regional Coordination Centres (RCCs) must optimise the use of interconnectors, ensuring security of supply and market efficiency.

4.1.2. Flexibility needs assessments

The assessment of flexibility needs, as outlined in Article 19e of the revised Electricity Regulation (EU) 2019/943, as amended by Regulation (EU) 2024/1747, is a crucial component of the EU's approach to increase flexibility. Each Member State, through its regulatory authority or a designated entity, is required to prepare a national flexibility needs report within one year of ACER's approval of a common methodology¹¹⁰ and every two years thereafter. The assessment should project the flexibility needs for a period of at least five to ten years.

The national report must evaluate the flexibility needs on a seasonal, daily and hourly basis, considering various assumptions about electricity prices, generation and demand. It also analyses the potential of non-fossil flexibility resources such as demand response, energy storage, interconnections and aggregation to meet these needs. It should furthermore identify barriers to flexibility and propose measures to address them, including regulatory adjustments and improvements to market design and system operation. The role of digitalisation in enhancing transmission and distribution networks and the potential contribution of cross-border flexibility will also be integrated in the assessment.

The national reports should rely on data and analysis provided by the electricity TSOs and DSOs. These inputs are coordinated with the natural gas and hydrogen system operators to ensure coverage of cross-sectoral needs, and ultimately submitted to a designated national authority/NRA, who adds an assessment on market barriers and adopts the report. ENTSO-E and the EU DSO Entity play a key role in standardising data formats and methodologies for the national assessments, facilitating consistent and robust evaluations across Member States. A draft methodology was published in November 2024 by these entities for public consultation¹¹¹, and updated, following the comments received, with some further guiding criteria and specifics on national implementation, availability issues, market barriers and others. By April 2025, ENTSO-E and the EU DSO entity will submit a final methodology to ACER for approval, outlining the approach for assessing flexibility needs, including planned investments and the full range of available flexibility sources across different timeframes.

EU-wide flexibility assessment platform

Article 19e of Electricity Regulation (EU) 2019/943, as amended by Regulation (EU) 2024/1747 requires ACER to review the national reports, providing recommendations to address cross-border issues and remove barriers to the entry of non-fossil flexibility resources. Its analysis should include a European level perspective, estimating flexibility needs and the economically available potential over the next five to ten years.

¹¹⁰ ENTSO-E and EU DSO Entity have in November 2024 published a draft for approval by ACER. Available at: https://consultations.entsoe.eu/system-development/public-consultation-on-flexibility-needs-assessment/supporting_documents/1_Flexibility_Needs_Assessment_Methodology.pdf.

¹¹¹ ENTSO-E & EU DSO Entity, 2024, *Definition of the type and format of data and the methodology for the analysis by transmission system operators and distribution system operators of the flexibility needs at national level*. Available at: https://consultations.entsoe.eu/system-development/public-consultation-on-flexibility-needs-assessment/supporting_documents/1_Flexibility_Needs_Assessment_Methodology.pdf.

In this context, ACER will set up an EU-wide flexibility assessment platform. The platform will represent a significant step in improving system-wide coordination and planning for flexibility resources across the European energy market. A core element of the EU-wide flexibility assessment is the development of an open-source modelling framework. An expert group, including industry professionals, academics and regulatory specialists, will advise ACER, providing recommendations to ensure that the platform remains aligned with the Union strategy on flexibility¹¹².

4.1.3. Indicative national objectives

As outlined in Article 19f of Electricity Regulation (EU) 2019/943, as amended by Regulation (EU) 2024/1747, Member States are required to establish indicative national objectives for non-fossil flexibility within six months of submitting their flexibility needs reports. The objectives should be integrated into the National Energy and Climate Plans (NECPs). These objectives will define the contributions of demand response and energy storage, to be achieved through unlocking this flexibility potential.

The determination of these objectives must be informed by the flexibility needs assessments outlined in section 4.1.2, ensuring a consistent and data-driven approach. The flexibility needs assessments must provide critical insights into seasonal, daily and hourly variations in residual load, identify the role of interconnections and evaluate the contributions of non-fossil flexibility resources. These assessments will form the foundation upon which Member States can establish realistic and achievable targets, factoring in the barriers and opportunities identified at the national and regional levels.

The indicative national objectives also serve as a bridge between national and EU-wide energy strategies, fostering alignment with the 2030 renewable energy and energy efficiency targets and the 2050 climate neutrality objective. By requiring Member States to define these objectives, the Regulation ensures that flexibility measures are not developed in isolation but contribute to the broader EU flexibility needs.

The European Parliament and Council play a key role in overseeing this process. After Member States submit their indicative objectives, the European Commission evaluates these reports and presents its findings to the European Parliament and Council. They assess these findings, ensuring transparency and accountability in the implementation of flexibility strategies. Furthermore, their oversight ensures that national objectives account for critical elements identified in the flexibility needs assessments, such as addressing barriers to demand response, improving market design and leveraging interconnections for cross-border flexibility. For example, barriers such as inappropriate legal frameworks, inadequate incentives for demand response and limited access to balancing markets persist in many Member States. These issues must be addressed in the national approaches to unlock the full potential of non-fossil flexibility resources.

By integrating these objectives into the NECPs, Member States are encouraged to adopt a holistic approach to flexibility planning. This approach should not only align with EU-wide goals but also address specific national and regional challenges identified in the flexibility needs assessments. For instance, the assessment should emphasise the importance of both short- and long-term flexibility solutions, interconnections to mitigate regional variability and technological and geographical complementarities of variable renewable energy sources.

¹¹² ACER, 2025, *Open call for experts to join ACER's new Expert Group on EU-wide flexibility needs assessment*. Available at: <https://www.acer.europa.eu/news-and-events/news/open-call-experts-join-acers-new-expert-group-eu-wide-flexibility-needs-assessment>.

To further strengthen coherence across EU Member States, the European Commission is tasked with developing a Union strategy on flexibility, based on the national flexibility needs reports and indicative objectives. This framework will provide a coordinated framework for enhancing demand response and energy storage across all Member States, ensuring their deployment aligns with EU-wide flexibility requirements. The strategy will emphasise cross-border cooperation, promoting the use of interconnections to cost-efficiently cover flexibility needs at a European level.

4.1.4. Support mechanisms for non-fossil flexibility

To promote investments in flexibility resources, Member States can implement non-fossil flexibility support schemes as outlined in Article 19g and 19h of Electricity Regulation (EU) 2019/943, as amended by Regulation (EU) 2024/1747. These schemes address gaps in investments and help achieve the indicative or provisional national objectives for non-fossil flexibility.

However, policy makers must carefully consider how their decisions impact system flexibility. For example, providing support for renewable electricity generation that does not align with real-time system needs or locking consumers into fixed-price contracts without considering different energy usage patterns can reduce flexibility. Such approaches may limit the ability of generators or consumers to respond dynamically to grid tariff or market price signals. As highlighted by ACER and CEER, it is essential to design support mechanisms that enhance flexibility¹¹³.

To further enhance market integration and ensure system stability at least cost, participation of cross-border flexibility resources in support mechanisms should be included when feasible. Cross-border cooperation is especially important for balancing differences and complementarities in variable renewable energy production between regions.

4.1.5. Demand Response Network Code

On 8 May 2024, the EU DSO Entity and ENTSO-E submitted a joint proposal for the Network Code on Demand Response (NC DR)¹¹⁴ to ACER. This Code should facilitate a cost-efficient integration of demand-side flexibility, including load, storage and distributed generation, into the European energy system by establishing a harmonised framework across all EU Member States. By creating market-based processes for the procurement of flexibility services by TSOs and DSOs, the network code aims to enhance system flexibility in a cost-efficient manner. The proposal also focuses on harmonising rules to enable resource aggregation, a crucial step in allowing smaller generators, storage operators and consumers/prosumers to actively participate in electricity and ancillary services' markets. The proposal not only addresses the integration of demand-side flexibility but also includes amendments to related electricity network codes on balancing, system operation and demand connection.

Since the submission of the draft NC DR in May 2024, ACER has thoroughly reviewed the draft and published in September 2024 a revised version¹¹⁵ for the public consultation, which ran from September to October 2024. After finalising its revision of the Network Code, ACER will submit the amended proposal to the European Commission by March 2025. The Commission will then propose a final implementing act, which will go through the comitology procedure for approval. Once adopted,

¹¹³ ACER & CEER, 2023, *ACER-CEER Reaction to the European Commission's public consultation on electricity market design*. Available at: https://acer.europa.eu/sites/default/files/documents/Media/News/Documents/ACER-CEER_Response_EC_PC_EMD.pdf.

¹¹⁴ EU DSO Entity & ENTSO-E, 2024, *Proposal for a network Code on Demand Response*. Available at: [https://eepublicdownloads.blob.core.windows.net/public-cdn-container/clean-documents/Network codes documents/NC DR/NC DR_DSO ENTITY_ENTSO-E.pdf](https://eepublicdownloads.blob.core.windows.net/public-cdn-container/clean-documents/Network%20codes%20documents/NC%20DR/NC%20DR_DSO%20ENTITY_ENTSO-E.pdf).

¹¹⁵ ACER, 2024, *Public consultation on the draft network code on demand response*. Available at: <https://acer.europa.eu/documents/public-consultations/pc2024e07-public-consultation-draft-network-code-demand-response>.

the final Network Code will establish a binding framework across all EU Member States, ensuring a harmonised approach to demand-side flexibility.

4.2. Energy system flexibility in recent EU strategic documents

Flexibility is becoming an increasingly central topic in strategic discussions on EU energy policy and market design. Besides having a central place in the EU electricity market design reform, it is a key theme in the Letta and Draghi reports (detailed further below), and is also addressed in the European Commission's Competitiveness Compass:

- The **Letta report**¹¹⁶, which focuses on strengthening the Single Market, dedicates a section to the Single Energy Market. It highlights the role of interconnectivity in improving flexibility. Eliminating barriers to interconnection and ensuring that at least 70% of the interconnection capacity is available for trading are key recommendations.
- The **Draghi Report**¹¹⁷, centred on EU competitiveness, comprises a specific section on energy and recognises the growing flexibility needs. It emphasises the need for higher investments in flexibility solutions, stronger locational price signals and harmonised capacity remuneration mechanisms.
- The **Competitiveness Compass**¹¹⁸ mentions among others the need to incentivise demand flexibility from industrial customers and to accelerate investments in energy infrastructure, scheduling the EU Electrification Action Plan and European Grids Package for the first quarter of 2026. Activating the flexibility potential in the industry can reduce the electricity cost and hence improve the competitiveness of industrial electricity users.

4.2.1. Flexibility in the Letta Report

The Letta report emphasises that as the EU advances towards a more decarbonised energy system, and aims to increase the RES share in the overall electricity mix to 70% by 2030, the need for flexibility and market integration becomes even more critical. Interconnectivity plays a fundamental role in ensuring that the growing share of renewable energy is efficiently integrated into the system. By optimising resource sharing across Member States, cross-border electricity trading allows renewable electricity generation to be deployed in the most efficient locations, such as wind-rich regions for wind energy and sunny areas for solar PV. This reduces the overall need for redundant capacity and flexibility resources, as variability in one region can be offset by production in another. However, many Member States are still falling short of the EU's target to make at least 70% of their interconnector capacity available for electricity trading by the end of 2025. Eliminating barriers to interconnection are therefore essential for increasing the flexibility in the electricity system.

The report also highlights that cross-border cooperation is crucial for reducing the costs associated with procuring flexibility. Currently, most flexibility procurement schemes and Contracts for Differences (CfDs) for renewable energy projects remain limited to national borders. Establishing cross-border flexibility procurement schemes and organising joint renewable energy auctions between

¹¹⁶ Letta, E., 2024, *Much more than a market*. Available at: <https://www.consilium.europa.eu/media/ny3j24sm/much-more-than-a-market-report-by-enrico-letta.pdf>.

¹¹⁷ Draghi, M., 2024, *The future of European competitiveness*. Available at: https://commission.europa.eu/document/download/ec1409c1-d4b4-4882-8bdd-3519f86bbb92_en?filename=The%20future%20of%20European%20competitiveness_%20In-depth%20analysis%20and%20recommendations_0.pdf.

¹¹⁸ European Commission, 2025, *A Competitiveness Compass for the EU. COM(2025) 30 final*. Available at: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:52025DC0030>.

multiple Member States could significantly reduce the overall costs to reach the renewable electricity targets while ensuring system stability.

To accelerate progress, the report outlines a roadmap that includes implementing a cross-border cost-benefit allocation methodology for offshore wind energy projects and developing joint procurement schemes for flexibility and renewable energy auctions by 2025.

4.2.2. Flexibility in the Draghi Report

The Draghi report places significant emphasis on the need to reinforce energy system integration, storage and demand flexibility to keep the total system costs in check as the uptake of renewable energy sources accelerates. As the flexibility requirements are expected to grow significantly, rising from 11% of total electricity demand in 2021 to 24% in 2030 and reaching 30% by 2050, the report stresses that investments in flexibility solutions must accelerate alongside the expansion of renewable energy.

A key recommendation is the development of an EU-wide strategy to map flexibility needs and support investments in flexibility assets. Without sufficient flexibility, the costs of integrating variable renewable energy sources will increase, and curtailment will become more frequent, particularly in distribution grids with limited flexibility options. Hybrid power purchase agreements (PPAs), which incorporate flexibility assets like energy storage, allow for more adaptable offtake structures. These contracts enable energy consumers to better align their demand with fluctuating renewable electricity generation, thus enhancing the overall flexibility of the system. Additionally, competitive bidding procedures for renewable energy auctions should not only focus on price but also include non-price criteria that enhance system integration, thus maintaining competition while managing system costs efficiently.

The report also highlights the role of electricity interconnections in reducing flexibility needs by facilitating cross-border electricity trade. However, while stronger interconnections reduce flexibility needs, they do not eliminate them, reinforcing the importance of investments in domestic flexibility solutions.

Recognising the economic benefits of demand-side participation, the Draghi report calls for the creation of a standardised compensation mechanism for industrial demand flexibility. Industrial consumers can play a significant role in balancing supply and demand by shifting their energy consumption based on market price signals, but their participation in flexibility markets remains underdeveloped in the EU compared to other global markets. Therefore, establishing a common framework for industrial demand response would enhance the competitiveness of EU industries.

To further strengthen flexibility, the report advocates for the acceleration of capacity remuneration mechanisms and flexibility instruments, ensuring they are standardised and well-integrated into electricity markets. The authorisation process for such mechanisms should be streamlined to encourage investments in storage, demand response and grid digitalisation. Additionally, the report highlights the need for stronger locational price signals to better reflect network constraints, incentivising flexibility where it is most needed and reducing curtailment and redispatch measures.

Bidirectional charging for electric vehicles (EVs) is in the Draghi report identified as another key flexibility resource. With the EU's EV fleet rapidly growing, incentivising vehicle-to-grid (V2G) charging could transform EVs into valuable flexibility assets, reducing peak demand pressure and optimising renewable energy use.

4.2.3. Flexibility in the Clean Industrial Deal

The Clean Industrial Deal¹¹⁹ announced in February 2025, along with the Affordable Energy Action Plan¹²⁰, both sign clear directions to advance towards electrification and a fully integrated single EU market for energy, and see flexibility as a key enabler of integrating decarbonised energy sources. The Clean Industrial Deal highlights the need for demand-side flexibility through digitalisation, including AI-driven smart grids and internet-of-things-based monitoring, as essential for energy system integration, managing grids in real-time, improving demand-side flexibility, and enabling predictive maintenance for critical infrastructure.

According to the document, the Commission plans to consult Member States on a clean flexibility instrument based on Power Purchase Agreements (PPAs), encouraging industry to commit to consuming clean electricity while minimising competition distortions and subsidy races within the Single Market. By Q4 2025, the Commission intends to develop guidance to promote flexibility remuneration in retail contracts, enabling industrial consumers to adjust their demand according to the availability of cheap clean energy, thereby making immediate savings on their energy bills.

Furthermore, the EU set out to streamline State aid approvals to support non-fossil flexibility measures and capacity mechanisms, accelerating the decarbonisation of the energy system. This includes allowing separate support schemes for specific technologies like wind and solar, and updating rules to support additional manufacturing.

4.3. Additional policies and measures suggested in the literature

This section compiles potential additional policies and measures complementing the existing or planned policies, as presented in Table 4. The focus is on potential initiatives at the EU level, while also emphasising the role of Member States (e.g. in cooperation with EU institutions) to foster flexibility. Member States should of course close any remaining implementation gaps on already existing EU legislation, and properly implement the provisions of the EMDR and the future NC DR and FNA methodology, once approved.

We present hereafter non-exhaustive suggestions from the literature to improve the flexibility market functioning, including by providing innovative incentives for increasing flexibility and mitigating grid congestion, taking into account the socio-economic costs.

The suggestions are regrouped in the following four categories:

- **Planning and operational measures**, e.g. to more accurately assess flexibility needs and available solutions;
- **Cross-border issues**, e.g. that enable sharing flexibility across borders;
- **Flexibility services and marketplaces** that allow seamless trading and cost-effective deployment of flexibility assets;
- **Support mechanisms** that help make flexibility solutions financially viable.

¹¹⁹ European Commission, 2025, *The clean industrial deal: a joint roadmap for competitiveness and decarbonisation*. COM(2025) 85 final. Available at: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52025DC0085>.

¹²⁰ European Commission, 2025, *Action Plan for Affordable Energy*. COM(2025) 79 final. Available at: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52025DC0079&qid=1741349807435>.

Table 5: Policies and measures proposed in the literature to facilitate the deployment of flexibility solutions

Title	Description	Source
Planning and operational measures		
Improved forecasting on TSO level	ENTSO-E points to limited accuracy of forecasts pertaining to solar and wind energy, especially in the short-term, due to uncertainties in vRES availability and electricity demand after the day-ahead planning stage. To mitigate the impact of forecasting errors, the accuracy of forecasting models can be enhanced through advanced data analytics and machine learning. ENTSO-E quantified the forecast errors, and found that for vRES they increase over time in absolute terms, corresponding with the anticipated growth in installed wind and solar energy capacities.	ENTSO-E, 2024, <i>System Flexibility Needs for the Energy Transition</i>
TSO-DSO cooperation	Collaboration between TSOs and DSOs is crucial for the cost-efficient deployment of flexibility assets, and to manage pooled resources efficiently and avoid conflicting control actions. Key elements of TSO-DSO coordination are flexibility solutions' validation (making sure activations do not breach network limits) and dispatch (deciding on which flexibility sources to activate). Both actions can be undertaken either centrally or decentrally. Next to national and regional data provision, the localised use of flexibility services is expected to require additional data from DSOs. Any further exchange of information or best practices between system operators regarding technical options to optimise the use of the electricity grid is beneficial.	Magnus Energy, 2024, <i>Maximising distributed flexibility with effective TSO-DSO coordination</i> Trinomics and Artelys, 2022, <i>Power System Flexibility in the Penta region – Current State and Challenges for a Future Decarbonised Energy System</i>
Holistic, system-wide planning and cost-benefit assessments	Electricity network planning should gradually evolve to integrate other energy vectors, such as low-carbon gases, at the transmission and distribution level, and consider new flexibility solutions. Due to a rapidly changing energy landscape, new methodologies for cost-and benefit assessments are needed to support policy decisions and project development. These assessments should take into account system costs incl. network and flexibility needs. Investment decisions should not solely be based on the levelised cost of electricity (LCOE) associated with certain projects or technologies, but should also account for the increasing overall system cost associated with the development of variable renewable electricity.	Mario Draghi, 2024, <i>The future of European competitiveness</i> Trinomics and Artelys, 2022, <i>Power System Flexibility in the Penta region – Current State and Challenges for a Future Decarbonised Energy System</i>
Cross-border issues		
EU level auctions for cross-border flexibility and RES	Implementing cross-border schemes for flexibility procurement, statistical transfers and conducting joint Member State auctions for renewable electricity projects could significantly lower costs and improve the efficiency of cross-border electricity flows.	European Commission, 2025, <i>The future of European competitiveness</i> ACER & EEA, 2024, <i>Flexibility solutions to support a decarbonised and secure EU electricity system</i>

Title	Description	Source
Compensation for industrial DSF	To reduce grid congestion, more flexibility can be contracted with large consumers that often have potential for unlocking flexibility. This would allow these industrial or commercial sectors to continue to grow while reducing network congestion. A Dutch report refers to such deals with water boards and the upcoming public transport management companies, as well as SMEs. The creation of a standardised compensation mechanism for industrial demand flexibility could reduce overall system costs while improving grid stability. Standardising mechanisms across Member States could encourage broader participation in demand response programs by industrial players.	Gerard Schouw, 2024, <i>Slim met Stroom voor Groene Groei</i> European Commission, 2025, <i>The future of European competitiveness</i>
Simplify and streamline permitting and administrative processes	Streamlining and simplifying permitting processes can help accelerate the development of renewable energy sources, flexibility uptake and grid deployment, on both TSO and DSO level. Existing EU legislation on renewable energy projects' permitting needs to be properly transposed and implemented, and stronger legislative action at the EU level can be taken to accelerate permitting, including for RES projects to be connected at the distribution level.	European Commission, 2025, <i>The future of European competitiveness</i>
Make use of market price signals	To better incentivise consumers to participate in flexibility markets, the potential of market signals must be exploited in a way that network users can properly benefit from adapting their consumption patterns. Implementing capacity-based and time-of-use network tariffs, for example, can enhance the utilisation of network capacity and minimise the need for investment in reinforcements. However, the ideal design and structure of these tariffs should take into account national and local characteristics. Location-differentiated price signals could better value flexibility resources, however their implementation through zonal wholesale pricing in the current bidding zone system would be difficult for Europe. An alternative option would be having locational signals along a wholesale market, as proposed in the UK.	ACER & EEA, 2024, <i>Flexibility solutions to support a decarbonised and secure EU electricity system</i> Trinomics and Artelys, 2022, <i>Power System Flexibility in the Penta region – Current State and Challenges for a Future Decarbonised Energy System</i> UK Energy Research Centre, 2024, <i>Locational Signals for Electricity System Flexibility Alongside a National Wholesale Market</i>
Higher granularity of bidding zones	Continue the bidding zone review process, investigating alternative bidding zone configurations and adequately assessing impacts for structurally congested market bidding zones.	TenneT, 2023, <i>Bidding Zone Review: TSOs investigate alternative bidding zone configurations</i>

Title	Description	Source
Services and Marketplaces		
Offer flexible connection agreements	Flexible connection agreements between TSOs/DSOs and electricity generators or consumers can be considered to accelerate the connection of new RES installations or load, if the available network capacity is not sufficient. An adequate legal framework should be in place that obliges the concerned grid operators to remunerate flexible grid users for curtailments in injection or off-take that exceed an agreed level. An adequate regulatory scheme should stimulate grid operators to make timely investments in capacity reinforcements and incentivise increased market participation of flexibility providers (especially on the demand side). To apply and scale such solutions, appropriate legal frameworks are needed, and authorities including NRAs and grid operators need to collaborate.	Trinomics and Artelys, 2022, <i>Power System Flexibility in the Penta region – Current State and Challenges for a Future Decarbonised Energy System</i> Gerard Schouw, 2024, <i>Slim met Stroom voor Groene Groei</i>
Allow for value stacking	Value stacking refers to the provision of multiple flexibility services by the operator of a flexibility resource. This can encompass balancing and other ancillary services, distribution grid congestion management services, or arbitrage in wholesale energy markets. By combining revenues from various value streams, operators can enhance their earnings and achieve profitability for flexibility solutions that would otherwise be unprofitable if dependent on a single value stream.	Trinomics and Artelys, 2022, <i>Power System Flexibility in the Penta region – Current State and Challenges for a Future Decarbonised Energy System</i>
Hybrid PPAs	Direct Power Purchase Agreements (PPAs) between electricity generators/suppliers and off-takers are an adequate contract type to reduce exposure of project developers to merchant risk and ensure participation of diverse RES off-takers (corporate and utility customers). A solution to simultaneously drive maturity in these PPA markets and increase flexibility includes pooling supply and demand, and developing hybrid PPAs that incorporate flexibility assets, hence allowing for more tailored offtake patterns and mitigate price risk.	European Commission, 2025, <i>The future of European competitiveness; The Clean Industrial Deal</i>
Support mechanisms		
Subsidies for flexibility assets (such as heat pumps and EVs)	Evaluate and adapt where appropriate the support schemes for electricity producers and end users to ensure they also provide adequate incentives for flexibility. Policy measures should encourage, or at the very least, not discourage the deployment of flexibility. Support for renewable energy, end-use electrification (such as heat pumps), capacity remuneration mechanisms, and other support measures should enable flexibility and certainly not obstruct it. Additional subsidy to boost demand response and storage systems, based on the EMDR. Encourage the widespread adoption of bidirectional charging for electric vehicles (EVs) by implementing appropriate consumer compensation mechanisms.	Trinomics and Artelys, 2022, <i>Power System Flexibility in the Penta region – Current State and Challenges for a Future Decarbonised Energy System</i> Magnus Energy, 2024, <i>Maximising distributed flexibility with effective TSO-DSO coordination</i> European Commission, 2025, <i>The future of European competitiveness</i>

Source: Authors' own elaboration.

5. KEY FINDINGS AND RECOMMENDATIONS

Flexibility in the energy system is acknowledged as being crucial to the energy transition. As electricity is more difficult to store than other energy vectors (e.g., gas, oil or thermal energy), **flexibility is particularly a critical issue in the electricity system**, which is the focus of this study. Nonetheless, resources in or at the interface with other energy sectors such as the gas/hydrogen and heat sectors, can provide significant contributions to the electricity system's flexibility. Therefore, to cost-efficiently cover the flexibility needs in the electricity system, a **multi-vector approach (sector coupling)** is required.

The **flexibility needs of the EU electricity system will significantly increase in the current and coming decades**, as the integration of variable RES and electrification of end-uses continues, compounding current challenges. The increase in flexibility needs in the (sub-)daily timeframe will be the largest, but flexibility needs in the weekly and seasonal timeframes will also be substantial.

The relative and absolute contribution of historical sources of flexibility will gradually decrease. The flexibility offered by dispatchable thermal power generation is currently high but will decrease due to the gradual phasing out of fossil-fuel based power plants. The current and future contribution of nuclear power plants to flexibility is limited, and the contribution of hydropower (including pumped hydro) to flexibility is at present substantial but further large-scale expansion is for technical and environmental reasons not feasible.

New solutions will thus have to be developed and upscaled to fulfil most of the flexibility needs in the 2030-2050 timeframe. There is high uncertainty about the exact potential and cost of each solution, and flexibility needs will substantially vary per Member State. Nonetheless, a mix of different solutions will be needed in all Member States. Demand response and energy storage (mobile and stationary batteries) will become major contributors of (sub-)daily and weekly flexibility. Flexible operation (demand response) of electrolyzers may also provide some short-term flexibility, while for seasonal flexibility hydrogen storage and other novel long-duration energy storage solutions are expected to contribute (complementing RES curtailment and reservoir/pumped storage hydropower). However, although the seasonal flexibility needs will be smaller than the daily/weekly ones, they may constitute a significant challenge as there are currently fewer mature solutions. Biomass/biogas-based power generation and system-friendly wind and solar energy could to some extent contribute to the flexibility puzzle.

Besides developing and implementing adequate flexibility solutions at the supply and demand side, the EU will need to **reinforce and modernise the cross-border and domestic transmission and distribution networks**. Such investments will not only reduce the flexibility needs but should also enable cost-efficient provision and (cross-border) sharing of flexibility resources. This is not just a future challenge. Investments in electricity networks need to urgently and substantially accelerate, as **renewable electricity development and electrification are facing significant challenges** across the EU, with several grid operators being unable to timely connect new electricity producers or users due to limited grid capacity. Structural congestion is increasingly occurring at national borders and in domestic grids with high resulting congestion management costs and welfare losses.

The **costs to ensure sufficient system flexibility and to expand and modernise the electricity networks will be significant**, while the recent energy crisis has brought to the fore affordability and competitiveness concerns. This highlights the need to continue showcasing the benefits of decarbonising and integrating the EU energy system, as well as adequately considering the

distributional impacts. Several policies related to for example providing adequate electricity price and network tariff signals to consumers require the consideration of impacts on affordability and fairness.

The deployment of new flexibility solutions still faces significant challenges, including arising from the market design and overall regulatory framework, in particular small-scale assets connected to distribution grids. For example, despite the significant demand response potential, it still faces restrictive requirements to provide congestion management services or retail price interventions in several Member States. The increasing number of negative clearing prices in several EU wholesale electricity markets illustrates that the **flexibility market is not yet adequately reacting to market price signals.**

Policy makers, regulators, network operators, and market players are already working on developing and implementing several flexibility-related policy measures. These derive from the ongoing implementation of the Clean Energy package and EU electricity market design reform provisions as well as the finalisation of the demand response network code. Nonetheless, continuous improvement of the EU electricity market design will be necessary, including with the adoption of new measures – while balancing the need not to compromise solid foundations nor risk a proliferation of incompatible national approaches.

Recommendations

With this context in mind, the following main **specific recommendations** are made to complement already ongoing efforts to promote and enable flexibility in the EU:

- Given the significant role of electricity interconnectors in reducing flexibility needs and enabling the sharing of flexibility resources, a **more integrated, top-down assessment of flexibility needs can be considered**. The flexibility needs assessment procedure which will be implemented is largely bottom-up, with national FNAs feeding into ACER's EU-wide flexibility report and the electricity TYNDP. Despite the mandated coordination between ENTSO-E, the EU DSO Entity, TSOs and DSOs, this could lead to difficulties in aggregating and comparing national flexibility needs. ENTSO-E and the EU DSO Entity will provide data input and methodological guidance to TSOs and DSOs on their assessments on the national level (incl. coordinated granularity regarding time, spatial and voltage factors), but have no mandate to outline the tasks of the designated national authorities/NRAs. ACER will conduct an EU-wide flexibility assessment employing modelling which could address this specific point, but further details were not made public yet. **ACER could also provide guidance to NRAs and network operators on the contents of the national FNAs.**
- FNAs and ACER's report on flexibility needs should also take into account and explicitly document the contribution of system integration (interactions between electricity and heat, methane, hydrogen) and regional cooperation (cross-border sharing of flexibility capacity and exchange of energy). **ACER could be mandated to include an explicit assessment of these issues in its report analysing submitted FNAs.**
- The points mentioned above also increase the **need for consistency and risk of overlaps between the European or National Resource Adequacy Assessments and the National Flexibility Needs Assessments**. A number of inputs and outputs of the ERAA will be considered in the FNAs, including the assumed installed capacity of flexibility resources, and ACER raised a number of recommendations regarding the representation of demand-side response resources in the 2023 ERAA¹²¹. Hence, adequate representation of new flexibility resources, system integration aspects and consistency between the EU (ENTSO-E's ERAA and ACER's flexibility report) and national analyses (national resource adequacy and flexibility assessments) will become even more important. We thus stress the need for **ACER continuing to provide recommendations across all these dimensions**. ACER is explicitly required to provide recommendations on this issue in its flexibility report as per Article 19e(7) of the revised Electricity Directive.
- Given the increasing impact of variable renewable electricity sources on the wholesale price formation and the occurrence of negative prices in several bidding zones, as well as the lack of visibility on the profitability of flexibility solutions, **national FNAs and the subsequent ACER report should conduct an evaluation of the economic feasibility of key or archetypal flexibility solutions**. Such an exercise is already part of the European Resource Adequacy Assessment and would provide guidance to policy makers on the need of the introduction of specific non-fossil flexibility support mechanisms to achieve the national flexibility targets, including for storage and demand response. This exercise could also include an estimation of

¹²¹ See section 6.2.8 of ACER, 2024, Decision No 06/2024 of 2 May 2024 on the European Resource Adequacy Assessment for 2023. Available at: <https://www.acer.europa.eu/news/acer-approves-european-resource-adequacy-assessment-eraa-marking-milestone-security>.

the market price capture for key renewable electricity sources and its impact on the profitability of renewable electricity production projects.

- Based on the European and national network development plans, the flexibility needs assessment and the specific national energy and climate targets, **integrated/coordinated infrastructure investment plans** should be developed for the different energy vectors (electricity, methane, hydrogen, heat). The investment plans should be consistent with the commitments endorsed in the NECPs and with the national/regional spatial planning. A coordinated approach should facilitate a smoother realisation of investments projects and reduce risks for public opposition and delays.
- Capacity remuneration mechanisms (CRM) mainly focus on system adequacy, but most assets supported by them can also provide flexibility. To ensure system adequacy and flexibility at least cost, all relevant technologies, including flexible power generation assets, energy storage and demand response **should be eligible for CRM participation, with adequate technology dependent parameters for considering their adequacy contributions**. Smaller assets should be enabled to participate via aggregation.
- EU and national authorities should stimulate **cross-border participation in CRMs, procurement of flexibility services by TSOs, and in non-fossil flexibility support schemes**. While specific provisions already exist at the EU level to stimulate this to some extent, cross-border participation in CRMs and procurement of flexibility services remain in practice limited, with the roll-out of the EU balancing platforms slow. Moreover, to avoid competition distortion and to facilitate cross-border trade of flexibility services, national CRMs and non-fossil flexibility support schemes should be further **harmonised**.
- National authorities should better value the flexibility in renewable energy support schemes by **considering this aspect in auction price criteria (by monetising expected flexibility contributions) or in system integration non-price criteria**¹²². National authorities could also require developers of large solar PV and wind energy projects to assess the feasibility of integrating co-located storage. This solution can reduce investment needs in grid capacity and contribute to system adequacy and flexibility.
- National authorities and grid operators should **enable the provision of flexibility by consumers/prosumers connected to the distribution grid** by a full roll-out of low voltage smart meters (if needed by conducting a new cost-benefit analysis for Member States which have not decided for a full roll-out in the past), adequate metering data management and exchange, and enabling administrative processes (e.g. aggregation). To incentivise the provision of implicit flexibility by end-users/prosumers in provision, time-differentiated grid tariff and market-based electricity price signals should be provided. Fixed retail prices can reduce the price volatility risks for vulnerable consumers (but retail price regulation should be phased-out in any case for non-vulnerable consumers), but supply contracts with variable or dynamic price formulas should at least be offered to active consumers and prosumers.
- EU and national authorities and system operators should **improve energy data management and flexibility solutions transparency**, by promoting the development of registers of flexibility resources (with adequate considerations for data privacy) as well as publishing national and EU statistics. The assumptions and results regarding the contribution of different

¹²² European Commission, 2024, *Recommendation on auction design for renewable energy*. Available at: <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32024H1344>.

flexibility solutions in the upcoming EU Reference Scenario 2025 should also be published. More transparency will enhance the visibility for investors in flexibility resources.

- In line with Article 6a of Electricity Directive (EU) 2024/1711, national authorities in Member States where structural congestion is becoming a critical issue should develop a legal framework for TSOs and DSOs offering them the possibility of establishing **flexible connection agreements** in areas where there is limited or no network capacity availability for new connections. Such agreements are already being implemented in a few Member States and should be considered as a temporary solution to avoid delays in the connection of (renewable) electricity generation projects and new industrial or commercial users. An adequate regulation should comprise appropriate connection tariffs and consider compensation to the concerned grid users, at least if the availability of the connection falls under pre-agreed thresholds, in order to balance the attractiveness of such agreements to network users against eventual costs to network operators.

REFERENCES

- ACER & CEER, 2023, *ACER-CEER Reaction to the European Commission's public consultation on electricity market design*. Available at: https://acer.europa.eu/sites/default/files/documents/Media/News/Documents/ACER-CEER_Response_EC_PC_EMD.pdf.
- ACER & CEER, 2024, *Energy retail - Active consumer participation is key to driving the energy transition: how can it happen?* Available at: https://www.acer.europa.eu/sites/default/files/documents/Publications/ACER-CEER_2024_MMR_Retail.pdf.
- ACER & EEA, 2023, *Flexibility solutions to support a decarbonised and secure EU electricity system*. Available at: https://www.acer.europa.eu/sites/default/files/documents/Publications/EEA-ACER_Flexibility_solutions_support_decarbonised_secure_EU_electricity_system.pdf.
- ACER, 2020, *ACER Decision on the ERAA methodology: Annex I Methodology for the European resource adequacy assessment*. Available at: https://www.acer.europa.eu/sites/default/files/documents/Individual%20Decisions_annex/ACER%20Decision%2002-2022%20-%20Annex%20I%20-%20Detailed%20ERAA%202021%20review_0.pdf.
- ACER, 2023, *Demand response and other distributed energy resources: what barriers are holding them back? 2023 Market Monitoring Report*. Available at: <https://acer.europa.eu/monitoring/MMR>.
- ACER, 2024, Decision No 06/2024 of 2 May 2024 on the European Resource Adequacy Assessment for 2023. Available at: <https://www.acer.europa.eu/news/acer-approves-european-resource-adequacy-assessment-eraa-marking-milestone-security>.
- ACER, 2024, *Electricity infrastructure development to support a competitive and sustainable energy system - 2024 Monitoring Report*. Available at: https://www.acer.europa.eu/monitoring/MMR/electricity_infrastructure_2024.
- ACER, 2024, *European Resource Adequacy Assessment*. Available at: <https://www.acer.europa.eu/electricity/security-of-supply/european-resource-adequacy-assessment>.
- ACER, 2024, *Progress of EU electricity wholesale market integration*. Available at: https://www.acer.europa.eu/sites/default/files/documents/Publications/ACER_2024_MMR_Market_Integration.pdf.
- ACER, 2024, *Public consultation on the draft network code on demand response*. Available at: <https://acer.europa.eu/documents/public-consultations/pc2024e07-public-consultation-draft-network-code-demand-response>.
- ACER, 2024, *Security of EU electricity supply*. Available at: https://www.acer.europa.eu/sites/default/files/documents/Publications/Security_of_EU_electricity_supply_2024.pdf.
- ACER, 2024, *Transmission capacities for cross-zonal trade of electricity and congestion management in the EU - 2024 Market Monitoring Report*. Available at: https://www.acer.europa.eu/monitoring/MMR/crosszonal_electricity_trade_capacities_2024.
- ACER, 2025, *Open call for experts to join ACER's new Expert Group on EU-wide flexibility needs assessment*. Available at: <https://www.acer.europa.eu/news-and-events/news/open-call-experts-join-acers-new-expert-group-eu-wide-flexibility-needs-assessment>
- Ackermann, L., et al., 2024, *Cost-benefit analysis and comparison of grid-stabilizing energy flexibility options and their applications in relation to the German energy system*. Available at: <https://www.sciencedirect.com/science/article/pii/S1755008424000346>.

- ACM, 2024, *ACM publishes overview of measures against grid congestion*. Available at: <https://www.acm.nl/en/publications/acm-publishes-overview-measures-against-grid-congestion>.
- ARENA, 2022, *DEIP Dynamic Operating Envelopes Workstream: Outcomes Report*. Available at: <https://arena.gov.au/knowledge-bank/deip-dynamic-operating-envelopes-workstream-outcomes-report/>.
- Bamboo energy, 2022, *Europastry*. Available at: <https://bambooenergy.tech/en/europastry/>.
- Bloomberg BNEF, 2024, *Lithium-Ion Battery Pack Prices See Largest Drop Since 2017, Falling to \$115 per Kilowatt-Hour*. Available at: <https://about.bnef.com/blog/lithium-ion-battery-pack-prices-see-largest-drop-since-2017-falling-to-115-per-kilowatt-hour-bloombergnef/>.
- CEER, 2023, *CEER Paper on Alternative Connection Agreements*. Available at: https://www.ceer.eu/wp-content/uploads/2024/04/ACA_2023.pdf.
- Draghi, M., 2024, *The future of European competitiveness*. Available at: https://commission.europa.eu/document/download/ec1409c1-d4b4-4882-8bdd-3519f86bbb92_en?filename=The%20future%20of%20European%20competitiveness_%20In-depth%20analysis%20and%20recommendations_0.pdf.
- EEA et al., 2023, *Flexibility solutions to support a decarbonised and secure EU electricity system*. Available at: <https://www.eionet.europa.eu/etcs/etc-cm/products/flexibility-solutions-to-support-a-decarbonised-and-secure-eu-electricity-system>.
- EEAS, 2024, *Joint Statement by the European Commission and the High Representative on the Investigation into Damaged Electricity and Data Cables in the Baltic Sea*. Available at: https://www.eeas.europa.eu/eeas/joint-statement-european-commission-and-high-representative-investigation-damaged-electricity-and_en.
- Elering, s.d., *Flexible connections*. Available at: <https://elering.ee/en/flexible-connections>.
- Elia, 2023, *Adequacy & flexibility study for Belgium (2024-2034)*. Available at: <https://www.elia.be/en/electricity-market-and-system/adequacy/adequacy-studies>.
- Ember, 2023, *Draft NECPs show EU just falling short of REPowerEU*. Available at: https://ember-energy.org/app/uploads/2023/11/Report_Draft-NECPs-show-EU-just-falling-short-of-REPowerEU.pdf.
- Ember, 2024, *Clean flexibility is the brain managing the clean power system*. Available at: <https://ember-energy.org/latest-insights/clean-flexibility-is-the-brain-managing-the-clean-power-system/>.
- Ember, 2024, *Making Clean Power Flexy*. Available at: <https://ember-energy.org/latest-insights/making-clean-power-flexy/>.
- Ember, 2025, *European Electricity Review 2025*. Available at: <https://ember-energy.org/latest-insights/european-electricity-review-2025/>.
- Energy Brainpool, 2017, *Flexibility Needs and Options for Europe's Future Electricity System*. Available at: <https://www.eugene.eu/category/brochures/>.
- ENTSO-E & ENTSG, 2025, *TYNDP 2024 - Final Scenarios Report*. Available at: <https://2024.entsos-tyndp-scenarios.eu/>.

- ENTSO-E & EU DSO Entity, 2024, *Definition of the type and format of data and the methodology for the analysis by transmission system operators and distribution system operators of the flexibility needs at national level*. Available at: https://consultations.entsoe.eu/system-development/public-consultation-on-flexibility-needs-assessment/supporting_documents/1_Flexibility_Needs_Assessment_Methodology.pdf.
- ENTSO-E, 2024, *Overview of Frequency Control in the Nordic Power System*. Available at: <https://www.entsoe.eu/publications/system-operations-reports/>.
- ENTSO-E, 2024, *PICASSO*. Available at: https://www.entsoe.eu/network_codes/eb/picasso/.
- ENTSO-E, 2024, *System Flexibility Needs for the Energy Transition*. Available at: <https://www.entsoe.eu/system-flexibility/>.
- ENTSO-E, 2025, *Opportunities for a more efficient European power system by 2050: Infrastructure Gaps Report TYNDP 2024 - Version for public consultation*. Available at: <https://tyndp.entsoe.eu/>.
- EPEX Spot, 2024, *EPEX SPOT & Local Flexibility - GOPACS developments*.
- EU DSO Entity & ENTSO-E, 2024, *Proposal for a network Code on Demand Response*. Available at: https://eepublicdownloads.blob.core.windows.net/public-cdn-container/clean-documents/Network_codes_documents/NC_DR/NCDR_DSO_ENTITY_ENTSO-E.pdf.
- Eurelectric, 2022, *The Coming Storm - Building electricity resilience to extreme weather*. Available at: <https://resilience.eurelectric.org/>.
- Eurelectric, 2024, *Understanding ultra-low and negative power prices: causes, impacts and improvements*. Available at: <https://www.eurelectric.org/wp-content/uploads/2024/11/Eurelectric-explainer-on-negative-prices.pdf>.
- European Commission, 2022, *Non paper on complementary economic modelling undertaken by DG ENER analysing the impacts of overall renewable energy target of 45% to 56% in the context of discussions in the European Parliament on the revision of the Renewable Energy Directive*. Available at: https://energy.ec.europa.eu/system/files/2022-06/2022_06_20_RED_non-paper_additional_modelling.pdf.
- European Commission, 2023, *Commission sets out actions to accelerate the roll-out of electricity grids*. Available at: https://ec.europa.eu/commission/presscorner/api/files/document/print/en/ip_23_6044/IP_23_6044_EN.pdf.
- European Commission, 2023, *EU action plan for grids*. Available at: <https://data.europa.eu/doi/10.2775/919160>.
- European Commission, 2024, *Quarterly report on European electricity markets, Volume 17*. Available at: https://energy.ec.europa.eu/data-and-analysis/market-analysis_en.
- European Commission, 2024, *Recommendation on auction design for renewable energy*. Available at: <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32024H1344>.
- European Commission, 2024, *SWD/2024/63 final: Impact assessment 2040 climate target*. Available at: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52024SC0063>.
- European Commission, 2025, *2050 long-term strategy*. Available at: https://climate.ec.europa.eu/eu-action/climate-strategies-targets/2050-long-term-strategy_en.

- European Commission, 2025, *A Competitiveness Compass for the EU. COM(2025) 30 final*. Available at: https://ec.europa.eu/commission/presscorner/detail/en/ip_25_339.
- European Commission, 2025, *Action Plan for Affordable Energy. COM(2025) 79 final*. Available at: https://energy.ec.europa.eu/publications/action-plan-affordable-energy-unlocking-true-value-our-energy-union-secure-affordable-efficient-and_en.
- European Commission, 2025, *The clean industrial deal: a joint roadmap for competitiveness and decarbonisation. COM(2025) 85 final*. Available at: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52025DC0085>.
- European Commission, n.d., *Electricity interconnection targets*. Available at: https://energy.ec.europa.eu/topics/infrastructure/electricity-interconnection-targets_en.
- European Environmental Agency, Trinomics et al., 2022, *Adaptation challenges and opportunities for the European energy system*. Available at: <https://www.eea.europa.eu/en/analysis/publications/adaptation-in-energy-system>
- Flexitricity, 2024, *Load management helps Norish Cold Storage reduce GB carbon emissions and generate revenue*. Available at: <https://www.flexitricity.com/resources/case-studies/load-management-helps-norish-cold-storage-reduce-gb-carbon-emissions-and-generate-revenue>.
- Frontier Economics & ENTSO-E, 2022, *Potential of P2H technologies to provide system services*. Available at: <https://www.entsoe.eu/2022/06/28/entso-e-publishes-a-study-on-flexibility-from-power-to-hydrogen-p2h2/>.
- Gimeno-Gutierrez, M., et al., 2015, *Assessment of the European potential for pumped hydropower energy storage based on two existing reservoirs*. Available at: <https://www.sciencedirect.com/science/article/abs/pii/S096014811400706X>.
- GOPACS, s.d., main website. Available at: <https://www.gopacs.eu/en/>.
- Henning Schuster & E-Bridge, 2017, *Sichere und effiziente Koordinierung von Flexibilitäten im Verteilnetz: Beitrag zur weiteren Ausgestaltung der Rolle des Verteilnetzbetreibers in der Energiewende*. Available at: <https://e-bridge.de/wp-content/uploads/2022/11/E-Bridge-Abschlussbericht-Sichere-und-effiziente-Koordinierung-von-Flexi....pdf>.
- IEA, 2021, *Technologies for Flexible Bioenergy*. Available at: <https://www.ieabioenergy.com/wp-content/uploads/2021/08/Task-44-Technologies-for-Flexible-Bioenergy.pdf>.
- IEA, 2024, *Batteries and Secure Energy*, Available at: <https://iea.blob.core.windows.net/assets/cb39c1bf-d2b3-446d-8c35-aae6b1f3a4a0/BatteriesandSecureEnergyTransitions.pdf>.
- IEA, 2024, *Managing seasonal variability of electricity demand and supply*, Figure 12. Available at: <https://www.iea.org/reports/managing-the-seasonal-variability-of-electricity-demand-and-supply>.
- IEA, 2025, *Electricity 2025*. Available at: <https://iea.blob.core.windows.net/assets/0f028d5f-26b1-47ca-ad2a-5ca3103d070a/Electricity2025.pdf>.
- JRC, 2023, *Flexibility requirements and the role of storage in future European power systems*. Available at: <https://publications.jrc.ec.europa.eu/repository/handle/JRC130519>.
- JRC, 2023, *Hydropower and pumped hydropower storage in the European Union*. Available at: https://setis.ec.europa.eu/hydropower-and-pumped-hydropower-storage-european-union-0_en.

- Letta, E., 2024, *Much more than a market*. Available at: <https://www.consilium.europa.eu/media/ny3j24sm/much-more-than-a-market-report-by-enrico-letta.pdf>.
- Magnus Energy, 2024, *Disentangling flexibility and its needs assessment methodology*. Available at: <https://www.magnusenergy.com/insights/working-paper-flexibility-needs-assessment>.
- Magnus Energy, 2024, *Maximising distributed flexibility with effective TSO-DSO coordination*. Available at: <https://www.magnusenergy.com/insights/TSO-DSO-coordination>.
- Next, 2023, *FlexAbility: How Battery Storage is Advancing the Energy Transition*. Available at: <https://www.next-kraftwerke.com/energy-blog/flexibility-battery-storage>.
- Next-kraftwerke, 2023, *Buffering the energy crisis with wind power and storage by Peleman industries*. Available at: <https://www.next-kraftwerke.com/vpp/case-studies/battery-bess-fcr>.
- Ondřej, Š., et al., 2024, *Electric vehicles as facilitators of grid stability and flexibility: A multidisciplinary overview*. Available at: <https://wires.onlinelibrary.wiley.com/doi/10.1002/wene.536>.
- PV Magazine, 2024, *Polish grid operator switches off gigawatts of PV*. Available at: <https://www.pv-magazine.com/2024/03/27/polish-grid-operator-switches-off-gigawatts-of-pv>.
- Pwtech, 2024, *Transmission grid congestion could hinder renewable power deployment in Spain – DNV*. Available at: <https://www.pv-tech.org/transmission-grid-congestion-could-hinder-renewable-power-deployment-spain-dnv/>.
- Science and Technology Committee House of Lords UK, 2024, *Long-duration energy storage: get on with it*. Available at: <https://publications.parliament.uk/pa/ld5804/ldselect/ldsctech/68/68.pdf>.
- SIA Partners, 2024, *Stationary battery storage*. Available at: <https://www.sia-partners.com/en/insights/publications/stationary-battery-storage>.
- SmartEn & DNV, 2022, *Demand-side flexibility: Quantification of benefits in the EU*. Available at: <https://www.dnv.com/publications/demand-side-flexibility-quantification-of-benefits-in-the-eu-232342/>.
- SmartEn, 2024, *Map on Wholesale Markets: Enabling demand-side flexibility in Europe*. Available at: <https://smarten.eu/reports/2024-smarten-map-on-wholesale-markets-enabling-demand-side-flexibility-in-europe/>.
- Trinomics & Artelys, 2022, *Study on flexibility options to support decarbonization in the Energy Community*.
- Trinomics & Artelys, 2023, *Power System Flexibility in the Penta region*. Available at: <https://www.benelux.int/nl/etude/power-system-flexibility-in-the-penta-region-current-state-and-challenges-for-a-future-decarbonised-energy-system-2/>.
- Trinomics, 2024, *Advancing cross-system solutions to address electricity network challenges*. Available at: https://www.acer.europa.eu/sites/default/files/documents/Publications/Consultancy_study_Electricity_Network_Solutions_2024.pdf.
- Wehrmann, B., 2024, *Germany delays gas plant decision, 2030 state coal phase-out uncertain*, Clean Energy Wire. Available at: <https://www.cleanenergywire.org/news/germany-delays-gas-plant-decision-2030-state-coal-phase-out-uncertain>.

- WindEurope, 2024, *Maximising the power of wind through grid flexibility*, based on *Power for 127 Hours: The Economics of Long-Duration Energy Storage*. Available at: <https://environment-review.yale.edu/power-127-hours-economics-long-duration-energy-storage>.
- Xoserve, 2023, 'Blog: Energy storage—the missing piece of the net zero puzzle?'. Available at: www.xoserve.com/news/blog-energy-storage-the-missing-piece-of-the-net-zero-puzzle.
- Zambotti, 2024, *ETPA Whitepaper – GOPACS*. Available at: <https://www.etpa.nl/en/whitepapers/gopacs>.

This study explores the current and future flexibility needs in the electricity system as well as existing and potential solutions to efficiently integrate renewable energy and facilitate electrification of end-uses. Key findings highlight the importance of flexible electricity generation, demand response and energy storage solutions for daily, weekly, and seasonal flexibility provision. However, challenges persist due to inadequate market rules and regulatory frameworks, and economic issues. The study recommends additional policy measures to enhance system flexibility, complementing ongoing policy initiatives at the EU and national level.

This document was provided by the Policy Department for Transformation, Innovation and Health for the Committee on Industry, Research and Energy (ITRE).
